The Asymmetric Metric-Tensor and the Thermodynamic Equation of State

Binyamin Tsadik Bair-Mosheh

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Mathematical Concordance: All calculations, tensor contractions, and dimensional scaling relations in this manuscript have been machine-checked across 180+ theorems with zero custom axioms in the Lean 4 theorem prover.

Abstract

This document resumes Einstein's effort to unify gravitation and electromagnetism by generalizing the classical symmetric metric into a complete Asymmetric Metric-Tensor ($\hat{U}_{\mu\nu}$). By relaxing the assumption of an irrotational vacuum baseline, the tensor forward-models both macroscopic spacetime curvature (General Relativity) and discrete quantum kinematics (Electromagnetism and Mass) as orthogonal, coupled strain components of a continuous manifold.

The scope of this manuscript constructs the primary tensor, computes its principal eigenvalues, and applies symmetry-breaking Lie derivatives to resolve the trace contractions of the matrix. These operations derive the four fundamental forces as boundary conditions of a single continuous pressure gradient. Beyond recovering the admittance-modified Navier-Stokes equation, the Kinematic Lagrangian and the Schwarzschild metric, the tensor extracts the cosmological constant (Λ), the physical Planck mass yield limit, the Compton cavitation boundary, anomalous galactic kinematics without dark matter, and cosmological redshift via thermodynamic metric capacity.

The capability of a single tensor to reproduce these scale-invariant phenomena validates the asymmetric continuous baseline, while advanced phenomenological applications emerge as predictive corollaries. This verification establishes the tensor as a forward-modeling engine for continuous physics, confirming Einstein's historical intuition that physical unification requires an asymmetric geometric extension.

1. Kinematic Parametrization (Clebsch Variables)

A continuous metric governed by a scalar potential (dh) is irrotational ($d(dh) = 0$). In order to capture the dynamics of topological defects and rotational shear, we first parameterize the macroscopic velocity field as a 1-form ($\mathbf{v}$) by using Clebsch variables [13].

Spatial vorticity is described by using a continuous velocity 1-form via orthogonal conjugate components:

$$\mathbf{v}(\mathbf{x}, t) = dh + \mathbf{v}_\perp + \lambda_\Omega d\beta_\Omega \tag{1}$$

Where dh represents the irrotational and longitudinal phase (compression and translation), whereas $\mathbf{v}_\perp$ represents propagating transverse shear, and $\lambda_\Omega d\beta_\Omega$ represents the internal rotational phase (topological twisting). The exterior derivative (d) is applied with respect to the flat Euclidean background (δ_{ij} in $\mathbb{R}^3$).

Taking the exterior derivative of the rotational velocity field extracts the internal topological vorticity 2-form ($\mathbf{\Omega}$):

$$\begin{aligned}\mathbf{\Omega} &= d(\lambda_\Omega d\beta_\Omega) \\ \mathbf{\Omega} &= d\lambda_\Omega \wedge d\beta_\Omega\end{aligned} \tag{2}$$

This parametrization represents the kinematic behavior of the metric: the irrotational phase represents longitudinal pressure, while the Clebsch wedge product represents spatial vorticity and the shear stress required for invariant rest mass [54].

2. The Thermodynamic Budget (Energy Conservation)

The metric sustains volumetric compression under applied kinematic load. To determine the total capacity of the baseline vacuum, the kinetic energy density of the metric is maximized to the acoustic phase limit (c). This defines the finite, background yield-stress, establishing the Critical Pressure ($P_c = \frac{1}{2}\rho_\tau c^2$). By the conservation of energy, the sum of remaining static pressure (P_{static}) and all applied kinematic loads equates to this limit.

Parsing the two scale domains of structural load:

- **Macroscopic Baseline Load (ΔP_{macro}):** The depletion of ambient capacity induced via cosmic expansion and gravitational bodies. Integrating the macroscopic Hubble parameter ($v = H_0 r$) with characteristic scale $R_h = c/H_0$ yields the velocity profile ($v_0(r) = c(1 - e^{-r/R_h})$). This background pressure produces $\Delta P_{macro} = \frac{1}{2}\rho_\tau v_0^2$, constructing an equation of state with maximum metric compliance at the origin ($r = 0, v_0 = 0$), $v_0(R_h) = c(1-e^{-1})$ at $r = R_h$, and asymptotic saturation ($v_0 \to c, \Delta P_{macro} \to P_c$) at large optical depth ($r \gg R_h$) [2] (derived in Section 15).
- **Microscopic Load ($P_{dyn} + P_{shear}$):** The kinetic translation and structural vorticity of a boundary.

The Nested Ledger:
The ambient capacity of a frame, pre-strained by the background, constructs the Ambient Pressure ($P_{ambient}$):

$$P_{ambient} = P_c - \Delta P_{macro} = P_c - \frac{1}{2}\rho_\tau v_0^2 \tag{3}$$

This macroscopic pressure deficit translates into the Gravitational Compliance (α_g):

$$\alpha_g^2 = \frac{P_{ambient}}{P_c} = 1 - \frac{v_0^2}{c^2} \implies P_{ambient} = P_c \alpha_g^2 \tag{4}$$

Kinematics consume the available ambient capacity. Dynamic pressure (P_{dyn}) is quantified by taking the root-sum-square of orthogonal velocities, relative to the background density (ρ_τ):

$$P_{dyn} = \frac{1}{2}\rho_\tau \left(v_\parallel^2 + v_\perp^2 + v_{\circlearrowleft}^2\right) \tag{5}$$

3. Assembly of the Asymmetric Metric-Tensor ($\hat{U}_{\mu\nu}$)

The corresponding thermodynamic loads hold representation in metric geometry; the parameters map to a dimensionless matrix. An asymmetric second-rank tensor decomposes into a symmetric component (volumetric strain) and an antisymmetric component (deviatoric shear strain). This summation is isomorphic to the Clifford product ($uv = u \cdot v + u \wedge v$), formulating spacetime as a continuous multivector and defining the metric as a micropolar Cosserat continuum [15, 25, 18]:

$$\hat{U}_{\mu\nu} = \hat{U}_{(\mu\nu)} + \hat{U}_{[\mu\nu]} \equiv S_{\mu\nu} + A_{\mu\nu} \tag{6}$$

To satisfy covariance, the metric is constructed via the continuous 4-velocity (u_μ) and the spatial projection tensor ($h_{\mu\nu}$). The symmetric sector ($S_{\mu\nu}$) and the antisymmetric sector ($A_{\mu\nu}$) split as:

$$\hat{U}_{\mu\nu} = \underbrace{-\alpha_s^2 u_\mu u_\nu + \frac{1}{\alpha_s^2} h_{\mu\nu}}_{S_{\mu\nu}} + \underbrace{\alpha \left(\frac{u_\mu v_{\perp,\nu} - u_\nu v_{\perp,\mu}}{c} \right) + \alpha(t_p)\omega_{\mu\nu}}_{A_{\mu\nu}} \tag{7}$$

Where the covariant tensor variables are:

- u_μ: **Continuum 4-Velocity.** The normalized, timelike vector field ($u_\mu u^\mu = -1$) representing the local rest frame of the coordinate metric. In this comoving frame, the spatial components evaluate to zero ($u^i = 0$) and the temporal component evaluates to $u^0 = 1$, isolating bulk geometric strain to the symmetric diagonals.
- $h_{\mu\nu}$: **Spatial Projection Tensor.** The symmetric tensor ($h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$) that projects the metric onto the orthogonal spatial hypersurface, where $g_{\mu\nu}$ is the unperturbed Minkowski background signature $(-, +, +, +)$.
- α_s: **Total Scalar Admittance.** The dimensionless phase capacity bounded by thermodynamic saturation, representing the scalar contraction of the symmetric sector.
- α: **Viscoelastic Modulus Ratio.** The dimensionless transverse coupling constant. Tracks the fractional capacity of the metric to sustain transverse shear relative to bulk compression.
- $v_{\perp,\mu}$: **Transverse Slip Vector.** The orthogonal spatial velocity representing solenoidal slip, coupling to the time-space components via the antisymmetric wedge product.
- $\omega_{\mu\nu}$: **Vorticity Tensor.** The antisymmetric component representing the curl of the Clebsch rotational phase.
- t_p: **Structural Relaxation Time.** The temporal phase limit ($t_p = l_p/c$) that normalizes the vorticity into dimensionless strain.

3.1 The Symmetric Diagonals (Isotropic Volumetric Strain)

The symmetric components construct the bulk geometric packing of the coordinate grid. Because the background metric responds to scalar thermodynamic pressure (P_{dyn}), the resulting spatial compression lacks an independent directional vector. A single scalar constant occupies all three spatial directions simultaneously, constituting isotropic volumetric strain ($\hat{U}_{11} = \hat{U}_{22} = \hat{U}_{33}$).

- **Temporal Component ($\hat{U}_{00}$):** Phase update capacity. The time-time component equates to the negative Total Scalar Admittance ($\hat{U}_{00} = -\alpha_s^2 = -\alpha_g^2\alpha_a^2$). This component acts as the product of gravitational and kinematic time dilation, reducing to $\hat{U}_{00} = -1$ in the unperturbed Minkowski vacuum ($\alpha_s = 1$).

- **Spatial Components ($\hat{U}_{ii}$):** Effective volumetric density and spatial packing of the coordinate grid. The isotropic spatial metric coefficients operate as the inverse of the Total Scalar Admittance ($\hat{U}_{ii} = 1/\alpha_s^2$, preserving the reciprocal invariant $\hat{U}_{00}\hat{U}_{ii} = -1$). The Riemannian metric volume element is governed by the symmetric sector determinant $g \equiv \det(S_{\mu\nu}) = -\alpha_s^{-4}$ (with 4D volume density $\sqrt{-g} = \alpha_s^{-2}$, spatial determinant $\det(S_{ij}) = \alpha_s^{-6}$, and 3D spatial volume density $\sqrt{\det(S_{ij})} = \alpha_s^{-3}$), whereas the full asymmetric determinant $\det(\hat{U}_{\mu\nu})$ includes off-diagonal shear terms (Section 9.7). As $\alpha_s \to 0^+$ toward the cavitation boundary, the spatial compliance diagonals diverge ($\hat{U}_{ii} \to \infty$).

3.2 The Antisymmetric Off-Diagonals (Transverse Shear)

The antisymmetric components capture the dynamic sliding and twisting of the metric. Translating these physical observables into a dimensionless strain matrix is achieved via normalization:

- **Viscoelastic Throttling (α):** The metric capacity to sustain transverse shear acts as a geometric fraction of its bulk compressive capacity. The off-diagonals are attenuated by the Viscoelastic Modulus Ratio (α).
- **Transverse Velocity ($\hat{U}_{0i}$):** The temporal rate of transverse displacement ($v_{\perp,i}$). Normalized by the acoustic phase limit (c), this defines the continuous strain of the Electric Field: $\hat{U}_{0i} = -\alpha\left(\frac{v_{\perp,i}}{c}\right)$.
- **Rotational Vorticity ($\hat{U}_{ij}$):** The exterior derivative of the metric's rotational phase ($d\lambda_\Omega \wedge d\beta_\Omega$). Because vorticity is a physical frequency (Ω_k), it is scaled by the structural relaxation time of the metric ($t_p \equiv l_p/c$) to yield a dimensionless strain. This determines the Deborah Number of the metric, generating the Magnetic Field: $\hat{U}_{ij} = \alpha(t_p\Omega_k)$.

3.3 The Final Tensor Matrix

Superimposing the symmetric and antisymmetric components yields the 4×4 Asymmetric Metric-Tensor for a Cartesian coordinate referenced against the unperturbed vacuum (c):

$$\hat{U}_{\mu\nu} = \begin{bmatrix} -\alpha_s^2 & -\alpha\left(\frac{v_{\perp,x}}{c}\right) & -\alpha\left(\frac{v_{\perp,y}}{c}\right) & -\alpha\left(\frac{v_{\perp,z}}{c}\right) \\ \alpha\left(\frac{v_{\perp,x}}{c}\right) & \frac{1}{\alpha_s^2} & -\alpha(t_p\Omega_z) & \alpha(t_p\Omega_y) \\ \alpha\left(\frac{v_{\perp,y}}{c}\right) & \alpha(t_p\Omega_z) & \frac{1}{\alpha_s^2} & -\alpha(t_p\Omega_x) \\ \alpha\left(\frac{v_{\perp,z}}{c}\right) & -\alpha(t_p\Omega_y) & \alpha(t_p\Omega_x) & \frac{1}{\alpha_s^2} \end{bmatrix} \tag{8}$$

Where the Cartesian matrix variables are specified as:

- $v_{\perp,i}/c$: **Dimensionless Transverse Slip.** The temporal rate of spatial sliding along the i-axis, normalized by the acoustic phase limit. Generates the electric field vector (E_i).
- $t_p\Omega_i$: **Dimensionless Vorticity.** The spatial curl of the Clebsch phase around the i-axis, normalized by the structural relaxation limit. Generates the magnetic field vector (B_i).

This architecture enables cross-coupling. As gravity, velocities, or internal vorticities increase, the scalar capacity (α_s^2) approaches zero. The spatial diagonals expand, generating a dense boundary that limits the metric's capacity to sustain transverse waves in the off-diagonals (enforcing the coordinate limits of the speed of light).

4. The Continuous Field Equation

Because $\hat{U}_{\mu\nu}$ represents dimensionless strain, it balances against the physical stress that caused the deformation. Applying generalized Hooke's Law, the dimensionless tensor is multiplied by the thermodynamic stiffness of the baseline vacuum (P_c).

Macroscopic field propagation is governed by the global background. The resulting field equates to the continuous asymmetric Cauchy stress tensor ($\hat{T}_{\mu\nu}$) of the overarching metric. Topological defects emerge as steady-state circulation structures embedded within this continuous medium.

The Unified Field Equation is defined as the continuum's stress deviation from the flat Minkowski baseline ($\eta_{\mu\nu}$):

$$\hat{T}_{\mu\nu} = P_c\Big(\hat{U}_{\mu\nu} - \eta_{\mu\nu}\Big) \tag{9}$$

To maintain covariance near topological boundaries where dynamic pressure approaches the baseline capacity ($P_{dyn} \to P_c$), this linear relation transitions to a non-linear hyperelastic

extension. To evaluate this limit, we define the second invariant constraint (I_2) not as the trace, but as the full scalar contraction of the strain tensor ($I_2 = \hat{U}_{\alpha\beta}\hat{U}^{\alpha\beta}$).

At the unperturbed Minkowski baseline ($\eta_{\mu\nu}$), this invariant evaluates to 4:

$$I_2^{(vac)} = \eta_{\mu\nu}\eta^{\mu\nu} = (-1)(-1) + (1)(1) + (1)(1) + (1)(1) = 4 \tag{10}$$

This invariant bounds the stress tensor to prevent infinite deformation at the boundary:

$$\hat{T}_{\mu\nu} = P_c \left(\hat{U}_{\mu\nu} - \eta_{\mu\nu} \right) + 2C_1 \left(I_2 - 4 \right) \hat{U}_{\mu\nu} \tag{11}$$

Because weak-field coordinates operate far below the saturation threshold ($I_2 \approx 4$), the hyperelastic coefficient (C_1, possessing units of thermodynamic pressure) term attenuates, recovering the linear baseline for standard field interactions.

4.1 Energy Density and Structural Stress

Isolating the temporal (00) slot computes the continuous equivalent of energy density. Given $\eta_{00} = -1$:

$$\begin{aligned} \hat{T}_{00} &= P_c \left(-\frac{P_{static}}{P_c} - (-1) \right) \\ \hat{T}_{00} &= P_c - P_{static} \\ \hat{T}_{00} &= \Delta P_{macro} + P_{dyn} + P_{shear} \end{aligned} \tag{12}$$

Energy density ($\hat{T}_{00}$) reduces to the sum of the macroscopic gravitational deficit and the internal kinematic pressure of the mass defect.

Evaluating the individual spatial components (no sum) determines the continuous structural stress. Given $\eta_{ii} = 1$:

$$\hat{T}_{ii} = P_c \left(\frac{P_c - P_{static}}{P_{static}} \right) = P_c \left(\frac{\Delta P_{macro} + P_{dyn} + P_{shear}}{P_{static}} \right) \tag{13}$$

The physical coordinate stress maps to the total applied pressure load scaled by the inverse of the available static capacity. As structural loads saturate the metric ($P_{static} \to 0$), the physical spatial stress diverges ($\hat{T}_{ii} \to \infty$), establishing the cavitation limit without a singularity.

4.2 The Cosmological Constant & Metric Density (ρ_τ)

Because the Unified Field Equation (9) evaluates the deviatoric stress applied by kinematic defects, an unperturbed coordinate space ($\hat{U}_{\mu\nu} = \eta_{\mu\nu}$) evaluates to zero ($\hat{T}_{\mu\nu} = 0$). To derive the baseline metric density (ρ_τ), the background stress tensor is defined as ($\Sigma_{\mu\nu}^{(vac)} = -P_c\eta_{\mu\nu}$), which establishes the isotropic tension of the manifold. We evaluate the spatial surface integral of this vacuum stress across the minimum resolvable coordinate boundary (a sphere of Planck radius, $A_p = 4\pi l_p^2$).

By divergence equilibrium, the total outward spatial force exerted by the static vacuum stress (P_c) on this boundary computes to $F = P_c(4\pi l_p^2)$. Because rest energy is evaluated over one full phase cycle (derived formally in Section 14 as $E = mc^2$), this boundary equates to the tension limit of the manifold ($F_p = c^4/G$).

Equating these bounds sets the microscopic yield pressure:

$$P_c(4\pi l_p^2) = \frac{c^4}{G} \implies P_c = \frac{c^4}{4\pi G l_p^2} \tag{14}$$

Substituting the geometric definition of the Planck area ($l_p^2 = \hbar G/c^3$) defines the metric stiffness:

$$P_c = \frac{c^7}{4\pi\hbar G^2} \tag{15}$$

Equating this to dynamic pressure ($P_c = \frac{1}{2}\rho_\tau c^2$) extracts the baseline metric density:

$$\begin{aligned} \frac{1}{2}\rho_\tau c^2 &= \frac{c^7}{4\pi\hbar G^2} \\ \rho_\tau &= \frac{2P_c}{c^2} = \frac{1}{2\pi}\left(\frac{c^5}{\hbar G^2}\right) \end{aligned} \tag{16}$$

The density of the metric is fixed as $1/2\pi$ of the Planck density.

The Macroscopic Limit (The Epoch-Dependent Cosmological Parameter Λ): The continuous manifold is bounded by the causal horizon ($R_h = c{\cdot}t$). Because the total tension of the spacetime manifold (c^4/G) acts as an invariant geometric constant, the force projected across any continuous 3D isotropic spherical boundary remains conserved ($F = P \cdot A = c^4/G$).

At the microscopic limit, this isolates the Planck stiffness ($P_c \cdot 4\pi l_p^2 = c^4/G$). As the universe expands, this invariant tension distributes over the causal area ($4\pi R_h^2$). The observed macroscopic vacuum pressure (P_{macro}) is therefore the geometric dilution of the baseline metric stiffness:

$$P_{macro}(4\pi R_h^2) = P_c(4\pi l_p^2) = \frac{c^4}{G} \tag{17}$$

Isolating the macroscopic pressure yields:

$$P_{macro} = \frac{c^4/G}{4\pi R_h^2} = \frac{c^4}{4\pi G R_h^2} \tag{18}$$

Substituting this scaled macroscopic pressure into the definition of the Cosmological Scalar ($\Lambda = 8\pi G P_{macro}/c^4$) yields:

$$\begin{aligned} \Lambda &= \frac{8\pi G}{c^4}\left(\frac{c^4}{4\pi G R_h^2}\right) \\ \Lambda &= \frac{2}{R_h^2} \end{aligned} \tag{19}$$

Substituting the present-day Hubble radius ($R_h = c/H_0 \approx 1.37 \times 10^{26}$ m) calculates the local snapshot of the metric:

$$\begin{aligned} \Lambda &= \frac{2}{(1.37 \times 10^{26}\ \mathrm{m})^2} \\ \Lambda &= 1.06 \times 10^{-52}\ \mathrm{m}^{-2} \end{aligned} \tag{20}$$

This reproduces the Planck 2018 empirical measurement to within 4%: ($\Lambda_{obs} \approx 1.1{\times}10^{-52}$ m^{-2}) [45].

The Poisson Limit:

Extracting the individual spatial components (no sum) of the continuous Cauchy stress tensor from the baseline field equation yields:

$$\hat{T}_{ii} = P_c\Big(\hat{U}_{ii} - \eta_{ii}\Big) \tag{21}$$

In the weak-field limit, the symmetric spatial diagonal acts as a linear perturbation of the volumetric packing. Because gravitational potential (Φ) represents a negative geometric energy

state ($P_{dyn} = -\rho_\tau \Phi$), the spatial strain expands inversely to the dropping scalar admittance ($\hat{U}_{ii} = 1/\alpha_s^2 \approx 1 - \frac{2\Phi}{c^2}$). Substituting this and the flat Minkowski baseline ($\eta_{ii} = 1$) isolates the continuous structural stress:

$$\hat{T}_{ii} = P_c \left(1 - \frac{2\Phi}{c^2} - 1\right)$$
$$\hat{T}_{ii} = P_c \left(-\frac{2\Phi}{c^2}\right) \tag{22}$$

Substituting the stiffness ($P_c = \frac{1}{2}\rho_\tau c^2$) eliminates the phase limits. This defines the restorative mechanical stress holding the mass defect in place. Because the potential is negative ($\Phi < 0$), the tensor generates a positive tensile stress with a negative spatial gradient:

$$\hat{T}_{ii} = \left(\frac{1}{2}\rho_\tau c^2\right)\left(-\frac{2\Phi}{c^2}\right)$$
$$\hat{T}_{ii} = -\rho_\tau \Phi \tag{23}$$

4.3 The Macroscopic Reduction (The Einstein Field Equation)

The classical formalism of General Relativity resolves as a macroscopic limit of the Unified Field Equation.

1. The Macroscopic Boundary Constraint:
At the cosmological scale, the restoring thermodynamic stiffness of the continuum is governed by the scaled macroscopic vacuum pressure (P_{macro}). Applying this boundary constraint to the Unified Field Equation limits the macroscopic Cauchy stress ($\hat{T}^{macro}_{\mu\nu}$):

$$\hat{T}^{macro}_{\mu\nu} = P_{macro}\Big(\hat{U}_{\mu\nu} - \eta_{\mu\nu}\Big) \tag{24}$$

2. The Defect Density and Thermodynamic Substitution:
As derived in Section 4.2, the restoring macroscopic pressure is parameterized by the expanding causal horizon ($P_{macro} = \frac{c^4}{4\pi G R_h^2}$). Macroscopic mass-energy introduces topological defects, mapping the defect density to the classical Stress-Energy Tensor ($T_{\mu\nu}$). Standard GR assumes a symmetric stress-energy tensor because macroscopic objects possess negligible net spin density.

3. The Volumetric Determinant (Jacobian Equivalence):
The symmetric diagonals ($S_{\mu\nu}$) measure how much a volume element is compressed or stretched, whereas the antisymmetric off-diagonals ($A_{\mu\nu}$) represent transverse shear and rotation (twisting). Therefore, pure volumetric compression is governed by the determinant of the symmetric metric sector ($g \equiv \det(S_{\mu\nu}) = -\alpha_s^{-4}$), whose square root defines the normalized spacetime volume factor ($J_{normalized}$):

$$\alpha_s^{-2} = \frac{\sqrt{-g}}{\sqrt{-g_0}} \equiv J_{normalized}, \qquad g \equiv \det(S_{\mu\nu}) = -\alpha_s^{-4} \tag{25}$$

Here $\sqrt{-g_0} = 1$ is the unperturbed vacuum baseline ($\alpha_s = 1$), and the 3D spatial block determinant $\det(S_{ij}) = \alpha_s^{-6}$ gives the corresponding spatial volume factor $\sqrt{\det(S_{ij})} = \alpha_s^{-3}$. Taking the determinant of the full asymmetric matrix $\det(\hat{U}_{\mu\nu})$ instead cross-multiplies the off-diagonal shear components into the result: as derived in Section 9.7 for orthogonal slip and vorticity ($\mathbf{E}\cdot\mathbf{B} = 0$), $\det(\hat{U}_{\mu\nu}) = -\alpha_s^{-4}(1-E^2) - B^2$, which equals $g = \det(S_{\mu\nu})$ under $\mathbf{E}\cdot\mathbf{B} = 0$ only when the electric and magnetic shear terms cancel ($E^2 = \alpha_s^4 B^2$, including $A_{\mu\nu} = 0$). Equation (25) thus maps the scalar vacuum admittance (α_s^2) to the volumetric compression of the symmetric manifold.

4. The Differential Incompatibility Mapping (The Kröner Formulation):
To map the zero-order asymmetric strain ($\hat{U}_{\mu\nu}$) to the second-order curvature tensor ($G_{\mu\nu}$) without manually stripping the antisymmetric components ($A_{\mu\nu}$), we apply the mathematics of continuum defect theory.

Macroscopic mass acts as a Volterra dislocation [55] within the continuous metric. The geometric consequence of the continuum warping to accommodate these spatial defects is measured by the Saint-Venant incompatibility tensor (Inc) [48, 33]. We apply this continuous differential operator to the covariant strain tensor:

$$G_{\mu\nu} \equiv \mathrm{Inc}(\hat{U}_{\mu\nu}) \approx \partial_\alpha \partial_\beta (\hat{U}_{\mu\nu} - \eta_{\mu\nu}) \epsilon^{\alpha\mu\rho} \epsilon^{\beta\nu\sigma} \eta_{\rho\sigma} \tag{26}$$

5. Torsion Nullification at the Macroscopic Limit (Coarse-Graining):
Because the strain tensor contains both symmetric volumetric strain ($S_{\mu\nu}$) and antisymmetric transverse shear ($A_{\mu\nu}$), the incompatibility operator inherently evaluates both. To prove that the antisymmetric sector decouples from the macroscopic large-scale curvature, we apply the mathematical formalism of spatial coarse-graining over a macroscopic volume V where $V \gg l_p^3$.

The macroscopic observable strain is defined as the volume average of the microscopic fields:

$$\langle \hat{U}_{\mu\nu} \rangle_V = \frac{1}{V} \int_V (S_{\mu\nu} + A_{\mu\nu})\, d^3x \tag{27}$$

In unpolarized mass distributions, transverse shear (quantum vorticity/spin and electromagnetic dipoles) are isotropically distributed. Their phase orientations are randomized, meaning their static vector sum across the macroscopic volume V statistically cancels:

$$\frac{1}{V} \int_V A_{\mu\nu}^{static} d^3x = 0 \implies \langle A_{\mu\nu}^{static} \rangle_V = 0 \tag{28}$$

Conversely, propagating electromagnetic waves exist as zero-mean oscillatory perturbations ($\Box A_{\mu\nu}^{dynamic} = 0$). While their kinematic energy propagates through the metric, their linear spatial average across macroscopic volumes evaluates to zero.

Because the Saint-Venant incompatibility tensor (Inc) is a linear differential operator with respect to strain, the incompatibility of the macroscopic average is equal to the average of the microscopic incompatibility:

$$\mathrm{Inc}(\langle A_{\mu\nu} \rangle_V) = \langle \mathrm{Inc}(A_{\mu\nu}) \rangle_V = 0 \tag{29}$$

Furthermore, this cancellation aligns with the established constraints of Einstein-Cartan-Sciama-Kibble (ECSK) theory [28, 49]. In ECSK spacetime, the antisymmetric torsion tensor is coupled to the spin density of matter. Unlike symmetric curvature, torsion is non-propagating; it vanishes outside the defect.

The smoothing of quantum properties ($\mathrm{Inc}(A_{\mu\nu}) \to 0$) does not apply to either a locally polarized dynamic or to macroscopic asymmetry ($\langle A_{\mu\nu} \rangle_V \neq 0$) such as cosmological torsion events.

However, General Relativity operates under the physical reality that most of the bulk mass distributions are randomly distributed. Under this assumption, the surviving equation relies solely on the symmetric incompatibility ($G_{\mu\nu} \approx \mathrm{Inc}(S_{\mu\nu})$) to generate classical large-scale gravitational curvature, preserving the continuous vacuum's capacity to propagate zero-mean transverse waves.

6. The Classical Limit and Scale Invariance:
As formalized by Kröner and extended to gravitational geometry by Katanaev and Volovich [27], the structural incompatibility of metric strain is directly proportional to the defect density tensor. When mapping it to spacetime, this defect density tensor functions identically to the

classical Stress-Energy Tensor ($T_{\mu\nu}$) [29]. Scaling this defect density by the thermodynamic stiffness limits of the vacuum ($8\pi G/c^4$) equates the geometric curvature to the physical stress-energy:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \tag{30}$$

The Einstein Field Equation resolves as the macroscopic, unpolarized compatibility condition of the full unified field. The antisymmetric components ($A_{\mu\nu}$) are never zeroed out or removed from the baseline tensor ($\hat{U}_{\mu\nu}$). While their net curvature footprint averages to zero for randomly oriented bulk mass, they remain active across all scales. They govern quantum mechanics and electrodynamics, and when coherent, they scale proportionally to dictate macroscopic rotational kinematics at the galactic level, demonstrating the universal scale-invariance of the overarching field equations.

5. The Equation of Motion (Divergence Equilibrium)

Energy-momentum conservation emerges via the spacetime divergence of the stress tensor ($\nabla_\mu \hat{T}^{\mu\nu} = 0$). The covariant derivative incorporates the non-linear affine connection ($\Gamma^\lambda_{\mu\nu}$) constructed in Section 12. At macroscopic boundaries ($r \gg r_c$), the antisymmetric torsion ($T^\lambda_{\mu\nu} = \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu}$) decays, reducing $\Gamma^\lambda_{\mu\nu}$ to the symmetric Levi-Civita connection. Under the weak-field flat-background approximation ($\Gamma^\lambda_{(\mu\nu)} \approx 0$ against $\eta_{\mu\nu}$), the covariant divergence reduces to the partial derivative ($\partial_\mu \hat{T}^{\mu\nu} \approx 0$).

To extract the equation of motion, Momentum density ($\hat{T}^{0i}$) is derived from the covariant parameters of the metric tensor ($\hat{U}^{\mu\nu}$). Evaluating the temporal-spatial component of Equation (7) with the normalized 4-velocity ($u^\mu = \gamma(1, v^i/c)$) and the spatial projection tensor ($h^{0i} = \eta^{0i} + u^0 u^i = \gamma^2 v^i/c$) isolates the symmetric field strain:

$$S^{0i} = -\alpha_s^2 u^0 u^i + \frac{1}{\alpha_s^2} h^{0i} = -\gamma^{-2}\left(\gamma^2 \frac{v^i}{c}\right) + \gamma^2\left(\gamma^2 \frac{v^i}{c}\right) = \frac{v^i}{c}(\gamma^4 - 1) \tag{31}$$

Applying the leading weak-field expansion ($\gamma^4 - 1 \approx 2v^2/c^2$) retains S^{0i} at $\mathcal{O}(v^3/c^3)$ (dropping the $\mathcal{O}(v^5/c^5)$ remainder). Multiplying by the baseline stiffness ($P_c = \frac{1}{2}\rho_\tau c^2$) translates this strain into the Cauchy stress momentum density ($\hat{T}^{0i} = P_c S^{0i}$):

$$\hat{T}^{0i} = \left(\frac{1}{2}\rho_\tau c^2\right)\frac{v^i}{c}\left(\frac{2v^2}{c^2}\right) = 2\left(\frac{1}{2}\rho_\tau \frac{v^2}{c^2}\right) c v^i \tag{32}$$

Defining density as the mass equivalent of pressure ($\rho_{dyn} = P_{dyn}/c^2 = \frac{1}{2}\rho_\tau v^2/c^2$), the tensor outputs momentum density flux ($\hat{T}^{0i} = 2c\rho_{dyn} v^i$).

Because ρ_{dyn} already carries a factor of v^2/c^2, retaining the convective flux ($\rho_{dyn} v^i v^j$) requires expanding the symmetric spatial components (S^{ij}) to $\mathcal{O}(v^4/c^4)$ using $\gamma^2 \approx 1 + v^2/c^2 + v^4/c^4$ and $\gamma^4 - 1 \approx 2v^2/c^2$ (dropping the $\mathcal{O}(v^6/c^6)$ remainder):

$$S^{ij} = -\alpha_s^2 u^i u^j + \frac{1}{\alpha_s^2} h^{ij} = \gamma^2 \delta^{ij} + (\gamma^4 - 1)\frac{v^i v^j}{c^2} \approx \left(1 + \frac{v^2}{c^2} + \frac{v^4}{c^4}\right)\delta^{ij} + 2\frac{v^2}{c^2}\frac{v^i v^j}{c^2} \tag{33}$$

Multiplying by the baseline stiffness translates this strain into the Cauchy stress ($\hat{T}^{ij} = P_c(S^{ij} - \delta^{ij}) - \tau^{ij}$):

$$\hat{T}^{ij} \approx P_c\left(\frac{v^2}{c^2} + \frac{v^4}{c^4}\right)\delta^{ij} + P_c\left(2\frac{v^2}{c^2}\frac{v^i v^j}{c^2}\right) - \tau^{ij} = \frac{1}{2}\rho_\tau v^2\left(1 + \frac{v^2}{c^2}\right)\delta^{ij} + \left(\rho_\tau \frac{v^2}{c^2}\right) v^i v^j - \tau^{ij} \tag{34}$$

Absorbing the isotropic compressibility factor into the effective pressure ($P_{total} = \frac{1}{2}\rho_\tau v^2(1 + v^2/c^2) \approx \frac{1}{2}\rho_\tau v^2$ for $v^2 \ll c^2$), the tensor produces the spatial flux:

$$\hat{T}^{ij} = P_{total}\delta^{ij} + 2\rho_{dyn} v^i v^j - \tau^{ij} \tag{35}$$

5.1 The Equation of Motion (Navier-Stokes)

To construct the equation of motion independent of scalar energy density variations, the 4D spacetime divergence is expanded across its temporal and spatial components ($\nabla_0 \hat{T}^{0i} + \nabla_j \hat{T}^{ji} = 0$). By applying the affine connection contraction ($\Gamma^{\lambda}_{\mu\lambda} = \partial_\mu \ln\sqrt{-g}$), the covariant momentum transport resolves across curved coordinates. In the previously established macroscopic weak-field limit, this expansion simplifies to the partial derivative domain.

Substituting the derived macroscopic stress components ($\hat{T}^{0i} = 2c\rho_{dyn}v^i$ and $\hat{T}^{ij} = P_{total}\delta^{ij} + 2\rho_{dyn}v^iv^j - \tau^{ij}$):

$$\frac{1}{c}\partial_t(2c\rho_{dyn}v^i) + \partial_j(P_{total}\delta^{ij} + 2\rho_{dyn}v^iv^j - \tau^{ij}) = 0 \tag{36}$$

Dividing the entire equilibrium equation by 2 isolates the dynamic density (ρ_{dyn}) relative to the structural pressure:

$$\partial_t(\rho_{dyn}v^i) + \partial_j\left(\frac{1}{2}P_{total}\delta^{ij} + \rho_{dyn}v^iv^j - \frac{1}{2}\tau^{ij}\right) = 0 \tag{37}$$

Applying the product rule to the convective momentum terms expands the temporal and spatial derivatives:

$$v^i\partial_t\rho_{dyn} + \rho_{dyn}\partial_t v^i + \frac{1}{2}\partial^i P_{total} + v^i\partial_j(\rho_{dyn}v^j) + \rho_{dyn}v^j\partial_j v^i - \frac{1}{2}\partial_j\tau^{ij} = 0 \tag{38}$$

Grouping the kinetic terms extracts the acceleration from the volumetric mass flux:

$$\rho_{dyn}(\partial_t v^i + v^j\partial_j v^i) + v^i\left[\partial_t\rho_{dyn} + \partial_j(\rho_{dyn}v^j)\right] = -\frac{1}{2}\partial^i P_{total} + \frac{1}{2}\partial_j\tau^{ij} \tag{39}$$

Assuming steady-state Eulerian mass conservation for the dynamic density envelope ($\partial_t\rho_{dyn} + \nabla\cdot(\rho_{dyn}\mathbf{v}) = 0$), the remaining kinetic term is the definition of the Eulerian Material Derivative ($\frac{D\mathbf{v}}{Dt} \equiv \partial_t\mathbf{v} + (\mathbf{v}\cdot\nabla)\mathbf{v}$), determining the coordinate acceleration of the coordinate frame:

$$\rho_{dyn}\frac{D\mathbf{v}}{Dt} = -\frac{1}{2}\nabla P_{total} + \frac{1}{2}\nabla\cdot\boldsymbol{\tau} \tag{40}$$

1. The Stress Tensor Partition:
The Cauchy stress on the right side of Equation (40) is driven by the components of the Tensor.

- **Symmetric Volumetric Stress ($-\frac{1}{2}\nabla P_{total}$):** The spatial diagonals ($\hat{U}_{ii}$) govern volumetric packing. A boundary condition upon the local frame identifies the total coordinate stress (P_{total}) as the superposition of the macroscopic force of the environment ($-\frac{1}{2}\nabla P_{ambient}$) and the compressive surface force of the defect ($-\frac{1}{2}\nabla P_{dyn}$).
- **Antisymmetric Deviatoric Shear ($\frac{1}{2}\nabla\cdot\boldsymbol{\tau}$):** The off-diagonals control transverse shear and the divergence of Clebsch vorticity ($\nabla\cdot\boldsymbol{\Omega}$). This transverse stress is attenuated by the Viscoelastic Modulus Ratio (α).

Substituting the partitioned stress components into the momentum equation yields:

$$\rho_{dyn}\frac{D\mathbf{v}}{Dt} = -\frac{1}{2}\nabla P_{ambient} - \frac{1}{2}\nabla P_{dyn} + \frac{1}{2}\alpha\big(\nabla\cdot\mathbb{T}_\perp - \mu\nabla\cdot(d\lambda_\Omega\wedge d\beta_\Omega)\big) \tag{41}$$

2. Kinematic Normalization:
To resolve the dimensional units from force density (N/m^3) into kinematic acceleration (m/s^2) the equation is divided by the baseline density (ρ_τ).

This defines the mass-independent kinematic pressure gradients and stresses (absorbing the geometric halving factor):

$$\mathcal{P}_{ambient} = \frac{P_{ambient}}{2\rho_\tau} \tag{42}$$

$$\mathcal{P}_{dyn} = \frac{P_{dyn}}{2\rho_\tau} \tag{43}$$

$$\mathbb{T}_{\perp,kin} = \frac{\mathbb{T}_\perp}{2\rho_\tau} \tag{44}$$

$$\nu = \frac{\mu}{2\rho_\tau} \tag{45}$$

Applying these invariant dimensional substitutions constructs the kinematic equation of motion:

$$\left(\frac{\rho_{dyn}}{\rho_\tau}\right)\frac{D\mathbf{v}}{Dt} = -\nabla\mathcal{P}_{ambient} - \nabla\mathcal{P}_{dyn} + \alpha\big(\nabla\cdot\mathbb{T}_{\perp,kin} - \nu\nabla\cdot(d\lambda_\Omega \wedge d\beta_\Omega)\big) \tag{46}$$

The divergence equilibrium conserves total angular momentum by transferring macroscopic deviatoric shear into the internal spin of the coordinate phase. The tensor establishes the Weak Equivalence Principle mechanically: macroscopic gravitational acceleration ($-\nabla\mathcal{P}_{ambient}$) operates independently of internal mass, while mechanical propulsion is bounded by the dynamic density limits of the coordinate frame.

5.2 The Incompatible Defect (Compliance-Modified Momentum)

Section 5.1 evaluates the compatible strain ($\mathrm{Inc}(\hat{U}_{\mu\nu}) = 0$) of the massless background metric, scaling momentum via the dynamic flow density ($\rho_{dyn} = \frac{1}{2}\rho_\tau v^2/c^2$). Conversely, when kinematic strain reaches the capacity limit ($v \to c$), the dynamic density climbs to its geometric maximum ($\rho_{dyn} \to \frac{1}{2}\rho_\tau$). The spatial diagonals of the tensor diverge ($\hat{U}_{ii} \to \infty$). The metric cavitates, generating a topological defect. The continuous strain field becomes incompatible ($\mathrm{Inc}(\hat{U}_{\mu\nu}) \neq 0$). This establishes two distinct mass regimes: ρ_{dyn} governs the kinetic flow of the unbroken continuous field, whereas the inertia of the yielded defect is governed by the displaced added mass of the metric.

1. The Defect Stress-Energy Tensor & Kinematic Inertia:
As established in Section 4.3, this strain incompatibility projects through the field equation ($G_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}$) to construct the stress-energy tensor of the defect ($T_{\mu\nu}$). For the extended boundary surrounding the core, the stress-energy tensor is parameterized by the invariant mass density (ρ_τ) and the total isotropic pressure (P) in the rest frame, supplemented by the transverse deviatoric shear tensor ($\mathbb{T}_\perp^{\mu\nu}$).

Because the defect is generated by the yielding of the continuum, it inherits the baseline density of the metric (ρ_τ), which acts as the density of volumetric strain. However, its macroscopic rest mass is not the passive coordinate volume of the yielded core (V_{r_c}). Adopting the generalized hydrodynamic added-mass identification ($m = C_m\rho_\tau V_\tau$) for the defect, consistency with the Compton cavitation boundary of Section 8.3 ($r_c = \hbar/mc$) fixes the effective strain volume (V_τ). Here, C_m represents the species-specific topological form factor determined by the geometric circulation of the given defect. While Section 8.3 derives the scalar magnitude of the rest mass (m) via the antisymmetric trace, the added-mass formulation dictates the spatial distribution of that mass across the boundary.

$$V_\tau := \frac{m}{C_m \rho_\tau} = \frac{1}{C_m \rho_\tau} \left(\frac{\hbar}{r_c c} \right)$$
$$V_\tau = \frac{\hbar}{C_m r_c c} \left(\frac{2\pi \hbar G^2}{c^5} \right) = \frac{2\pi \hbar^2 G^2}{C_m r_c c^6} = \frac{2\pi l_p^4}{C_m r_c} \tag{47}$$

The inverse-cube relation to the physical coordinate volume ($V_{r_c} \cdot V_\tau^3 = \frac{32\pi^4}{3C_m^3} l_p^{12}$) follows as a consequence of this equivalence constraint. The geometric integration of V_τ from the tensor strain field ($\hat{U}_{\mu\nu}$) is left to future work.

The relativistic inertial mechanics of this boundary are carried by the coordinate 4-velocity (u^μ). To produce energy density (J/m^3), the mass components are scaled by the phase limit (c^2). To project the geometric coupling, transverse shear is attenuated by the Viscoelastic Modulus Ratio (α):

$$T^{\mu\nu} = \left(\rho_\tau c^2 + P \right) \frac{u^\mu u^\nu}{c^2} + P g^{\mu\nu} + \alpha \mathbb{T}_\perp^{\mu\nu} \tag{48}$$

To preserve covariant consistency, the deviatoric shear tensor is transverse and traceless, satisfying the standard orthogonality constraints ($u_\mu \mathbb{T}_\perp^{\mu\nu} = 0$ and $g_{\mu\nu} \mathbb{T}_\perp^{\mu\nu} = 0$). Operating in the relaxation limit, it functions as the relativistic Israel-Stewart shear tensor [26], housing the viscous friction ($-2\mu\sigma^{\mu\nu}$), where $\sigma^{\mu\nu} = \frac{1}{2}(h^{\mu\alpha}h^{\nu\beta} + h^{\mu\beta}h^{\nu\alpha} - \frac{2}{3}h^{\mu\nu}h^{\alpha\beta})\nabla_\alpha u_\beta$ defines the standard relativistic shear.

2. The Covariant Divergence Expansion:
By the contracted Bianchi identity [6], the covariant divergence of the stress-energy tensor equates to zero ($\nabla_\mu T^{\mu\nu} = 0$). Projecting this divergence onto the spatial hypersurface via the projection tensor ($h_\nu^\lambda = \delta_\nu^\lambda + u^\lambda u_\nu / c^2$) produces the relativistic momentum equation:

$$h_\nu^\lambda \nabla_\mu T^{\mu\nu} = \left(\rho_\tau + \frac{P}{c^2} \right) u^\mu \nabla_\mu u^\lambda + h^{\lambda\mu} \nabla_\mu P + h_\nu^\lambda \nabla_\mu (\alpha \mathbb{T}_\perp^{\mu\nu}) = 0 \tag{49}$$

Because the convective derivative of the 4-velocity constructs the covariant 4-acceleration ($a^\lambda \equiv u^\mu \nabla_\mu u^\lambda$), the dynamic acceleration of the topological boundary is separated:

$$\left(\rho_\tau + \frac{P}{c^2} \right) a^\lambda = -h^{\lambda\mu} \nabla_\mu P - h_\nu^\lambda \nabla_\mu (\alpha \mathbb{T}_\perp^{\mu\nu}) \tag{50}$$

3. Pressure Partition & Equivalence:
The total macroscopic isotropic pressure (P) partitions into the ambient environmental pressure ($P_{ambient}$) and the localized dynamic pressure of the defect (P_{dyn}).

This separates the momentum into Body Forces and Surface Forces:

- **Body Forces (Gravity):** The macroscopic ambient gradient dictates the gravitational 4-acceleration vector (g^λ). Because the Weak Equivalence Principle mandates that inertial mass and passive gravitational mass are identical, the body force density computes as $\left(\rho_\tau + \frac{P}{c^2} \right) g^\lambda = -h^{\lambda\mu} \nabla_\mu P_{ambient}$.
- **Surface Forces ($\Sigma \mathcal{F}_{surface}^\lambda$):** Kinematic gradients ($-h^{\lambda\mu} \nabla_\mu P_{dyn}$) and transverse deviatoric shear ($-h_\nu^\lambda \nabla_\mu (\alpha \mathbb{T}_\perp^{\mu\nu})$) act upon the spatial geometry of the boundary, operating as independent force densities.

Substituting these partitioned forces into the momentum equilibrium (Equation 50) constructs the general balance:

$$\left(\rho_\tau + \frac{P}{c^2} \right) a^\lambda = \left(\rho_\tau + \frac{P}{c^2} \right) g^\lambda + \Sigma \mathcal{F}_{surface}^\lambda$$
$$\left(\rho_\tau + \frac{P}{c^2} \right) a^\lambda = \left(\rho_\tau + \frac{P}{c^2} \right) g^\lambda - h^{\lambda\mu} \nabla_\mu P_{dyn} - h_\nu^\lambda \nabla_\mu (\alpha \mathbb{T}_\perp^{\mu\nu}) \tag{51}$$

Dividing the equation by the relativistic density separates the covariant 4-acceleration (a^λ):

$$a^\lambda = g^\lambda + \left(\frac{1}{\rho_\tau + \frac{P}{c^2}}\right)\Big[- h^{\lambda\mu}\nabla_\mu P_{dyn} - h^\lambda_\nu \nabla_\mu(\alpha \mathbb{T}^{\mu\nu}_\perp)\Big] \tag{52}$$

The inertial density fraction cancels from the gravitational body force, ensuring gravity accelerates all topological defects identically regardless of mass or velocity. Conversely, 4-acceleration from mechanical surface forces remains divided by the inertial density ($\rho_\tau + P/c^2$). As dynamic pressure approaches the capacity limit ($\alpha_s \to 0$), the internal pressure (P) diverges to match the spatial stress ($\hat{T}_{ii}$). This diverging pressure drives the acceleration denominator to infinity. Consequently, finite surface forces cannot accelerate a boundary at the kinematic phase limit ($v \to c$), producing relativistic force suppression from the metric compliance limit.

4. Boundary Matching (The Phase Transition):
This incompatible momentum state is bounded by the metric. As the applied dynamic pressure dissipates and the ambient compliance returns to unity ($\alpha_s \to 1$), the incompatibility dissolves ($\mathrm{Inc}(\hat{U}_{\mu\nu}) \to 0$). The topological boundary returns to the compatible metric state, and the momentum carrier transitions from the compacted defect mass (ρ_τ) back to the continuous dynamic flow density (ρ_{dyn}) governed by Section 5.1.

5.3 The Archimedean Limit of the Geodesic Equation

A void submerged within a pressure gradient experiences a restorative buoyant force directed toward the lowest static pressure. Newtonian gravity emerges as macroscopic metric buoyancy directly from the geodesic equation of motion.

The equation of motion for a test coordinate in a continuous metric is governed by the geodesic equation ($\frac{d^2x^i}{d\tau^2} = -\Gamma^i_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau}$). This derivation isolates the Newtonian limit by evaluating a static metric ($\partial_0 \hat{U}_{\mu\nu} = 0$) and a non-relativistic test coordinate ($dx^i/d\tau \ll c$). While relativistic or time-varying geometries require the full non-linear evaluation of the metric connection, this boundary condition reduces the equation to the temporal Christoffel symbol ($a^i \approx -c^2\Gamma^i_{00}$).

For a static metric, the connection evaluates as $\Gamma^i_{00} \approx -\frac{1}{2}\eta^{ij}\partial_j \hat{U}_{00}$. Substituting this derivative determines the coordinate acceleration:

$$\mathbf{a} \approx \frac{1}{2}c^2 \nabla \hat{U}_{00} \tag{53}$$

Equating the perturbation to the kinematic metric definition ($\hat{U}_{00} = -P_{static}/P_c$) limits the acceleration as a function of the static pressure gradient:

$$\begin{aligned}
\mathbf{a} &= \frac{1}{2}c^2 \nabla\left(-\frac{P_{static}}{P_c}\right) \\
\mathbf{a} &= -\frac{c^2}{2\left(\frac{1}{2}\rho_\tau c^2\right)}\nabla P_{static} \\
\mathbf{a} &= -\frac{1}{\rho_\tau}\nabla P_{static}
\end{aligned} \tag{54}$$

The physical force ($\mathbf{F}_g$) acting upon a particle is derived by multiplying this coordinate acceleration against the particle's entrained hydrodynamic added mass (m_τ):

$$\mathbf{F}_g = m_\tau \mathbf{a} = -\left(\frac{m_\tau}{\rho_\tau}\right)\nabla P_{static} \tag{55}$$

The total phase capacity consumed within the continuous metric functions as an Effective Thermodynamic Displacement ($V_{eff} = m_\tau/\rho_\tau$). This effective volume represents the total

hydrodynamic added mass rather than the physical volume of the cavitation. Substituting this effective displacement constructs the equation of state:

$$\mathbf{F}_g = -V_{eff} \nabla P_{static} \tag{56}$$

To derive the static pressure gradient without external kinematic postulates, the overarching metric must be evaluated as a continuous 3D spatial geometry. In a static continuous medium, the macroscopic scalar strain generated by a spherically symmetric cavitation boundary resolves via the radial Laplace equation ($\nabla^2 P_{dyn} = 0$).

Expanding the Laplacian operator in spherical coordinates establishes the structural conservation of the spatial pressure gradient across the expanding coordinate area (r^2):

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial P_{dyn}}{\partial r} \right) = 0 \tag{57}$$

Integrating the equation once extracts the strain gradient. The constant of integration ($-k$) represents the structural scalar magnitude of the defect:

$$\begin{aligned} r^2 \frac{\partial P_{dyn}}{\partial r} &= -k \\ \nabla P_{dyn} &= -\frac{k}{r^2} \hat{\mathbf{r}} \end{aligned} \tag{58}$$

Integrating this gradient radially outward dictates the continuous scalar dynamic pressure field:

$$\begin{aligned} P_{dyn}(r) &= \int -\frac{k}{r^2} \, dr \\ P_{dyn}(r) &= \frac{k}{r} \end{aligned} \tag{59}$$

By Bernoulli conservation ($P_{static} = P_c - P_{dyn}$), the spatial gradient of the static pressure equates to the inverse dynamic gradient:

$$\nabla P_{static} = -\nabla P_{dyn} = -\left(-\frac{k}{r^2} \hat{\mathbf{r}} \right) = \frac{k}{r^2} \hat{\mathbf{r}} \tag{60}$$

Inserting this restorative pressure gradient into the metric buoyancy equation (56) calculates the physical force exerted upon an advecting boundary:

$$\begin{aligned} \mathbf{F}_g &= -V_{eff} \nabla P_{static} \\ \mathbf{F}_g &= -\left(\frac{m_\tau}{\rho_\tau} \right) \left(\frac{k}{r^2} \hat{\mathbf{r}} \right) \end{aligned} \tag{61}$$

To bridge this continuum mechanic to classical kinematics, empirical mass (M) maps to the geometry. The continuous metric does not treat mass as an independent input parameter; rather, it evaluates here as the structural magnitude of the defect (k) normalized by the metric density (ρ_τ) and the macroscopic displacement limit (G).

Applying the geometric limits of macroscopic measurement ($G = l_p c^2 / m_p$ and $\rho_\tau = m_p / 2\pi l_p^3$) strips the empirical constants. The hydrodynamic volume displacement defining this boundary evaluates in Section 8, but the scalar isomorphism resolves the boundary ratio as a Planck-scale hemisphere:

$$M \equiv \frac{k}{\rho_\tau G} \implies k = \frac{M}{2\pi t_p^2} \tag{62}$$

Substituting this normalized definition into the dynamic pressure field recovers the kinematic profile:

$$P_{dyn} = \frac{M}{2\pi t_p^2 r} = \rho_\tau \frac{GM}{r} \tag{63}$$

Substituting the identical definition into the buoyant force equation isolates the Newtonian limit:

$$\mathbf{F}_g = -\frac{GMm_\tau}{r^2}\hat{\mathbf{r}} \tag{64}$$

The tensor geodesic equation functions as the physical definition of Archimedes' Principle. Universal gravitation manifests as the symmetric spatial gradients of the continuous metric, adopting macroscopic mass as a consequence of dynamic pressure.

6. Unified Action & Singularity Resolution

The total unified action emerges as a consequence of the tensor contraction.

6.1 Unified Action

To evaluate the global energy equilibrium of the tensor, action is extracted via the Lorentzian trace contraction of the Metric-Tensor against itself ($\hat{U}_{\mu\nu}\hat{U}^{\mu\nu} = \eta^{\mu\alpha}\eta^{\nu\beta}\hat{U}_{\mu\nu}\hat{U}_{\alpha\beta}$, with indices raised by the inverse Minkowski metric $\eta^{\mu\nu}$). Because the tensor decomposes into symmetric ($S_{\mu\nu}$) and antisymmetric ($A_{\mu\nu}$) components, the expansion produces:

$$\hat{U}_{\mu\nu}\hat{U}^{\mu\nu} = S_{\mu\nu}S^{\mu\nu} + A_{\mu\nu}A^{\mu\nu} + 2S_{\mu\nu}A^{\mu\nu} \tag{65}$$

By the rules of tensor calculus, the inner product of a symmetric matrix and an antisymmetric matrix reduces to zero ($S_{\mu\nu}A^{\mu\nu} = 0$).

Because macroscopic expansion ($v_0(r)$) acts as radial volumetric strain, the strain partitions into independent scalars:

$$\hat{U}_{\mu\nu}\hat{U}^{\mu\nu} = S_{\mu\nu}S^{\mu\nu} + A_{\mu\nu}A^{\mu\nu} \tag{66}$$

These two macroscopic actions emerge from a single constraint:

- **The Gravitational Action ($S_{\mu\nu}S^{\mu\nu}$):** The contraction of the symmetric diagonals captures the volumetric compression and spatial packing of the coordinate grid. Because the unified stress tensor is divergence-free ($\nabla_\mu \hat{T}^{\mu\nu} = 0$), Lovelock's Theorem [37] mandates that its symmetric geometric representation must map to the Einstein Tensor ($\hat{T}^{\mu\nu} \propto G^{\mu\nu}$). Applying the background boundary scaling derived in Section 4, this proportionality defines the macroscopic curvature limit:

$$P_{macro}(S_{\mu\nu} - \eta_{\mu\nu}) = \frac{c^4}{8\pi G}G_{\mu\nu} \implies S_{\mu\nu} = \frac{c^4}{8\pi G P_{macro}}G_{\mu\nu} + \eta_{\mu\nu} \tag{67}$$

  Because the cosmological constant equates to this ratio ($\Lambda = 8\pi G P_{macro}/c^4$), substituting this limit ($S_{\mu\nu} = \frac{1}{\Lambda}G_{\mu\nu} + \eta_{\mu\nu}$) resolves the strain against classical curvature invariants:

$$\begin{aligned} S_{\mu\nu}S^{\mu\nu} &= \left(\frac{1}{\Lambda}G_{\mu\nu} + \eta_{\mu\nu}\right)\left(\frac{1}{\Lambda}G^{\mu\nu} + \eta^{\mu\nu}\right) \\ S_{\mu\nu}S^{\mu\nu} &= \frac{1}{\Lambda^2}G_{\mu\nu}G^{\mu\nu} + \frac{2}{\Lambda}G_{\mu\nu}\eta^{\mu\nu} + \eta_{\mu\nu}\eta^{\mu\nu} \end{aligned} \tag{68}$$

  Because the Einstein Tensor is defined as $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}R\eta_{\mu\nu}$, expanding the squared term ($G_{\mu\nu}G^{\mu\nu}$) cancels the scalar traces:

$$G_{\mu\nu}G^{\mu\nu} = \left(R_{\mu\nu} - \frac{1}{2}R\eta_{\mu\nu}\right)\left(R^{\mu\nu} - \frac{1}{2}R\eta^{\mu\nu}\right) = R_{\mu\nu}R^{\mu\nu} - R^2 + \frac{1}{4}R^2(4) = R_{\mu\nu}R^{\mu\nu} \tag{69}$$

Substituting this identity, the symmetric contraction maps to a quadratic gravitational Lagrangian:

$$S_{\mu\nu}S^{\mu\nu} = \frac{1}{\Lambda^2}R_{\mu\nu}R^{\mu\nu} - \frac{2}{\Lambda}R + 4 \tag{70}$$

Therefore, volumetric strain contraction generates the Ricci scalar (R) alongside the higher-order quadratic curvature ($R_{\mu\nu}R^{\mu\nu}$) required for topological boundary regularization.

- **The Electromagnetic Action ($A_{\mu\nu}A^{\mu\nu}$):** The contraction of the antisymmetric off-diagonals quantifies propagating transverse shear ($\mathbf{E} \equiv \alpha \mathbf{v}_\perp/c$) and Clebsch vorticity ($\mathbf{B} \equiv \alpha t_p \boldsymbol{\Omega}$). Because raising indices with $\eta^{\mu\nu} = \text{diag}(-1,1,1,1)$ reverses the sign of the mixed time-space slip components ($A^{0i} = -A_{0i}$) while preserving the spatial vorticity components ($A^{ij} = +A_{ij}$), the Lorentzian contraction evaluates to $A_{\mu\nu}A^{\mu\nu} = 2\alpha^2(t_p^2\Omega^2 - v_\perp^2/c^2) = 2(B^2 - E^2)$, so that $-\frac{1}{4}A_{\mu\nu}A^{\mu\nu} = \frac{1}{2}(E^2 - B^2)$ recovers the Maxwell Lagrangian invariant attenuated by the viscoelastic coupling α^2 (whereas the positive-definite Euclidean Frobenius sum $\|A\|_F^2 = 2(E^2 + B^2)$ in Section 8 measures the total stored shear load).

Gravity and electromagnetism manifest as orthogonal mechanical responses (longitudinal compression versus transverse shear) originating from the same thermodynamic budget.

6.2 Strain Incompatibility & Riemann Curvature

Macroscopic curvature of the metric is derived via the continuous deformation of the Tensor ($\hat{U}_{\mu\nu}$) against the geometric constraints of a Euclidean background.

1. Cauchy Strain Extraction:
The symmetric spatial components of the tensor ($S_{ij} = \delta_{ij}/\alpha_s^2$) constrain the effective volumetric packing. The symmetric Cauchy strain tensor (ε_{ij}) maps to the deviation of these spatial diagonals from the flat unperturbed background (δ_{ij}):

$$\varepsilon_{ij} = \frac{1}{2}(S_{ij} - \delta_{ij}) = \frac{1}{2}\left(\frac{1}{\alpha_s^2} - 1\right)\delta_{ij} = \frac{P_{dyn}}{2P_c\alpha_s^2}\delta_{ij} \tag{71}$$

In the weak-field limit ($\alpha_s^2 \to 1$), this reduces to the linear gravitational strain $\varepsilon_{ij} \approx \frac{P_{dyn}}{2P_c}\delta_{ij}$.

2. The Saint-Venant Compatibility Operator:
An arbitrary strain field (ε_{ij}) is constrained within a continuous medium via the Saint-Venant Compatibility Condition ($\text{Inc}(\boldsymbol{\varepsilon})_{ijkl} = 0$).

The Incompatibility Tensor measures the failure of the continuum to satisfy this closure. It is constructed as the linear combination of the second-order spatial derivatives of the strain field:

$$\text{Inc}(\boldsymbol{\varepsilon})_{ijkl} = \partial_j\partial_l\varepsilon_{ik} + \partial_i\partial_k\varepsilon_{jl} - \partial_j\partial_k\varepsilon_{il} - \partial_i\partial_l\varepsilon_{jk} \tag{72}$$

3. Topological Dislocation & The Burgers Vector:
In an unperturbed vacuum, the Clebsch vorticity is zero ($\boldsymbol{\Omega} = d\lambda_\Omega \wedge d\beta_\Omega = 0$), the spatial displacement (u_i) is single-valued, and the incompatibility tensor reduces to zero ($\text{Inc}(\boldsymbol{\varepsilon}) = 0$). The metric operates as flat Euclidean space.

The generation of discrete mass corresponds to the formation of a rotational Clebsch vortex. Because the metric circulates around the topological core, the displacement field (u_i) becomes multi-valued. Integrating the spatial displacement around a closed contour (C) of the vortex core produces a phase gap, defined as the Burgers vector (b_i) [42]:

$$\oint_C du_i = b_i \tag{73}$$

This functions physically as a Volterra dislocation. Because the continuous streamlines cannot close around the rotational core without a phase shift ($b_i \neq 0$), the strain field is structurally incompatible ($\text{Inc}(\boldsymbol{\varepsilon}) \neq 0$).

4. Macroscopic Riemann Curvature:
To conserve continuum stability across this dislocation, the surrounding spatial metric yields from the flat Euclidean plane to accommodate the missing phase.

In the weak-field limit of infinitesimal strain, the Saint-Venant incompatibility tensor ($\text{Inc}(\boldsymbol{\varepsilon})_{ijkl}$) generates the combination of second-order spatial derivatives comprising the spatial components of the linearized Riemann Curvature Tensor (R_{ijkl}) [30]:

$$R_{ijkl}^{(lin)} = \text{Inc}(\boldsymbol{\varepsilon})_{ijkl} \tag{74}$$

As structural loads approach the cavitation boundary from the subcritical elastic domain ($r \to r_c^+$, $\alpha_s \in (0, 1]$), the Eulerian Almansi strain ($e_{ij} = \frac{1}{2}[1 - \alpha_s^2]\delta_{ij}$) remains bounded while the Lagrangian Cauchy compliance ($S_{ij} = \alpha_s^{-2}\delta_{ij}$) diverges. Evaluating this non-linear regime maps the continuous deformation to the Kröner incompatibility tensor [30]. The fully non-linear Riemann curvature, including the self-interacting connection terms ($\Gamma\Gamma$), recovers the finite strain incompatibility required to resolve the topological rotational defect.

6.3 Volumetric Cavitation & The Metric Lagrangian

1. The Symmetric Boundary (Volumetric Cavitation):
The spatial diagonals ($S_{ii} = 1/\alpha_s^2$) represent the compliance of the coordinate grid. Setting the dynamic pressure ($P_{dyn}(r) = \rho_\tau GM/r$) equal to the critical vacuum capacity ($P_c = \frac{1}{2}\rho_\tau c^2$) collapses the Total Scalar Admittance ($\alpha_s \to 0$):

$$P_{dyn}(r_c) = P_c \implies \cancel{\rho_\tau}\frac{GM}{r_c} = \frac{1}{2}\cancel{\rho_\tau}c^2 \tag{75}$$

Calculating the cavitation radius outputs the Schwarzschild boundary ($r_c = 2GM/c^2$).

2. The Symmetric Lagrangian Components:
The total action partitions into symmetric (gravitational) and antisymmetric (gauge) sectors. Targeting the symmetric tensor ($S_{\mu\nu}$) evaluates the macroscopic strain.

The kinematic gradient energy formulates the square of the 4-gradient, mapping to the bosonic scalar propagation term:

$$\mathcal{L}_{grad} = -(\partial_\alpha S_{\mu\nu})(\partial^\alpha S^{\mu\nu}) \equiv -|D\phi|^2 \tag{76}$$

The static potential resolves the trace contraction ($S_{\mu\nu}S^{\mu\nu} \equiv m^2\phi^2$). Volumetric contraction reduces the equation to the quadratic curvature invariant ($R_{\mu\nu}R^{\mu\nu}$). Under the harmonic gauge condition ($\partial_\mu S^{\mu\nu} = \frac{1}{2}\partial^\nu S_\mu^\mu$), the Ricci curvature scales proportionally to the d'Alembertian ($\Box S_{\mu\nu}$):

$$\mathcal{L}_{static} \propto -R_{\mu\nu}R^{\mu\nu} \propto -(\Box S_{\mu\nu})(\Box S^{\mu\nu}) \tag{77}$$

To satisfy dimensional analysis, the kinetic gradient ($1/L^2$) and the higher-order curvature potential ($1/L^4$) are balanced by multiplying the resulting potential with the square of the cavitation radius (r_c^2). This constructs the raw Lagrangian prior to integration:

$$\mathcal{L}_{sym} \propto -(\partial_\alpha S_{\mu\nu})(\partial^\alpha S^{\mu\nu}) - r_c^2(\Box S_{\mu\nu})(\Box S^{\mu\nu}) \tag{78}$$

3. Integration by Parts (The d'Alembertian Operator):
Because the tensor evaluates dynamic strain over a flat Minkowski background ($\sqrt{-\eta} = 1$), the continuous action ($\mathcal{S} = \int \mathcal{L}_{sym}\, d^4x$) transforms the gradient term via integration by parts using

standard partial derivatives. Bounding the surface terms to zero at infinity generates the linear 4D d'Alembertian:

$$\int -(\partial_\alpha S_{\mu\nu})(\partial^\alpha S^{\mu\nu})\, d^4x = \int S_{\mu\nu}\Box S^{\mu\nu}\, d^4x \tag{79}$$

Substituting this identity constructs the requisite geometric form required for higher-order gravity:

$$\mathcal{L}_{sym} \propto S_{\mu\nu}\Box S^{\mu\nu} - r_c^2(\Box S_{\mu\nu})(\Box S^{\mu\nu}) \tag{80}$$

6.4 Euler-Lagrange Variation & Singularity Resolution

To evaluate the macroscopic field equations, the principle of least action is applied to the total Lagrangian, summing the derived geometric strain ($\mathcal{L}_{geom}$) and the internal dynamic mass ($\mathcal{L}_{matter}$). Applying the higher-order Euler-Lagrange variation to this total action ensures structural equilibrium across the continuum:

$$\frac{\delta \mathcal{L}_{total}}{\delta S_{\mu\nu}} = \frac{\partial \mathcal{L}}{\partial S_{\mu\nu}} - \partial_\alpha \frac{\partial \mathcal{L}}{\partial(\partial_\alpha S_{\mu\nu})} + \Box \frac{\partial \mathcal{L}}{\partial(\Box S_{\mu\nu})} = 0 \tag{81}$$

Because the physical mass density (ρ_{dyn}) is explicitly extracted from the trace contraction of the tensor itself (Section 8.3), the matter action does not require external parameters. Classical mechanics filters directly through the symmetric spatial diagonals (S_{ii}), where the variation of the matter term defines the source coupling ($\delta\mathcal{L}_{matter}/\delta S_{ii} = -\kappa\rho_{dyn}$). Formulating the total classical Lagrangian yields:

$$\mathcal{L}_{classical} = \frac{1}{2} S_{ii}\Box S_{ii} - \frac{r_c^2}{2}(\Box S_{ii})^2 - \kappa\rho_{dyn} S_{ii} \tag{82}$$

At the static weak-field limit ($P_{dyn} = 0$), the scalar potential (Φ) drives the volumetric contraction linearly:

$$S_{ii} \approx 1 - \frac{2\Phi}{c^2} \tag{83}$$

Evaluating the variation converts the d'Alembertian to the 3D Laplacian ($\Box \to \nabla^2$). Separating the strain into coupled differential operators:

$$\nabla^2 S_{ii} - r_c^2 \nabla^4 S_{ii} = \kappa\rho_{dyn} \tag{84}$$

Substituting Equation (83) into the variation yields $-\frac{2}{c^2}(\nabla^2\Phi - r_c^2\nabla^4\Phi) = \kappa\rho_{dyn}$. Multiplying by $-\frac{c^2}{2}$ isolates the scalar potential. The resulting factor cancels against the source coupling constant ($\kappa \equiv -8\pi G/c^2$), producing the 4th-order Modified Poisson Equation [46, 51]:

$$\nabla^2\Phi - r_c^2\nabla^4\Phi = 4\pi G\rho_{dyn} \tag{85}$$

The Laplacian (∇^2) dominates the geometric curvature. At macroscopic coordinate boundaries where the spatial radius exceeds the boundary radius ($r \gg r_c$), the r_c^2 coefficient forces the $1/L^4$ biharmonic operator (∇^4) to attenuate to zero, restoring the Newtonian field limit ($\nabla^2\Phi \approx 4\pi G\rho_{dyn}$). Conversely, the negative biharmonic operator governs microscopic boundaries ($r \to r_c$). This strain-gradient elasticity restricts infinite compression, resolving the singularity. Because the cavitation limit operates as a cutoff, substituting the mass limit extracts the Schwarzschild radius ($r_c = 2GM/c^2$), truncating the depth of the metric and bridging continuum mechanics to macroscopic gravitational collapse.

6.5 The Unified Wave Operator & Phase Propagation

Applying the 4D d'Alembert wave operator ($\Box = \partial_\alpha \partial^\alpha$) to the full asymmetric tensor evaluates the continuous propagation of geometric strain across the Minkowski baseline. Because the derivative distributes linearly, the operator separates the tensor into orthogonal wave equations:

$$\Box \hat{U}_{\mu\nu} = \Box S_{\mu\nu} + \Box A_{\mu\nu} \tag{86}$$

1. The Symmetric Wave (Longitudinal Propagation):
Applying the operator to the spatial and temporal diagonals specifies the propagation of volumetric pressure perturbations. Taking the d'Alembertian of the macroscopic field equation (Equation 24) constructs the linearized gravitational wave equation:

$$\Box S_{\mu\nu} = \frac{8\pi G}{\Lambda c^4} \Box \hat{T}^{macro}_{(\mu\nu)} \tag{87}$$

This operator drives the longitudinal geometric strain (metric compression and expansion) through the continuum.

2. The Antisymmetric Wave (Transverse Propagation):
Applying the operator to the off-diagonals calculates the propagation of rotational shear. Because the antisymmetric matrix tracks the kinematic Clebsch fields, the geometric derivative forms the full transverse wave:

$$\Box A_{\mu\nu} = \alpha \Box \left(\frac{u_\mu v_{\perp,\nu} - u_\nu v_{\perp,\mu}}{c} \right) + \alpha t_p \Box \omega_{\mu\nu} \tag{88}$$

This operator guides the physical propagation of the electromagnetic gauge field.

3. The Source-Free Vacuum Limit:
In a source-free vacuum, the macroscopic propagation of the stress tensor equates to zero ($\Box \hat{T}_{\mu\nu} = 0$). By the unified field equation, this sets the geometric wave operator to zero:

$$\Box \hat{U}_{\mu\nu} = 0 \implies \left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2 \right) \hat{U}_{\mu\nu} = 0 \tag{89}$$

Because both the symmetric diagonals and antisymmetric off-diagonals are bounded by the identical geometric wave operator, longitudinal volumetric strain (gravity) and transverse rotational strain (electromagnetism) are forced to propagate at the identical acoustic phase limit of the continuum (c).

7. Conformal Symmetry & The Lie Derivative Constraints

The Conformal Group $SO(4,2)$ bounds the causal structure of the metric. Applying the 15 vector fields (ξ^μ) to the Asymmetric Strain Tensor via the Lie derivative ($\mathcal{L}_\xi$) identifies generic symmetry-breaking conditions.

Standard Conformal Killing Equations operate on symmetric metrics; therefore, the Lie derivative distributes across the asymmetric tensor. The tensor partitions into symmetric volumetric diagonals ($S_{\mu\nu}$) and antisymmetric shear/vorticity off-diagonals ($A_{\mu\nu}$):

$$\hat{U}_{\mu\nu} = S_{\mu\nu} + A_{\mu\nu} \tag{90}$$

The Lie derivative produces the kinematic transport along a flow path. Substituting the decomposed tensor isolates the symmetric and antisymmetric transformation rules:

$$\mathcal{L}_\xi \hat{U}_{\mu\nu} = \xi^\alpha \partial_\alpha (S_{\mu\nu} + A_{\mu\nu}) + (S_{\alpha\nu} + A_{\alpha\nu}) \partial_\mu \xi^\alpha + (S_{\mu\alpha} + A_{\mu\alpha}) \partial_\nu \xi^\alpha \tag{91}$$

7.1 The Poincare Subgroup (10 Generators)

The Poincare subgroup tests for volume-preserving transport. For Poincare symmetry to hold, the Lie derivative of the full tensor must equal zero ($\mathcal{L}_\xi \hat{U}_{\mu\nu} = 0$).

1. Translations (P_μ, 4 Generators):
For a constant translation parameter ϵ^μ, the vector field assigns $\xi^\mu = \epsilon^\mu$. The coordinate derivative vanishes ($\partial_\nu \xi^\alpha = 0$). Substituting this into Equation (91) extracts the position-dependent gradient of the tensor:

$$\mathcal{L}_\xi \hat{U}_{\mu\nu} = \epsilon^\alpha (\partial_\alpha S_{\mu\nu} + \partial_\alpha A_{\mu\nu}) \tag{92}$$

For translation symmetry, the gradient of both conjugate fields must vanish ($\partial_\alpha \hat{U}_{\mu\nu} = 0$), enforcing equilibrium. Substituting the macroscopic Hubble velocity field into the temporal diagonal ($\hat{U}_{00} \propto \Delta P_{macro}$) generates a temporal gradient ($\partial_0 \hat{U}_{00} \neq 0$). Furthermore, an ambient pressure gradient ($-\nabla \mathcal{P}_{ambient}$) leaves a spatial gradient ($\partial_i \hat{U}_{\mu\nu} \neq 0$). This demonstrates that macroscopic expansion and ambient gravity break translation symmetry.

2. Lorentz Transformations ($M_{\mu\nu}$, 6 Generators):
The vector field is constructed by an antisymmetric rotation matrix ($\omega_{\mu\nu} = -\omega_{\nu\mu}$) such that $\xi^\mu = \omega^\mu{}_\alpha x^\alpha$. The spatial derivative computes the constant transformation matrix ($\partial_\nu \xi^\mu = \omega^\mu{}_\nu$).

Substituting this into the distributed Lie derivative expands the rotational shear acting upon the full tensor:

$$\mathcal{L}_\xi \hat{U}_{\mu\nu} = (\omega^\alpha{}_\beta x^\beta)\partial_\alpha \hat{U}_{\mu\nu} + (S_{\alpha\nu} + A_{\alpha\nu})\omega^\alpha{}_\mu + (S_{\mu\alpha} + A_{\mu\alpha})\omega^\alpha{}_\nu \tag{93}$$

To separate the geometric couplings, we contract the symmetric tensor components ($S_{\mu\nu}$). Using the covariant formulation $S_{\mu\nu} = -\alpha_s^2 u_\mu u_\nu + \frac{1}{\alpha_s^2} h_{\mu\nu}$, the unperturbed isotropic spatial metric cancels the antisymmetric transformation ($\omega_{\mu\nu} + \omega_{\nu\mu} = 0$). The symmetric contraction outputs the 4-velocity projection:

$$S_{\alpha\nu}\omega^\alpha{}_\mu + S_{\mu\alpha}\omega^\alpha{}_\nu = \left(\frac{1}{\alpha_s^2} - \alpha_s^2\right)(u_\nu u_\alpha \omega^\alpha{}_\mu + u_\mu u_\alpha \omega^\alpha{}_\nu) \tag{94}$$

This residual vanishes for spatial rotations ($SO(3)$) because the temporal 4-velocity is orthogonal to spatial vorticity, preserving the volumetric trace. However, the 3 Lorentz boosts (M_{0i}) mix the temporal and spatial capacities, generating a residual for the time-space components:

$$S_{\alpha i}\omega^\alpha{}_0 + S_{0\alpha}\omega^\alpha{}_i = -\omega_{0i}\left(\frac{1}{\alpha_s^2} - \alpha_s^2\right) \tag{95}$$

Because the Lie derivative preserves index symmetry, the symmetric and antisymmetric constraints must be satisfied independently ($\mathcal{L}_\xi S_{\mu\nu} \propto S_{\mu\nu}$ and $\mathcal{L}_\xi A_{\mu\nu} \propto A_{\mu\nu}$).

Computing the symmetric sector for a Lorentz boost (M_{0i}), the time-space components ($0i$) target the residual:

$$(\mathcal{L}_\xi S)_{0i} = \xi^\alpha \partial_\alpha S_{0i} - \omega_{0i}\left(\frac{1}{\alpha_s^2} - \alpha_s^2\right) \tag{96}$$

For Lorentz symmetry to hold across the symmetric sector, this residual must vanish. Because this symmetric time-space residual cannot algebraically cancel against the antisymmetric transverse sector ($A_{\mu\nu}$), Lorentz boost symmetry is broken unless the continuum is static ($\alpha_s = 1$).

Assessing the antisymmetric sector independently exposes the rotational couplings:

$$(\mathcal{L}_\xi A)_{\mu\nu} = \xi^\alpha \partial_\alpha A_{\mu\nu} + (A_{\mu\alpha}\omega^\alpha{}_\nu - \omega_{\mu\alpha}A^\alpha{}_\nu) \tag{97}$$

This demonstrates that while spatial rotations (M_{ij}) correspond to the antisymmetric internal Clebsch vorticity ($\omega_{\mu\nu}$), kinematic translation through the continuum ($v \to c$, $\alpha_s \to 0$) introduces a symmetric residual that structurally breaks the Lorentz invariance of the metric.

7.2 Dilation & Special Conformal Transformations (5 Generators)

The Dilation (D) and Special Conformal (K_μ) vector fields scale the 4D coordinate volume. Because these paths compress or expand the metric, the constraint utilizes the Conformal Killing Equation. The Lie derivative resolves to a scalar multiple of the tensor:

$$\mathcal{L}_\xi \hat{U}_{\mu\nu} = f(x)\hat{U}_{\mu\nu} \tag{98}$$

1. The Dilation Generator (D):
The vector field is parameterized by a constant scalar σ, constructed as $\xi^\mu = \sigma x^\mu$. The coordinate derivative yields the Kronecker delta ($\partial_\nu \xi^\mu = \sigma \delta^\mu_\nu$).

Substituting these definitions into the Lie derivative unfolds the operator:

$$\begin{aligned}\mathcal{L}_\xi \hat{U}_{\mu\nu} &= (\sigma x^\alpha)\partial_\alpha \hat{U}_{\mu\nu} + \hat{U}_{\alpha\nu}(\sigma\delta^\alpha_\mu) + \hat{U}_{\mu\alpha}(\sigma\delta^\alpha_\nu) \\ \mathcal{L}_\xi \hat{U}_{\mu\nu} &= \sigma x^\alpha \partial_\alpha \hat{U}_{\mu\nu} + 2\sigma \hat{U}_{\mu\nu}\end{aligned} \tag{99}$$

To satisfy scale invariance ($f(x) = 2\sigma$), the position-dependent gradient must vanish ($\sigma x^\alpha \partial_\alpha \hat{U}_{\mu\nu} = 0$). This constraint requires a homogeneous metric field. Discrete topological boundaries manifest as Volterra dislocations. Because the rotational Clebsch phase generates a Burgers vector, the metric violates the Saint-Venant compatibility condition ($\mathrm{Inc}(\boldsymbol{\varepsilon}) \neq 0$). This strain incompatibility ensures the metric varies with coordinate position ($\partial_\alpha \hat{U}_{\mu\nu} \neq 0$), structurally breaking scale invariance.

2. Special Conformal Transformations (K_μ, 4 Generators):
The vector field operates via a constant parameter (b^μ) with dimensions of inverse length (L^{-1}):

$$\xi^\mu = 2(x_\beta b^\beta)x^\mu - (x_\beta x^\beta)b^\mu \tag{100}$$

Expanding the spatial derivative retrieves the local scaling and rotational shift:

$$\partial_\alpha \xi^\mu = 2b_\alpha x^\mu + 2(x_\beta b^\beta)\delta^\mu_\alpha - 2x_\alpha b^\mu \tag{101}$$

Substituting this transformation into the Lie derivative distributes the non-inertial strain terms:

$$\mathcal{L}_\xi \hat{U}_{\mu\nu} = \xi^\alpha \partial_\alpha \hat{U}_{\mu\nu} + 4(x_\beta b^\beta)\hat{U}_{\mu\nu} + 2x^\alpha(b_\mu \hat{U}_{\alpha\nu} + b_\nu \hat{U}_{\mu\alpha}) - 2b^\alpha(x_\mu \hat{U}_{\alpha\nu} + x_\nu \hat{U}_{\mu\alpha}) \tag{102}$$

Contracting the trailing antisymmetric covariant combinations against the metric reveals that the spatial components (i, j) mutually cancel due to spatial isotropy. However, mixing the time-space components produces a residual strain:

$$2(x^0 b^1 - x^1 b^0)\left(\frac{1}{\alpha_s^2} - \alpha_s^2\right) \tag{103}$$

This residual vanishes only in a static vacuum ($\alpha_s = 1$). Under dynamic pressure ($\alpha_s < 1$), the Special Conformal transformation induces non-inertial shear across the metric boundaries.

To calculate the remaining volumetric trace $4(x_\beta b^\beta)\hat{U}_{\mu\nu}$, the global parameter ($b^\mu$) must align dimensionally to a physical field without violating the product rule of the constant Lie generator. We designate b^μ as the global symmetry generator bounding a locally uniform kinematic acceleration field (a^μ, where $\partial_\alpha a^\mu = 0$) across the phase limit ($b^\mu \equiv a^\mu/c^2$). A non-uniform local field ($\partial_\alpha a^\mu \neq 0$) generates tidal shearing, which breaks the conformal symmetry and sets the structural stability limit of the translating defect.

Substituting the spatial material derivative for an accelerating volume ($a^i = -\frac{1}{\rho_\tau}\nabla^i P_{dyn}$) derives the thermodynamic scaling constraint:

$$\begin{aligned} 4(x_\beta b^\beta) &= 4\left(x_i \frac{a^i}{c^2}\right) \\ 4(x_\beta b^\beta) &= -\frac{4}{\rho_\tau c^2}\left(x_i \nabla^i P_{dyn}\right) \\ 4(x_\beta b^\beta) &= -\frac{2}{P_c}\left(x_i \nabla^i P_{dyn}\right) \end{aligned} \tag{104}$$

This geometric scaling factor operates proportionally to the directional derivative of dynamic pressure relative to the critical vacuum capacity ($P_c = \frac{1}{2}\rho_\tau c^2$). A conformal transformation manifests as a topological boundary propagating through a scalar pressure gradient, where the convergence of the coordinate lines governs the refraction of the trajectory.

3. Derivation of the Gordon Optical Metric:
This directly alters the apparent phase velocity of transverse shear waves (light) propagating through the medium. The continuous metric acts as a Gradient-Index (GRIN) optical medium.

The macroscopic index of refraction (n) of the vacuum is formulated as the ratio of the unperturbed baseline capacity (P_c) to the available static capacity (P_{static}):

$$n(x) = \frac{P_c}{P_c - P_{dyn}} = \frac{1}{\alpha_s^2} \tag{105}$$

As dynamic pressure increases ($P_{dyn} \to P_c$), the scalar admittance collapses ($\alpha_s^2 \to 0$) and the refractive index diverges ($n \to \infty$), reproducing the black hole horizon as an optical phase limit.

To formalize the trajectory of a shear wave through this pressure gradient, the conformal scaling constraint (Equation 104) maps to the Gordon optical metric [23]. If the defect translates with a normalized four-velocity ($\hat{u}_\mu \hat{u}^\mu = -1$), the effective optical spacetime ($\hat{U}^{opt}_{\mu\nu}$) experienced by the propagating wave is constructed by superimposing the refractive penalty onto the baseline metric:

$$\hat{U}^{opt}_{\mu\nu} = \hat{U}_{\mu\nu} + \left(1 - \frac{1}{n^2}\right)\hat{u}_\mu \hat{u}_\nu \tag{106}$$

Substituting the thermodynamic formulation of the refractive index from Equation (105) determines the acoustic metric:

$$\hat{U}^{opt}_{\mu\nu} = \hat{U}_{\mu\nu} + \left(1 - \alpha_s^4\right)\hat{u}_\mu \hat{u}_\nu \tag{107}$$

Under the conformal transformation (K_μ), acceleration (b^μ) generates a dynamic pressure gradient, affecting scalar admittance (α_s). The continuous coordinate transformation causes null geodesics ($ds^2 = 0$) to bend, reproducing gravitational lensing as acoustic refraction of a wave traversing a density gradient.

7.3 Kinematic Implications of Symmetry Breaking

The Lie derivative assessments confirm that Lorentz and conformal symmetries hold for a static continuum ($\alpha_s = 1$). Under applied kinematic strain ($\alpha_s \in (0, 1)$), the asymmetric tensor generates residuals proportional to the dynamic pressure gradient (∇P_{dyn}) and the temporal capacity variance ($1/\alpha_s^2 - \alpha_s^2 > 0$). Expressing this variance in terms of the refractive index ($n = 1/\alpha_s^2$) and the dynamic pressure ratio ($P_{dyn}/P_c = 1 - \alpha_s^2$) bridges the Section 7.1 boost residual directly to the Gordon optical metric (Equation 106):

$$\frac{1}{\alpha_s^2} - \alpha_s^2 = \frac{n^2 - 1}{n} = \frac{P_{dyn}}{P_c}\left(\frac{2 - P_{dyn}/P_c}{\alpha_s^2}\right) > 0 \tag{108}$$

Substituting these conformal boundary constraints into the Gordon metric demonstrates that vacuum impedance dictates the local phase velocity ($v_c = c/n = c\alpha_s^2 < c$). Consequently, null geodesic deflection and acoustic horizons ($\alpha_s \to 0^+$) emerge from the thermodynamic limits.

7.4 Conserved Noether Currents

By Noether's theorem, the invariances of the Poincaré Lie derivatives determine the conserved kinematic currents.

For a continuous unified Lagrangian action ($\mathcal{L}_{metric}$) invariant under a generalized coordinate transformation ($\delta x^\mu = \xi^\mu$), the conserved Noether current density (J^μ) expands as:

$$J^\mu = \frac{\partial \mathcal{L}_{metric}}{\partial(\partial_\mu \hat{U}_{\alpha\beta})} \mathcal{L}_\xi \hat{U}_{\alpha\beta} - \xi^\mu \mathcal{L}_{metric} \tag{109}$$

Substituting the zero-residual Lie symmetries from Section 7.1 isolates the orthogonal conserved currents of the asymmetric metric:

1. Translational Symmetry (Energy-Momentum Conservation):
Under uniform spacetime translation ($\xi^\mu = \epsilon^\mu$), the Lie derivative yields zero gradient ($\partial_\alpha \hat{U}_{\mu\nu} = 0$). The Noether current equates to the macroscopic continuous stress-energy tensor ($\hat{T}^\mu{}_\nu = P_c(\hat{U}^\mu{}_\nu - \eta^\mu{}_\nu)$). The continuity equation constrains the 4-divergence to zero:

$$\partial_\mu \hat{T}^{\mu\nu} = P_c \partial_\mu \hat{U}^{\mu\nu} = 0 \tag{110}$$

This operation conserves continuous energy and linear momentum.

2. Lorentz Symmetry (Total Angular Momentum & Spin):
Under spatial rotation ($\xi^\mu = \omega^\mu{}_\alpha x^\alpha$), the macroscopic energy-momentum tensor defines orbital angular momentum ($L^{\mu\alpha\beta} = x^\alpha \hat{T}^{\mu\beta} - x^\beta \hat{T}^{\mu\alpha} = -L^{\mu\beta\alpha}$). Because the constitutive stress tensor incorporates an antisymmetric transverse sector ($\hat{T}^{\alpha\beta} - \hat{T}^{\beta\alpha} = 2P_c A^{\alpha\beta}$), orbital angular momentum is not conserved in isolation ($\partial_\mu L^{\mu\alpha\beta} = 2P_c A^{\alpha\beta}$), and the Noether current generates an internal Clebsch spin tensor ($\Sigma^{\mu\alpha\beta}$):

$$J^\mu_{(Lorentz)} = M^{\mu\alpha\beta} = L^{\mu\alpha\beta} + \Sigma^{\mu\alpha\beta} \tag{111}$$

Because the antisymmetric rotational couplings resolve the Lie residual ($A_{\mu\alpha}\omega^\alpha{}_\nu - \omega_{\mu\alpha}A^\alpha{}_\nu = 0$), the internal spin divergence balances the antisymmetric shear torque ($\partial_\mu \Sigma^{\mu\alpha\beta} = -\partial_\mu L^{\mu\alpha\beta}$), and the divergence of the total angular momentum converges to zero ($\partial_\mu M^{\mu\alpha\beta} = 0$). Internal Clebsch vorticity ($A_{ij} = \alpha t_p (d\lambda_\Omega \wedge d\beta_\Omega)_{ij}$) recovers conserved quantum spin.

7.5 The Dressing Field Method (Gauge-Invariant Composites)

The continuous extraction of physical degrees of freedom from the metric avoids spontaneous symmetry breaking and gauge-fixing artifacts via the Dressing Field Method (DFM) [21]. In general relativistic gauge field theories, the topological obstruction to global gauge-fixing (the Gribov-Singer ambiguity [24, 50]) is bypassed by constructing composite, gauge-invariant variables directly from the fields.

The Asymmetric Metric-Tensor ($\hat{U}_{\mu\nu}$) operates as a dressed field space, utilizing the Clebsch parameters to extract the invariant content of the continuum without requiring an absolute background manifold.

1. $U(1)$ Gauge Dressing (Electromagnetism):
The Clebsch parametrization (Equation 1) executes an Abelian DFM operation. In standard DFM formalism, a bare gauge connection (A) and a dressing field (u) combine to form a gauge-invariant composite ($\hat{A} = A + du$).

Within the asymmetric tensor, the transverse slip 1-form ($v_\perp$) operates as the bare gauge connection (A). The irrotational phase scalar (h) operates as the compensatory dressing field (u). Under a local $U(1)$ gauge transformation parameterized by χ, the bare fields transform as:

$$v_\perp^g = v_\perp - d\chi, \quad h^g = h + \chi \tag{112}$$

The composite kinematic field ($\hat{\mathbf{v}}$) is constructed via the DFM addition:

$$\hat{\mathbf{v}}^g = v_\perp^g + dh^g = (v_\perp - d\chi) + d(h + \chi) = v_\perp + dh = \hat{\mathbf{v}} \tag{113}$$

The composite resolves as gauge-invariant ($\delta_\chi \hat{\mathbf{v}} = 0$). Consequently, the antisymmetric off-diagonals ($\hat{U}_{0i} \propto \hat{\mathbf{v}}_i$) representing electromagnetic transverse shear are dressed physical observables rather than gauge-dependent potentials.

2. Diffeomorphism Invariance (Diff(M)) and Relational Spacetime:
To satisfy relational physics, the continuous metric is not evaluated against a fixed Euclidean stage. DFM for diffeomorphisms requires a dressing field that transforms under Diff(M) to counteract the Lie derivative ($\mathcal{L}_\xi$) of the bare coordinate manifold.

The Clebsch rotational phases ($\lambda_\Omega, \beta_\Omega$) act as these relational dressing fields. Under a diffeomorphism generated by a vector field ξ, the fields co-define the invariant geometric vorticity ($\hat{\Omega}$):

$$\hat{\Omega}_{\mu\nu} = \partial_{[\mu}\lambda_\Omega \partial_{\nu]}\beta_\Omega \tag{114}$$

By establishing internal spatial coordinates through this exterior product intersection ($d\lambda_\Omega \wedge d\beta_\Omega$), the metric defines its own reference frame.

This relational structure evaluates physical states as boundary transitions. Internal composition resolves not as a discrete volume, but as nested geometric yield limits scaling down to the Planck boundary. A phase-locked transition layer (defined via Clebsch vorticity) acts as a Quantum Reference Frame (QRF) [22, 47]. Because the tensor provides an invariant mapping across these nested scales, evaluating covariance across distinct boundaries (ranging from microscopic cavitation limits to macroscopic shear layers) utilizes transformations of the second kind (active gauge transformations that switch the physical reference boundary). This maps kinematic strain and physical action between relational interfaces, establishing force as a geometric boundary condition rather than an intrinsic property.

3. Invariant Path Integral Quantization:
Because the Asymmetric Metric-Tensor ($\hat{U}_{\mu\nu}$) is assembled from these dressed, composite variables, it constitutes an invariant field space. The geometric components evaluated in the Lagrangian (Section 6.3) are devoid of redundant gauge orbits. By evaluating continuous physical states utilizing these dressed variables, an invariant path integral quantization prevents Gribov obstructions, permitting deterministic convergence at the yield limits.

8. The Mass Mechanism (Topological Tensor Yield)

While the conformal scaling constraints are restricted by the Lie derivatives (Section 7), rest mass emerges as a phase transition where kinematic loads decouple metric continuity.

8.1 The Tensor Yield Criterion (The VEV Origin)

The symmetric spatial diagonals of the tensor define the volumetric density of the coordinate grid. Because the spatial diagonals are derived via the inverse scalar admittance ($S_{ii} = 1/\alpha_s^2$), substituting the thermodynamic ratio ($\alpha_s^2 = P_{static}/P_c$) equates the strain to the available static capacity:

$$S_{ii} = \frac{P_c}{P_{static}} \tag{115}$$

Substituting the Bernoulli equilibrium ($P_{static} = P_c - P_{dyn}$) and the kinematic pressure of metric circulation ($P_{dyn} = \frac{1}{2}\rho_\tau v^2$) translates the coordinate expansion into a velocity-dependent boundary:

$$S_{ii} = \frac{P_c}{P_c - \frac{1}{2}\rho_\tau v^2} \tag{116}$$

Because the total static capacity of the vacuum equates to the phase limit of the continuum ($P_c = \frac{1}{2}\rho_\tau c^2$), the diagonals evaluate as a function of the velocity ratio:

$$\begin{aligned} S_{ii} &= \frac{\frac{1}{2}\rho_\tau c^2}{\frac{1}{2}\rho_\tau c^2 - \frac{1}{2}\rho_\tau v^2} \\ S_{ii} &= \frac{1}{1 - \frac{v^2}{c^2}} \end{aligned} \tag{117}$$

Taking the limit as velocity approaches the speed of light ($v \to c$) forces the spatial component to diverge:

$$\lim_{v \to c} S_{ii} = \lim_{v \to c} \frac{1}{1 - \frac{v^2}{c^2}} = \infty \tag{118}$$

Because the overarching continuum cannot sustain an infinite spatial coordinate geometry, the internal structure yields. The metric cavitates, thereby conserving the thermodynamic boundary (P_c).

This critical threshold acts as the Vacuum Expectation Value (v_{vev}) [54]. It is the specific velocity where dynamic pressure equals the total ambient capacity, forcing a geometric phase transition from a continuous manifold into a localized topological defect:

$$P_{dyn} = P_{ambient} \implies v_{vev}^2 = \frac{2P_{ambient}}{\rho_\tau} \tag{119}$$

Redefining the kinematic state relative to the baseline defines the longitudinal acoustic breathing mode ($\delta v(x)$):

$$v(x) = v_{vev} + \delta v(x) \tag{120}$$

8.2 The Topological Phase Lock (Mass Generation)

Before structural yield ($\alpha_s \in (0, 1]$), the metric sustains continuous wave states across its symmetric bulk diagonals ($S_{\mu\nu}$) and antisymmetric transverse shear off-diagonals ($A_{\mu\nu}$).

1. Tangential Phase-Lock ($v_{core} = r_c\Omega_{core} = c$) and Invariant Rest Mass:
The transition from continuous metric strain to discrete rest mass proceeds in three mechanical steps:

- **Symmetric Collapse vs. Nonsingular Vorticity:** On the subcritical domain ($\alpha_s \in (0, 1)$), the spatial compliance diagonals ($S_{ii} = 1/\alpha_s^2$) are counterbalanced by the temporal admittance ($S_{00} = -\alpha_s^2$, preserving $S_{00}S_{ii} = -1$). At the VEV cavitation threshold ($v \to c$, $\alpha_s \to 0^+$), the static pressure reserve depletes ($P_{static} = \alpha_s^2 P_c \to 0$) and the temporal diagonal vanishes ($\hat{U}_{00} = -\alpha_s^2 \to 0$). However, the antisymmetric Clebsch vorticity off-diagonals ($A_{ij} = \alpha t_p [d\lambda_\Omega \wedge d\beta_\Omega]_{ij}$) contain no α_s denominators and remain finite across all 16 matrix entries.

- **Tangential Circulation at the Compton Boundary:** Because the symmetric bulk cannot compress radially past cavitation at r_c, phase conservation deflects the core strain into tangential circulation at the acoustic phase velocity ($v_{core} = r_c\Omega_{core} = c$). This locks the vacuum pressure capacity ($P_c = \frac{1}{2}\rho_\tau c^2$) into the boundary's rotational energy density, $P_{vortex} \equiv \frac{1}{2}\rho_\tau v_{core}^2 = \frac{1}{2}\rho_\tau (r_c\Omega_{core})^2 = P_c$. Across the displaced kinematic strain volume V_τ, the locked energy $E_{locked} = P_{vortex}V_\tau$ produces discrete invariant rest mass:

$$m_0 = \frac{E_{locked}}{c^2} = \frac{P_{vortex} V_\tau}{c^2} = \frac{1}{2}\rho_\tau V_\tau \left(\frac{r_c \Omega_{core}}{c}\right)^2 = \frac{1}{2}\rho_\tau V_\tau = \frac{P_c V_\tau}{c^2} > 0 \tag{121}$$

- **Compton Boundary Energy vs. Planck-Normalized Matrix Shear:** Whereas $P_{vortex} = P_c$ measures the macroscopic circulation energy density at the Compton radius r_c, the dimensionless matrix entries of $A_{\mu\nu}$ in Section 8.3 are scaled by the Planck relaxation time t_p and coupling α. Because each electric slip ($A_{0i} = -A_{i0}$) and magnetic vorticity ($A_{ij} = -A_{ji}$) component appears twice in the 4×4 matrix, the prefactor of $1/4$ cancels the double-counting factor of 2: the Euclidean Frobenius sum gives the nonnegative shear load $P_{shear} = \frac{1}{4} P_c \sum_{\mu,\nu} (A_{\mu\nu})^2 = \frac{1}{2} P_c \alpha^2 (v_\perp^2/c^2 + t_p^2 \Omega^2)$, while raising indices with the Minkowski metric $\eta^{\mu\nu}$ flips the sign of the A_{0i} slip terms to give the signed Lagrangian invariant $\frac{1}{4} P_c A_{\mu\nu} A^{\mu\nu} = \frac{1}{2} P_c \alpha^2 (t_p^2 \Omega^2 - v_\perp^2/c^2)$ (Equations 127–128).

2. Micro-Macro Horizon Duality ($r_c \Omega_{core} = c \leftrightarrow R_h H_0 = c$):
Clebsch vorticity (Ω_{core}) and the Hubble horizon strain rate (H_0) both operate as inverse-time strain rates (c/r). Comparing the cavitation boundary (r_c) to the cosmological horizon (R_h) establishes a duality across the symmetric ($S_{\mu\nu}$) and antisymmetric ($A_{\mu\nu}$) sectors of the tensor (Equation 122):

- **Inner Defect Boundary ($r = r_c$):** At the Compton cavitation radius $r_c = \hbar/(m_0 c)$, the radial compliance $S_{ii} = 1/\alpha_s^2$ diverges and deflects the core strain into a closed rotational loop inside the antisymmetric sector ($A_{\mu\nu}$), locking the core angular frequency to $r_c \Omega_{core} = c \implies \Omega_{core} = m_0 c^2/\hbar$.
- **Outer Cosmic Horizon (R_h):** At the characteristic Hubble scale $R_h = c/H_0$, the strain rate $H_0 = c/R_h$ operates in the symmetric sector ($S_{\mu\nu}$) as an irrotational radial gradient $v_0(r) = R_h H_0 (1 - e^{-r/R_h})$, matching the linear horizon scale $R_h H_0 = c$ and approaching the phase velocity limit asymptotically ($r \gg R_h$):

$$r_c \Omega_{core} = R_h H_0 = c \tag{122}$$

8.3 The Kinematic Lagrangian & Orthogonal Mass Extraction

The scalar energy state is determined by the trace contraction of the tensor ($\hat{U}_{\mu\nu}\hat{U}^{\mu\nu} = S_{\mu\nu}S^{\mu\nu} + A_{\mu\nu}A^{\mu\nu}$). Expanding this contraction against the covariant definitions extracts the kinematic wave velocities.

By applying the covariant 4-velocity orthogonality constraints ($u_\mu u^\mu = -1$, $u_\mu v_\perp^\mu = 0$, and $u_\mu h^{\mu\nu} = 0$) it reduces to:

$$S_{\mu\nu}S^{\mu\nu} = \left(-\alpha_s^2 u_\mu u_\nu + \frac{1}{\alpha_s^2} h_{\mu\nu}\right)\left(-\alpha_s^2 u^\mu u^\nu + \frac{1}{\alpha_s^2} h^{\mu\nu}\right) = \alpha_s^4 + \frac{3}{\alpha_s^4} \tag{123}$$

Applying a weak-field Taylor expansion ($\alpha_s^2 = 1 - \beta^2 \implies \alpha_s^4 \approx 1 - 2\beta^2$ and $\alpha_s^{-4} \approx 1 + 2\beta^2$), the trace simplifies to:

$$S_{\mu\nu}S^{\mu\nu} \approx (1 - 2\beta^2) + 3(1 + 2\beta^2) = 4 + 4\frac{v^2}{c^2} \tag{124}$$

This expansion operates as an approximation of the classical weak-field limit, not a physical law. As velocity approaches the phase limit ($v \to c$), the non-linear trace restricts the boundary. This approximation produces the total static capacity.

The antisymmetric spatial trace filters the transverse shear stress ($P_{shear} = \frac{1}{4} P_c A_{ij} A^{ij}$).

$$A_{\mu\nu}A^{\mu\nu} = \frac{\alpha^2}{c^2}(u_\mu v_{\perp,\nu} - u_\nu v_{\perp,\mu})(u^\mu v^{\perp,\nu} - u^\nu v^{\perp,\mu}) + \alpha^2 t_p^2 \omega_{\mu\nu}\omega^{\mu\nu}$$
$$A_{\mu\nu}A^{\mu\nu} = \frac{\alpha^2}{c^2}\Big((u_\mu u^\mu)(v_{\perp,\nu}v^{\perp,\nu}) - (u_\mu v^{\perp,\mu})(v_{\perp,\nu}u^\nu) - (u_\nu v^{\perp,\nu})(v_{\perp,\mu}u^\mu) + (u_\nu u^\nu)(v_{\perp,\mu}v^{\perp,\mu})\Big) + \alpha^2 t_p^2 \omega_{\mu\nu}\omega^{\mu\nu}$$
$$A_{\mu\nu}A^{\mu\nu} = \frac{\alpha^2}{c^2}\Big((-1)(v_\perp^2) - (0) - (0) + (-1)(v_\perp^2)\Big) + \alpha^2 t_p^2 \omega_{\mu\nu}\omega^{\mu\nu}$$
$$A_{\mu\nu}A^{\mu\nu} = -2\alpha^2 \frac{v_\perp^2}{c^2} + \alpha^2 t_p^2 \omega_{\mu\nu}\omega^{\mu\nu} \tag{125}$$

Because these traces measure dimensionless strain across four spacetime dimensions, subtracting the unperturbed Minkowski background trace ($\hat{U}_{\mu\nu}\hat{U}^{\mu\nu} - 4$) isolates the kinematic variance. Normalizing this trace by the spacetime dimensionality (1/4), mirroring standard classical field formulations ($\mathcal{L} \propto -\frac{1}{4} F_{\mu\nu}F^{\mu\nu}$), and multiplying by the background pressure (P_c) extracts the continuous kinematic Lagrangian.

Substituting the symmetric trace variance ($4\frac{v^2}{c^2}$) and expanding the thermodynamic coefficient ($P_c = \frac{1}{2}\rho_\tau c^2$):

$$\mathcal{L}_{kin} = \frac{1}{4} P_c \Big(\hat{U}_{\mu\nu}\hat{U}^{\mu\nu} - 4\Big)$$
$$\mathcal{L}_{kin} = \frac{1}{4}\left(\frac{1}{2}\rho_\tau c^2\right)\left(4\frac{v^2}{c^2} + A_{\mu\nu}A^{\mu\nu}\right)$$
$$\mathcal{L}_{kin} = \left(\frac{1}{2}\rho_\tau v^2\right) + \frac{1}{4} P_c \big(A_{\mu\nu}A^{\mu\nu}\big) \tag{126}$$

Because $\frac{1}{2}\rho_\tau v^2$ defines dynamic pressure (P_{dyn}), and the scaled antisymmetric trace produces the signed Lagrangian shear density ($\mathcal{L}_{shear} = \frac{1}{4} P_c A_{\mu\nu}A^{\mu\nu}$):

$$\mathcal{L}_{kin} = P_{dyn} + \mathcal{L}_{shear} \tag{127}$$

Mass density (ρ_{dyn}) is extracted from this tensor operation. Dividing the dynamic pressure component of the symmetric trace contraction by the squared phase boundary (c^2) outputs the physical density ($\rho_{dyn} = P_{dyn}/c^2$).

Because the Lorentzian contraction ($A_{\mu\nu}A^{\mu\nu}$) carries opposite metric signatures across temporal and spatial indices, $\mathcal{L}_{shear}$ is signed; under zero-vorticity slip ($\omega_{\mu\nu} = 0, v_\perp \neq 0$), it takes a negative value ($-\frac{1}{2} P_c \alpha^2 v_\perp^2/c^2$). In contrast, the nonnegative stored shear load (P_{shear}) that depletes the static budget ($P_c = P_{static} + P_{dyn} + P_{shear}$) is quantified by the Euclidean Frobenius contraction (Section 6.1, $\sum_{\mu,\nu} A_{\mu\nu}^2$):

$$P_{shear} = \frac{1}{4} P_c \sum_{\mu,\nu} (A_{\mu\nu})^2 = \frac{1}{2} P_c \alpha^2 \left(\frac{v_\perp^2}{c^2} + t_p^2 \Omega^2\right) \geq 0 \tag{128}$$

At $\Omega = 0$, this Frobenius norm gives $P_{shear} = \frac{1}{2} P_c \alpha^2 (v_\perp^2/c^2) \geq 0$, ensuring nonnegative budget depletion across all slip and vorticity configurations.

1. The Quantum Cavitation Boundary (Compton Limit):
While the symmetric trace bounds the macroscopic gravitational limit, the antisymmetric trace specifies the microscopic boundary. The term ($\alpha^2 t_p^2 \omega_{\mu\nu}\omega^{\mu\nu}$) represents internal shear, confining the rest mass energy of a topological defect ($E = mc^2$).

Because the cavitated boundary operates as an acoustic standing wave, its spatial circulation is phase-locked. Postulating rest-energy equivalence ($E = mc^2$) and the Planck-Einstein phase relation ($E = \hbar\omega$) as boundary inputs fixes the angular frequency ($\omega = mc^2/\hbar$); Section 14.3 subsequently applies this radius to identify the circulation constant ($\kappa = 2\pi\hbar$) without

circularity. The radius required to confine this rotational phase ($v = c$) is the phase velocity divided by the frequency:

$$r_c = \frac{c}{\omega} = \frac{c}{(mc^2/\hbar)} = \frac{\hbar}{mc} \tag{129}$$

This derives the reduced Compton wavelength as the cavitation radius of the antisymmetric tensor.

2. The Topological Yield Transition (Planck Mass):
Applying the Equivalence Principle identification of inertial and gravitational mass ($m = M$, Section 10.5), and equating the symmetric gravitational boundary ($r_s = 2GM/c^2$) to the antisymmetric shear boundary ($r_c = \hbar/(Mc)$), identifies the scale where volumetric compression (P_{dyn}) and rotational shear (P_{shear}) exert equal thermodynamic strain:

$$\frac{\hbar}{Mc} = \frac{2GM}{c^2} \implies \left(\frac{M}{M_P}\right)^2 = \frac{1}{2} \implies M = \frac{M_P}{\sqrt{2}} \tag{130}$$

This isolates the Planck mass scale ($M = M_P/\sqrt{2}$) as the transition threshold between continuous gravitation and discrete quantum localization. The $1/\sqrt{2}$ coefficient derives from the metric compliance limit governing inertia. As established in the derivation of kinematic inertia, the structural compliance (α_s) determines the metric's capacity to bound mass density:

$$\alpha_s = \left(1 - \frac{P_{dyn} + P_{shear}}{P_c}\right)^{1/2} \tag{131}$$

At the Planck scale ($r = l_p$), the normalized gravitational load scales with the squared mass ratio ($P_{load}/P_c = (M/M_P)^2 = 1/2$), while the complex eigenstate structure ($1 \pm i$, Section 9.6) reflects equal real and imaginary strain amplitudes. Substituting this derived load ratio ($P_{load}/P_c = 1/2$) into the compliance identity scales the structural yield limit of the tensor:

$$\alpha_{yield} = \left(1 - \left(\frac{M}{M_P}\right)^2\right)^{1/2} = \left(1 - \frac{1}{2}\right)^{1/2} = \frac{1}{\sqrt{2}} \tag{132}$$

Because a stable defect cannot possess a mass density exceeding the structural compliance of the metric bounding it, this partition projects the real bounded state. By mapping the metric compliance to the mass ratio ($M/M_P = \alpha_{yield} = 1/\sqrt{2} \implies M = M_P/\sqrt{2}$), this threshold extracts the same Planck mass limit at the intersection boundary.

9. Electromagnetic Gauge & Unitary Evolution

Eigenstate analysis reveals the parametric coupling and structural mechanics of the metric. Structural decoupling and macroscopic quantization are extracted via the principal eigenvalues (λ).

The principal axes of strain are calculated by evaluating the characteristic determinant:

$$\det(\hat{U}_{\mu\nu} - \lambda\delta_{\mu\nu}) = 0 \tag{133}$$

Specific components of the metric are evaluated by isolating individual parameters. Aligning the primary axis of rotation to the z-coordinate ($\Omega_z = \Omega$) induces an orthogonal transverse slip across the $x - y$ plane (v_x, v_y). This structures the matrix:

$$\hat{U}_{\mu\nu} - \lambda\delta_{\mu\nu} = \begin{bmatrix} -\alpha_s^2 - \lambda & -\alpha\frac{v_x}{c} & -\alpha\frac{v_y}{c} & 0 \\ \alpha\frac{v_x}{c} & \alpha_s^{-2} - \lambda & -\alpha t_p\Omega & 0 \\ \alpha\frac{v_y}{c} & \alpha t_p\Omega & \alpha_s^{-2} - \lambda & 0 \\ 0 & 0 & 0 & \alpha_s^{-2} - \lambda \end{bmatrix} \tag{134}$$

Expanding the characteristic determinant isolates the invariant longitudinal axis (z-axis). Within the coupled transverse dimensions, the antisymmetric cross-terms algebraically cancel ($v_x v_y \Omega - v_y v_x \Omega = 0$). The rotation couples to the Pythagorean magnitude of the transverse slip ($v_\perp^2 = v_x^2 + v_y^2$). This cancellation proves that the eigenstates are invariant under continuous transverse rotation, establishing $O(2)$ gauge symmetry:

$$(\alpha_s^{-2} - \lambda)\Big[(-\alpha_s^2 - \lambda)\big((\alpha_s^{-2} - \lambda)^2 + (\alpha t_p \Omega)^2\big) + \Big(\alpha \frac{v_\perp}{c}\Big)^2 (\alpha_s^{-2} - \lambda)\Big] = 0 \tag{135}$$

This polynomial expansion identifies the independent physical states of the boundary via its constituent roots.

9.1 The Longitudinal Axis (λ_1)

Extracting the decoupled spatial root ($\lambda_1 = \alpha_s^{-2}$) demonstrates invariant volumetric expansion along the primary axis of vorticity. Because this root factors completely from the transverse polynomial, the rotational axis is orthogonally independent from transverse shear. This decoupled strain constitutes the unperturbed inertial core.

9.2 Electromagnetic Cross-Coupling Equivalence

The remaining cubic polynomial restricts the temporal and transverse spatial states. Factoring the unperturbed volumetric strain roots from the active kinematic fields yields the following equality:

$$(\alpha_s^2 + \lambda)(\alpha_s^{-2} - \lambda)^2 = \Big(\alpha \frac{v_\perp}{c}\Big)^2 (\alpha_s^{-2} - \lambda) - (\alpha_s^2 + \lambda)(\alpha t_p \Omega)^2 \tag{136}$$

The spatial eigenvalue ($\alpha_s^{-2} - \lambda$) scales the electric strain ($v_\perp / c$), while the temporal eigenvalue ($\alpha_s^2 + \lambda$) scales the magnetic strain ($t_p \Omega$). Solving this equality for transverse slip determines the governing ratio:

$$\Big(\alpha \frac{v_\perp}{c}\Big)^2 = (\alpha_s^2 + \lambda)(\alpha_s^{-2} - \lambda) + \frac{\alpha_s^2 + \lambda}{\alpha_s^{-2} - \lambda}(\alpha t_p \Omega)^2 \tag{137}$$

Variation in vorticity causes a proportional shift in the transverse slip, establishing the pointwise impedance required to support Maxwell's equations.

9.3 Orthogonal Stability (Eigenstate Degeneracy)

As the transverse roots tend towards the longitudinal root ($\lambda \to \alpha_s^{-2}$), the spatial multipliers tend to zero and the polynomial collapses to:

$$0 = -(\alpha_s^2 + \alpha_s^{-2})(\alpha t_p \Omega)^2 \tag{138}$$

Because the scalar capacity terms are positive, this equality is satisfied when vorticity evaluates to zero ($\Omega = 0$). The matrix excludes transverse (electromagnetic) roots from degenerating into the longitudinal spatial axis. This boundary preserves the orthogonal geometry of propagating waves, preventing spatial collapse provided that internal vorticity is present.

9.4 Classical Volume Preservation ($\alpha_s = 1$)

Evaluating the polynomial at full background capacity, where longitudinal dynamic pressure is zero ($P_{dyn} = 0 \implies \alpha_s = 1$), reduces the polynomial to:

$$(1 + \lambda)(1 - \lambda)^2 = \Big(\alpha \frac{v_\perp}{c}\Big)^2 (1 - \lambda) - (1 + \lambda)(\alpha t_p \Omega)^2 \tag{139}$$

While the zero trace of the antisymmetric matrix ($\mathrm{Tr}(A_{\mu\nu}) = 0$) guarantees first-order volume preservation ($\delta \det(\hat{U}) = 0$), preserving the 4D determinant against the symmetric baseline ($\det(\hat{U}_{\mu\nu}) = \det(S_{\mu\nu}) = -1$ at $\alpha_s = 1$) is equivalent to a non-linear constraint between the electric slip and magnetic vorticity components:

$$\det(\hat{U}_{\mu\nu}) = -1 - (\alpha t_p \Omega)^2 + \left(\alpha \frac{v_\perp}{c}\right)^2 = -1 \iff \left(\alpha \frac{v_\perp}{c}\right)^2 = (\alpha t_p \Omega)^2 \tag{140}$$

By mapping the electric field ($E \propto v_\perp/c$) and magnetic field ($B \propto t_p\Omega$), this equivalence reproduces the impedance of free space (Z_0), and the proportional magnitude ($E = cB$).

9.5 Non-Linear Vacuum Polarization (The Schwinger Limit)

Electromagnetism assumes linear superposition, but the tensor restricts this to the weak-shear limit. For an aligned field configuration with electric slip $E_x = \alpha v_{\perp,x}/c$ and magnetic vorticity $B_z = \alpha t_p \Omega_z$ ($v_{\perp,y} = v_{\perp,z} = \Omega_x = \Omega_y = 0$, $\alpha_s \neq 0$), the 4×4 characteristic polynomial of $\hat{U}_{\mu\nu}$ expands as:

$$\det(\hat{U}_{\mu\nu} - \lambda \mathbf{I}) = (\alpha_s^{-2} - \lambda)\Big[(-\alpha_s^2 - \lambda)\big((\alpha_s^{-2} - \lambda)^2 + B_z^2\big) + E_x^2(\alpha_s^{-2} - \lambda)\Big] \tag{141}$$

At the unit-admittance vorticity threshold ($\alpha_s = 1, B_z = 1$), evaluating Equation 141 at $\lambda = 1 \pm i$ yields $\det(\hat{U}_{\mu\nu} - (1 \pm i)\mathbf{I}) = -E_x^2$ (for example, $E_x = 1/2$ gives $\det(\hat{U}_{\mu\nu} - (1+i)\mathbf{I}) = -1/4 \neq 0$). Consequently, $\lambda = 1 \pm i$ are eigenvalues of the full 4×4 tensor if and only if electric slip vanishes ($v_\perp = 0$, $E_x = 0$), where the polynomial factors into $(-1 - \lambda)(1 - \lambda)\big((1 - \lambda)^2 + 1\big) = 0$ with transverse 4-eigenvectors $\psi_\pm = (0, 1, \mp i, 0)^T$.

Substituting the decoupled transverse roots ($\lambda = 1 \pm i$ at $v_\perp = 0$) into the Mobius rational transformation $Z(\lambda) = (1 + \lambda)/(1 - \lambda)$ produces the spectral coefficient modulus:

$$|Z(\lambda_\pm)| = \left|\frac{1 + (1 \pm i)}{1 - (1 \pm i)}\right| = \frac{|2 \pm i|}{|\mp i|} = \sqrt{5} \tag{142}$$

Because $v_\perp = 0$ is required to decouple the spatial block, the physical slip-to-vorticity ratio $v_\perp/(ct_p\Omega)$ is zero in this state; $|Z(\lambda_\pm)| = \sqrt{5}$ is therefore a spectral coefficient modulus of the decoupled vorticity block rather than a simultaneous nonzero-slip electromagnetic impedance.

9.6 Continuous Unitary Evolution ($U(1)$ Symmetry)

Evaluating the characteristic polynomial (Equation 141) at the real spatial ($\lambda = \alpha_s^{-2}$) or real temporal ($\lambda = -\alpha_s^2$) roots under $E_x = 0$ isolates the longitudinal subspace, whereas the complex conjugate roots govern the decoupled 2×2 transverse spatial block ($v_\perp = 0$):

$$\hat{U}_\perp = \begin{bmatrix} \alpha_s^{-2} & -\alpha t_p \Omega \\ \alpha t_p \Omega & \alpha_s^{-2} \end{bmatrix} = \alpha_s^{-2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \alpha t_p \Omega \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{143}$$

A matrix possessing this structure ($a\mathbf{I} + b\mathbf{J}$, where $\mathbf{J}^2 = -\mathbf{I}$) is isomorphic to the complex plane ($a + ib$). The real scalar (α_s^{-2}) specifies the longitudinal volumetric amplitude. The antisymmetric generator ($\alpha t_p \Omega$) governs the rotational frequency.

Because the exponential of an antisymmetric generator structures the rotation group ($e^{\mathbf{J}\theta} = \cos\theta\mathbf{I} + \sin\theta\mathbf{J}$), the boundary operates as an acoustic standing wave. This connects the quantum mechanical $U(1)$ unitary phase rotation to the asymmetric strain components.

Evaluating the eigenvectors ($\mathbf{v}$) of this decoupled transverse sub-matrix for the complex conjugate roots ($\lambda = \alpha_s^{-2} \pm i\alpha t_p \Omega$ at $v_\perp = 0$) determines the spatial orientation of the strain:

$$\begin{bmatrix} \mp i\alpha t_p \Omega & -\alpha t_p \Omega \\ \alpha t_p \Omega & \mp i\alpha t_p \Omega \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \implies \mathbf{v}_\pm = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \mp i \end{bmatrix} \tag{144}$$

The unitary normalization of these orthogonal vectors scales by $1/\sqrt{2} = \cos(\pi/4)$, numerically matching the $\alpha_{yield} = 1/\sqrt{2}$ compliance factor derived independently from the Compton-Schwarzschild boundary intersection ($\hbar/Mc = 2GM/c^2$) in Section 8.3. These complex orthogonal vectors recover the Jones vectors for circular polarization in the decoupled transverse plane ($v_\perp = 0$).

9.7 The Generalized Determinant ($\det(\hat{U}) = 0$)

Whereas the Riemannian metric volume is governed by the symmetric determinant $g \equiv \det(S_{\mu\nu}) = -\alpha_s^{-4}$ (so that at $\alpha_s = 1, v_\perp = 0, \alpha t_p \Omega_z = 1$, $\det(S_{\mu\nu}) = -1$ while $\det(\hat{U}_{\mu\nu}) = -2$), setting the full asymmetric determinant $\det(\hat{U}_{\mu\nu})$ to zero on the subcritical domain ($\alpha_s \in (0, 1]$) identifies the singular characteristic threshold where electric slip balances magnetic vorticity and symmetric compliance:

$$\det(\hat{U}_{\mu\nu}) = -\alpha_s^{-4} - (\alpha t_p \Omega)^2 + \alpha_s^{-4} \left(\alpha \frac{v_\perp}{c}\right)^2 = 0 \tag{145}$$

Multiplying by the spatial capacity ($\alpha_s^4 > 0$) clears the denominator and isolates the hyperbolic boundary, which extends by continuity to $\alpha_s \to 0^+$:

$$\left(\alpha \frac{v_\perp}{c}\right)^2 - \alpha_s^4 (\alpha t_p \Omega)^2 = 1 \tag{146}$$

This equality ($E^2 - \alpha_s^4 B^2 = 1$) unites distinct yield limits across $\alpha_s \in [0, 1]$:

- **Hawking Radiation ($0 < \alpha_s < 1$):** As gravitational dynamic pressure increases, the magnetic term is suppressed by the fourth power of the scalar admittance (α_s^4). The boundary yields at exponentially lower energy thresholds, structuring spontaneous pair production near an event horizon.
- **The Kinematic Mass Limit ($\alpha_s \to 0^+$):** As dynamic pressure approaches saturation, the magnetic term vanishes ($\alpha_s^4 \to 0$). Evaluating the boundary (146) at $\alpha_s = 0$ yields $E^2 = 1 \implies v_\perp = c/\alpha$. Because longitudinal causality restricts energy transport (group velocity) to $v_g < c$, this variable operates as the orthogonal phase velocity of the boundary ($v_p = c/\alpha$). This satisfies the de Broglie kinematic wave relation ($v_p \cdot v_g = c^2$) for a ground-state defect oscillating at $v_g = \alpha c$.

9.8 The Temporal Cavitation Limit ($\alpha_s \to 0^+$)

In the one-sided cavitation limit as dynamic pressure approaches capacity ($P_{dyn} \to P_c^- \implies \alpha_s \to 0^+$), the spatial strain diagonals diverge ($\hat{U}_{ii} = 1/\alpha_s^2 \to +\infty$), while the inverse spatial compliance vanishes ($S^{ii} = \alpha_s^2 \to 0^+$) and the Eulerian Almansi strain remains finite ($e_{ij} = \frac{1}{2}(1 - \alpha_s^2)\delta_{ij} \to \frac{1}{2}\delta_{ij}$, Section 16.3). Simultaneously, the temporal diagonal collapses ($\hat{U}_{00} = -\alpha_s^2 \to 0^-$) and the temporal multiplier ($-\alpha_s^2 - \lambda$) simplifies to ($-\lambda$), driving the temporal root to zero ($\lambda_0 \to 0$).

This zero-root transitions the metric signature from $(-, +, +, +)$ to $(0, +, +, +)$. Kinematic transport halts across this boundary, identifying both the macroscopic event horizon and the microscopic rest mass boundary.

9.9 Complex Strain & The Deborah Number

Separating the transverse spatial components from the primary tensor constructs a 2×2 submatrix ($\hat{U}_\perp$). Section 9.6 identifies this structure as isomorphic to the complex plane:

$$\hat{U}_\perp = \alpha_s^{-2} \mathbf{I} + \alpha t_p \Omega \mathbf{J} \equiv \alpha_s^{-2} + i(\alpha t_p \Omega) \tag{147}$$

This partitions the strain into two orthogonal bases:

- **Real Axis (Elastic Storage):** The symmetric diagonal (α_s^{-2}) quantifies longitudinal volumetric packing.
- **Imaginary Axis (Viscous Flow):** The antisymmetric generator ($i\alpha t_p \Omega$) tracks Clebsch vorticity.

1. Derivation of the Rheological Phase Angle

The ratio of the antisymmetric off-diagonals (viscous flow) to the symmetric spatial diagonals (elastic storage) determines the continuous rheological phase angle ($\tan\delta$):

$$\begin{aligned} \tan\delta &= \frac{\mathrm{Im}(\hat{U}_\perp)}{\mathrm{Re}(\hat{U}_\perp)} \\ \tan\delta &= \frac{\alpha t_p \Omega}{\alpha_s^{-2}} \\ \tan\delta &= \alpha_s^2 \alpha t_p \Omega \end{aligned} \tag{148}$$

2. The Deborah Number

The Deborah number (De) scales inversely with the loss tangent ($De \propto 1/\tan\delta$). The tensor extracts the Deborah number from the inverted ratio:

$$De = \frac{\mathrm{Re}(\hat{U}_\perp)}{\mathrm{Im}(\hat{U}_\perp)} = \frac{\alpha_s^{-2}}{\alpha t_p \Omega} = \frac{1}{\alpha_s^2 \alpha t_p \Omega} \tag{149}$$

This derivation governs the thermodynamic state of the vacuum metric:

- **Fluid Phase ($De \to 0$):** As internal vorticity increases or background capacity remains unperturbed ($\alpha_s \to 1$), the ratio approaches zero. The matrix models as a compliant fluid.
- **Glass Transition ($De \to \infty$):** As dynamic pressure increases ($P_{dyn} \to P_c$), scalar admittance collapses ($\alpha_s \to 0$). The denominator goes to zero, causing the ratio to diverge. The coordinate geometry hardens, demonstrating solid elastic behavior.

3. Complex Magnitude

The total strain applied to the boundary calculates as the magnitude of the complex vector:

$$|\hat{U}_\perp| = \sqrt{(\alpha_s^{-2})^2 + (\alpha t_p \Omega)^2} = \sqrt{\alpha_s^{-4} + \alpha^2 t_p^2 \Omega^2} \tag{150}$$

Because the structural components operate orthogonally in the complex plane, the metric sustains volumetric compression and rotational shear simultaneously. The tensor prevents infinite coordinate compression by distributing kinematic load across the imaginary axis.

4. The Kinematic Crossover Constraint

To establish the boundary, the Deborah number is derived from the parameters of the matrix. The Deborah number represents the ratio of transverse coupling capacity to proper temporal compliance.

In the asymmetric tensor, the transverse capacity is bounded by the Viscoelastic Modulus Ratio (α). The proper temporal compliance is given by the square root of the time-time diagonal ($\sqrt{-\hat{U}_{00}} = \alpha_s$). Dividing these tensor components formulates the macroscopic kinematic boundary:

$$De_{macro} = \frac{\alpha}{\sqrt{-\hat{U}_{00}}} = \frac{\alpha}{\alpha_s} \tag{151}$$

Because the inverse scalar admittance equates to the Lorentz factor ($\alpha_s^{-1} = \gamma$), this substitution produces the time-dilated kinematic limit:

$$De_{macro} = \alpha\gamma \tag{152}$$

Setting this macroscopic boundary (De_{macro}) equal to the localized extraction (Equation 149) produces:

$$\begin{aligned} \frac{\alpha}{\alpha_s} &= \frac{\alpha_s^{-2}}{\alpha t_p \Omega} \\ \alpha^2 (t_p \Omega) &= \alpha_s^{-1} \\ t_p \Omega &= \frac{1}{\alpha^2 \alpha_s} = \frac{\gamma}{\alpha^2} \end{aligned} \tag{153}$$

This intersection dictates the physical scaling of the topological boundary. As dynamic pressure saturates the metric capacity ($P_{dyn} \to P_c$), scalar admittance collapses ($\alpha_s \to 0$). This substitution requires that the normalized internal Clebsch vorticity diverges proportionally to the Lorentz factor ($\gamma \to \infty$). The metric maintains the constraint ($\nabla \cdot \mathbf{v} = 0$) by transferring infinite longitudinal compression into unbounded transverse rotational shear.

5. The Phenomenological Energy Limit

In relativistic kinematics, the total energy (E) of a boundary increases from its unperturbed rest mass energy ($E_0 = mc^2$) via the Lorentz factor:

$$\gamma = \frac{E}{mc^2} \tag{154}$$

Substituting this energy ratio into the macroscopic boundary ($De_{macro} = \alpha\gamma$) translates the Deborah number into a thermodynamic energy scale:

$$De = \alpha \left(\frac{E}{mc^2} \right) \tag{155}$$

This formulation converts the transverse shear of the complex tensor into the observable kinematic energy threshold of the particle.

10. Tensor Covariance & Non-Inertial Reference Frames

Extending the boundary states defined in Section 8, the metric maintains structural equilibrium across relative observer geometries. Comparing states ($\hat{U}_{\mu\nu}$) across arbitrary reference frames reveals the coordinate transformations between separate observers.

10.1 Kinematic Classification of Observer Frames

Translating the tensor from a resting observer to a transformed observer decomposes into two mechanical categories: Scalar Translation and Tensor Covariance.

1. Scalar Translation (Gradient Isomorphism):

Comparisons between stationary observers separated by a thermodynamic gradient resolve without kinematic shear. Because the matrix defines macroscopic gravity as the static pressure gradient (∇P_{static}), comparing distinct gravitational reference frames operates via coordinate translation. The transformation matrices reduce to the Kronecker delta ($\Lambda_\alpha^\mu = \delta_\alpha^\mu \implies \hat{U}'_{\alpha\beta} = \hat{U}_{\alpha\beta}$). Modifying the observer coordinate translates the matrix along the continuous gradient, scaling the temporal and spatial diagonals inversely via the local scalar admittance ($\alpha_{s,1} \to \alpha_{s,2}$) while preserving the reciprocal invariant ($\hat{U}_{00}\hat{U}_{ii} = -1$):

$$\frac{\hat{U}_{00}(\alpha_{s,2})}{\hat{U}_{00}(\alpha_{s,1})} = \frac{\alpha_{s,2}^2}{\alpha_{s,1}^2}, \qquad \frac{\hat{U}_{ii}(\alpha_{s,2})}{\hat{U}_{ii}(\alpha_{s,1})} = \frac{\alpha_{s,1}^2}{\alpha_{s,2}^2} \tag{156}$$

This coordinate transformation governs stationary scalar variance (comparing gravitational elevations or distinct cosmological Hubble epochs), requiring no internal matrix cross-coupling.

2. Tensor Covariance (Kinematic Mixing):
Comparisons where one observer possesses relative velocity, applied rotation, or transverse shear demand matrix transformation. The tensor establishes equivalence between the observers via the Lorentz-Jacobian transformation matrix ($\Lambda^{\mu}_{\alpha} = \partial x^{\mu}/\partial x'^{\alpha}$) [41]:

$$\hat{U}'_{\alpha\beta} = \Lambda^{\mu}_{\alpha}\Lambda^{\nu}_{\beta}\hat{U}_{\mu\nu} = \Lambda^{\mu}_{\alpha}\Lambda^{\nu}_{\beta}S_{\mu\nu} + \Lambda^{\mu}_{\alpha}\Lambda^{\nu}_{\beta}A_{\mu\nu} \tag{157}$$

This operation directs the symmetric bulk diagonals ($S_{00} = -\alpha_s^2,\ S_{ii} = 1/\alpha_s^2$) to cross-multiply against the antisymmetric transverse shear components ($A_{\mu\nu}$), extracting the mechanics of relativistic and fictitious forces.

10.2 Active Kinematic Strain & The Invariant Phase Limit

When a topological boundary translates through the baseline metric, it generates a convective flow field (v_{rel}). Rather than a passive coordinate rotation, it represents active physical strain.

The convective velocity subjects the translating boundary to dynamic pressure ($P_{dyn} = \frac{1}{2}\rho_\tau v_{rel}^2$). By Bernoulli conservation [34] ($P_{static} = P_c - P_{dyn}$), the temporal diagonal ($\hat{U}_{00} = -\alpha_s^2$) yields the structural limit:

$$\begin{aligned}
\hat{U}'_{00} &= -\left(\frac{P_c - P_{dyn}}{P_c}\right) \\
\hat{U}'_{00} &= -\left(1 - \frac{\frac{1}{2}\rho_\tau v_{rel}^2}{\frac{1}{2}\rho_\tau c^2}\right) \\
\hat{U}'_{00} &= -\left(1 - \frac{v_{rel}^2}{c^2}\right)
\end{aligned} \tag{158}$$

As the relative velocity of the boundary approaches the acoustic limit ($v_{rel} \to c$), the dynamic pressure saturates ($P_{dyn} \to P_c$). The temporal diagonal collapses to zero ($\hat{U}'_{00} \to 0$), driving the symmetric spatial diagonals to diverge ($S'_{ii} \to \infty$).

Continuous temporal dilation and the invariant speed of light emerge from this operation. The tensor thus reveals Lorentz contraction as an active thermodynamic response which actively conserves the maximum static pressure capacity.

10.3 Rotational Covariance & Transverse Leakage

Comparing a stationary inertial observer to a rotating observer introduces multi-axis shear. For an observer rotating at angular velocity ω, the azimuthal coordinate transformation ($d\phi = d\phi' + \omega dt'$) defines the temporal Jacobian component [32]:

$$\Lambda^{\phi}_{0} = \frac{\partial \phi}{\partial t'} = \omega \tag{159}$$

Applying the transformation matrix (157), to evaluate the space-time off-diagonal ($\hat{U}'_{0i}$) in the rotating frame, produces the cross-coupling mechanism:

$$\begin{aligned}
\hat{U}'_{0i} &= \Lambda^{\alpha}_{0}\Lambda^{\beta}_{i}\hat{U}_{\alpha\beta} \\
\hat{U}'_{0i} &= \Lambda^{0}_{0}\Lambda^{\beta}_{i}\hat{U}_{0\beta} + \Lambda^{\phi}_{0}\Lambda^{\beta}_{i}\hat{U}_{\phi\beta} \\
\hat{U}'_{0i} &= (1)\Lambda^{\beta}_{i}\hat{U}_{0\beta} + (\omega)\Lambda^{\beta}_{i}\hat{U}_{\phi\beta}
\end{aligned} \tag{160}$$

The transformation causes the space-space off-diagonals (vorticity, $\hat{U}_{\phi\beta}$) to influence the space-time off-diagonals (transverse velocity, $\hat{U}'_{0i}$), scaled by the rotation rate (ω). The rotating coordinate frame results in apparent transverse velocity fields (Coriolis and Centrifugal strain) that resolve to zero for the resting frame. Fictitious forces manifest as the mechanical artifact of internal Clebsch vorticity across the rotating spatial boundary.

10.4 Conformal Scale Variance (The Micro-Macro Asymmetry)

The physical structure of the continuum breaks scale invariance. While the metric appears identical across arbitrary macroscopic scales, the generation of discrete topological mass (Section 8) sets a fixed physical boundary dimension ($r_c = \hbar/mc > 0$).

Applying the dilation vector field ($\xi^\mu = \sigma x^\mu,\ \partial_\mu \xi^\alpha = \sigma \delta^\alpha_\mu$) to compare a microscopic rest frame to a macroscopic observer exposes the broken symmetry. Evaluating the full three-term Lie derivative (Equation 91) decomposes the dilation response into an Eulerian directional gradient and a homogeneous conformal weight:

$$\mathcal{L}_\xi \hat{U}_{\mu\nu} = \sigma x^\alpha \partial_\alpha \hat{U}_{\mu\nu} + 2\sigma \hat{U}_{\mu\nu} \tag{161}$$

- **The Microscopic Observer:** Across the topological defect boundary ($r \sim r_c$), discrete spatial incompatibility generates a non-vanishing radial strain gradient ($\sigma x^\alpha \partial_\alpha \hat{U}_{\mu\nu} \neq 0$), so $\mathcal{L}_\xi \hat{U}_{\mu\nu} \neq 2\sigma \hat{U}_{\mu\nu}$, breaking conformal scale invariance.
- **The Macroscopic Observer:** Integrating the metric across a macroscopic volume ($V \gg r_c^3$) causes the counter-circulating transverse shear components ($A_{\mu\nu}$) and local boundary gradients to average to zero ($\iiint_V \partial_\alpha \hat{U}_{\mu\nu}\, dV = 0$), homogenizing the tensor to its symmetric bulk sector and recovering macroscopic conformal scaling ($\mathcal{L}_\xi \langle \hat{U}_{\mu\nu} \rangle_V = 2\sigma \langle S_{\mu\nu} \rangle_V$):

$$\iiint_V A_{\mu\nu}\, dV = 0 \implies \langle \hat{U}_{\mu\nu} \rangle_V = \langle S_{\mu\nu} \rangle_V \implies \text{Metric Homogenization} \tag{162}$$

The phase rotations of the microscopic tensor homogenize over expanded integration volumes, recovering the macroscopic dynamics of General Relativity. This distinguishes macroscopic scale invariance from the topological scale variance at the defect boundary.

10.5 The Equivalence Principle (Acceleration Isomorphism)

The equivalence principle postulates the observational equivalence between a uniform gravitational field and a uniformly accelerating reference frame [17]. The tensor derives this equivalence not as an independent assumption, but as a consequence of comparing distinct observer frames.

Evaluating the volumetric trace deformation ($\delta \hat{U}_{00} \propto \nabla P$) for both observers recovers the mechanical strain:

- **The Gravitational Observer:** A stationary frame submerged in a metric depression is governed by the ambient static pressure gradient (∇P_{static}).
- **The Accelerating Observer:** A non-inertial frame navigating a flat macroscopic vacuum increases its convective velocity. Substituting this kinematic load into the macroscopic metric generates a dynamic pressure gradient (∇P_{dyn}).

Because the continuous metric conserves total thermodynamic capacity ($P_{static} = P_c - P_{dyn}$):

$$\nabla P_{static} = -\nabla P_{dyn} \implies \mathbf{a}_{grav} = -\mathbf{a}_{kin} \tag{163}$$

Increasing the dynamic load extracts phase capacity from the identical structural ledger as an ambient gravitational deficit. Applying a continuous kinematic acceleration vector (∇P_{dyn}) and translating across a static gravitational potential (∇P_{static}) induce the identical volumetric deformation upon the Tensor.

10.6 Discrete Frame Inversion (CPT Symmetry & Antimatter)

Comparing a forward-time or spatially inverted observer partitions the discrete transformation invariants of the metric.

Part I: Time-Reversal (T-Symmetry)

Applying the time-reversal transformation matrix (Λ_T) inverts the temporal coordinate ($t \to -t$). The matrix is defined as $\Lambda_T = \text{diag}(-1, 1, 1, 1)$ [57]. Applying this operation to the tensor partitions orthogonal responses:

1. Symmetric Invariance (Mass/Gravity):

Transforming the temporal diagonal ($\hat{U}_{00}$) via the time-reversal matrix:

$$\hat{U}'_{00} = \Lambda^{\mu}_{0}\Lambda^{\nu}_{0}\hat{U}_{\mu\nu} = (-1)(-1)\hat{U}_{00} = \hat{U}_{00} \tag{164}$$

Because the negative inversion squares, the scalar diagonal is preserved. The inverted coordinate frame produces identical longitudinal inertia and gravitational attraction.

2. Antisymmetric Inversion (Kinematic Shear):

Transforming the space-time off-diagonal ($\hat{U}_{0i}$) defining transverse slip yields:

$$\hat{U}'_{0i} = \Lambda^{\mu}_{0}\Lambda^{\nu}_{i}\hat{U}_{\mu\nu} = (-1)(1)\hat{U}_{0i} = -\hat{U}_{0i} \tag{165}$$

The temporal index flips the sign of the transverse shear. The inverted coordinate frame outputs an identical phase with reversed transverse slip.

Executing these transformation operations verifies Time-Reversal (T) symmetry within the continuum.

Part II: Spatial Inversion (P-Symmetry) & Charge Conjugation (C)

To evaluate spatial Parity (P) and Charge Conjugation (C), the discrete spatial inversion matrix (Λ_P) is applied to the metric. The matrix inverts the spatial coordinates ($x^i \to -x^i$) and is defined as $\Lambda_P = \text{diag}(1, -1, -1, -1)$.

3. Volumetric Invariance (Mass/Pressure):

Transforming the spatial diagonals ($\hat{U}_{ii}$) yields:

$$\hat{U}'_{ii} = \Lambda^{\mu}_{i}\Lambda^{\nu}_{i}\hat{U}_{\mu\nu} = (-1)(-1)\hat{U}_{ii} = \hat{U}_{ii} \tag{166}$$

The symmetric spatial packing remains positive. The inverted coordinate frame produces an identical mass density and gravitational gradient.

4. Axial Inversion (Chirality & Parity):

Transforming the antisymmetric spatial off-diagonals ($\hat{U}_{ij}$) yields:

$$\hat{U}'_{ij} = \Lambda^{\mu}_{i}\Lambda^{\nu}_{j}\hat{U}_{\mu\nu} = (-1)(-1)\hat{U}_{ij} = \hat{U}_{ij} \tag{167}$$

While the spatial rotation is preserved as an axial vector, the spatial inversion of the coordinate axes reverses the chirality (geometric handedness) of the Clebsch vortex. The topological defect keeps its original flow magnitude but flips its spatial twist.

5. Polar Inversion (Winding Number & Charge):

Transforming the space-time off-diagonal ($\hat{U}_{0i}$) yields:

$$\hat{U}'_{0i} = \Lambda^{\mu}_{0}\Lambda^{\nu}_{i}\hat{U}_{\mu\nu} = (1)(-1)\hat{U}_{0i} = -\hat{U}_{0i} \tag{168}$$

The temporal-spatial cross term inverts as a polar vector. While classical kinematics treats this as boundary velocity ($v \to c$), this parameter acts as the directionality of the Clebsch phase. Charge polarity operates as the topological winding number of this boundary.

Because the metric preserves the axial rotation (U_{ij}) but inverts the transverse directionality (U_{0i}), the geometric handedness of the winding number is reversed. This permits spatial Parity inversion (P) to map to Charge Conjugation (C) for an isolated defect.

For the proposed direction for deriving CP violation, please see section 16.2.

11. Galactic Kinematics and the Cosmological Boundary

The continuous metric transitions from a collection of mass defects to the thermodynamic boundary of the expanding macroscopic background. Computing the total dynamic pressure state of the continuum secures a bounded geometric baseline for galactic kinematics.

11.1 The Total Dynamic Pressure State

The dynamic pressure depleted by a baryonic aggregate (mass) is parameterized as $P_{bar} = -\rho_\tau \Phi(r)$, where $\Phi(r)$ is the classical gravitational potential. Simultaneously, the continuum sustains the macroscopic load of longitudinal cosmological expansion. Attenuated by the causal horizon, this flow saturates exponentially: $v_0(r) = c(1 - e^{-r/R_h})$. The kinetic load equates to $P_{cosmo} = \frac{1}{2}\rho_\tau c^2(1 - e^{-r/R_h})^2$.

Because thermodynamic pressures superimpose as scalars, the total dynamic pressure sums to:

$$P_{dyn}(r) = -\rho_\tau \Phi(r) + \frac{1}{2}\rho_\tau c^2 \left(1 - e^{-r/R_h}\right)^2 \tag{169}$$

11.2 The Bounded Kinematic Baseline

In a continuous metric, orbital velocity responds to the spatial gradient of the dynamic pressure state. To maintain a circular orbit, inward centripetal acceleration balances the net continuum acceleration gradient ($g = -\frac{1}{\rho_\tau}\frac{\partial P_{dyn}}{\partial r}$). The kinematic state is modeled by the radial differential operator:

$$v^2 = -r\left(\frac{1}{\rho_\tau}\frac{\partial P_{dyn}}{\partial r}\right) \tag{170}$$

Executing the radial derivative across Equation 169 outputs the kinematic baseline of the continuum:

$$v_{base}^2(r) = r\frac{\partial \Phi}{\partial r} - \frac{c^2 r}{R_h}e^{-r/R_h}\left(1 - e^{-r/R_h}\right) \tag{171}$$

This derived baseline bounds the kinematic domain, correcting the coordinate divergences of the classical Schwarzschild-de Sitter (SdS) metric:

- **The Core Bound ($r \to 0$):** Inside a uniform baryonic bulge, the interior potential spatial derivative yields an outward gradient $\frac{\partial \Phi}{\partial r} \propto r$. Substituting this into the operator produces $v_{base}^2 \propto r^2$. As the cosmological gradient attenuates at local scales, the baseline reproduces the positive solid-body rotation profile ($v_{base} \propto r$) without generating imaginary roots.
- **The Finite Boundary (The Hill Sphere):** For exterior domains, the potential transitions to $\Phi_{out} = -GM/r$. Equation 171 evaluates to a finite root where inward baryonic acceleration equals outward cosmological advection ($v_{base}^2 = 0$). This marks the Jacobian capture radius, terminating the coordinate frame for bound rotational orbits.
- **The Macroscopic Asymptote ($r \to \infty$):** As the radial coordinate extends toward the macroscopic causal boundary, the longitudinal expansion velocity saturates ($v_0 \to c$). When the dynamic pressure maximizes to the capacity limit ($P_{dyn} \to P_c$) the orbital requirement drops to zero ($v_{base}^2 \to 0$). Geometric expansion is asymptotically inertial.

11.3 Isolating the Kinematic Discrepancy

The continuum requires a Keplerian decline terminating at a finite detachment limit. To map the geometric requirements of the missing mass anomaly, the unified kinematic state is formulated

as an empirically informed *ansatz*, rearranged to isolate the effective kinematic discrepancy (Γ_{anom}):

$$\Gamma_{anom}(r) = v_{obs}^2(r) - \left[\frac{GM_{bar}(r)}{r} - \frac{c^2 r}{R_h} e^{-r/R_h}\left(1 - e^{-r/R_h}\right)\right] \tag{172}$$

Where $v_{obs}^2(r)$ represents the empirical rotational velocity field. This separation bounds the structural parameters the anomaly must fulfill:

- **The Core Attenuation ($r \to 0$):** While High Surface Brightness (HSB) spirals exhibit baryon-dominated cores, Low Surface Brightness (LSB) and dwarf galaxies demand a non-zero anomaly at the origin. However, across all morphologies, empirical kinematics ($v_{obs}^2 \propto r^2$) require the enclosed anomalous mass to scale geometrically as $M_{anom}(r) \propto r^3$. Taking the spatial derivative dictates a constant volumetric density at the origin ($\rho \to$ const). Singular gravitational cusps are structurally prohibited; the anomaly is bounded to a flat-cored spatial profile.

- **The Effective Halo Plateau ($r \gg R_c$):** In the macroscopic halo, the observed rotation curve forms a plateau ($v_{obs}^2 \approx v_{flat}^2$). The ansatz isolates the discrepancy as $\Gamma_{anom}(r) \approx v_{flat}^2 - \frac{GM_{bar}(r)}{r}$. Sustaining this plateau requires a supplementary, positive dynamic pressure field scaling logarithmically ($P_{anom} \propto \ln(R_{max}/r)$) to dominate the decaying baryonic gradient.

- **The Detachment Limit (The Hill Sphere):** As established in Section 11.2, bound rotational orbits physically terminate at the Jacobian capture radius where total acceleration resolves to zero. Consequently, the logarithmic pressure well cannot extend to spatial infinity. The structural anomaly must truncate at this macroscopic coordinate boundary.

11.4 The Universal No-Go Theorem

A physical resolution to Γ_{anom} is constrained by two mechanical boundaries: it requires a supplementary pressure well ($P_{anom} \propto \ln(R_{max}/r)$) to provide a constant kinematic plateau, and its amplitude requires coupling to baryonic mass values to satisfy the Baryonic Tully-Fisher Relation ($M_{bar} \propto v_{flat}^4$).

By combining continuous geometric limitations with the causal coupling constraints of the ansatz, a Universal No-Go Theorem is constructed, falsifying current standard models:

- **Falsification of Modified Gravity (The Geometric Constraint):** In a continuous, empty vacuum topology ($\rho_{dyn} = 0$), the static pressure field satisfies Laplace's equation ($\nabla^2 P_{dyn} = 0$), permitting the Newtonian monopole solution in 3D space. Applying the 3D spherical Laplacian operator to a logarithmic well computes to $\nabla^2(-\ln r) = -1/r^2 \neq 0$. Utilizing non-linear modified operators violates the foundational axiom of the continuous tensor, which dictates that scalar dynamic pressures superimpose linearly.

- **Falsification of ΛCDM (The Causal Constraint):** While standard particulate mass (ΛCDM) supplies the required independent $1/r^2$ halo density, collisionless kinematic halos form hierarchically and independently of baryonic infall. They lack a dynamic thermodynamic mechanism to couple the required macroscopic logarithmic well to the baryonic mass, failing the causal constraints of the BTFR.

11.5 The Phenomenological Envelope

Consequently, the framework demonstrates that the missing mass anomaly cannot be a geometric illusion of empty space, nor can it be a decoupled primordial particulate halo. By restricting

convergence on existing standard models, the asymmetric tensor bounds the spacetime continuum.

To resolve the kinematic plateau, empirical observations specify that the continuum requires a structural modification exhibiting the following profile:

1. **Spatial Pressure Gradient:** A supplementary dynamic pressure field ($P_{anom} \propto \ln(R_{max}/r)$) generating the kinematic equivalent of a $1/r^2$ structural mass dispersion.

2. **Causal Coupling:** The amplitude of this pressure gradient requires a causal generation mechanism linked directly to the baryonic mass.

3. **Inertial Decoupling:** The gradient requires independent kinematic inertia to detach and coast ballistically during macroscopic cluster collisions.

The continuous metric derives the classical limit of General Relativity and extracts the phenomenological envelope of the kinematic anomaly. By satisfying the kinematic requirement for a $1/r^2$ structural mass dispersion while prohibiting the uncoupled variance of primordial ΛCDM particles, Dark Matter resolves as a dynamically spawned structural pressure gradient forged by thermodynamic strain, rather than a primordial cosmological remnant. The differential operators required to convert a centralized baryonic potential into this independent, stable pressure distribution remains an open problem for future works.

11.6 Proposed Resolution: Elastic Metric Tension (Scale-Invariant Vorticity Coupling)

As a resolution to satisfy the kinematic envelope without violating the density constraints, the rotational shear of the galactic disk translates as elastic tension upon the metric grid.

1. Microscopic Vorticity & The Baryon Count (N):
As derived in Section 8, a mass generates a cavitation boundary governed by internal vorticity ($\mathbf{\Omega}_{pr}$). The total baryonic mass (M_{bar}) of a galactic disk aggregates these defects ($N = M_{bar}/m_{pr}$).

2. Strain Polarization & Dimensional Reduction:
The macroscopic state of the metric is dependent on the volumetric expectation value of the aggregate micro-states. Thermodynamic randomness results in macroscopic cancellation ($\langle A_{\mu\nu} \rangle_V = 0$).

In a coordinate space, boundary conditions back-react upon microscopic states. A net galactic angular momentum ($\mathbf{J} \neq 0$) relative to the cosmic background applies a global rotational strain. This global strain breaks the degeneracy of the defects, making alignment with the bulk rotation energetically preferential.

Consequently, the thermal fluctuations of the baryonic vorticity cannot achieve isotropic cancellation. The global rotation induces a fractional alignment bias, formulating a non-zero macroscopic expectation value:

$$\langle A_{ij} \rangle = \frac{1}{V} \int_V A_{ij}^{(micro)} dV \neq 0 \tag{173}$$

Because this coherent residual correlates to the conserved angular momentum ($\mathbf{J}$), the aggregate torque becomes a steady-state axial field that is translationally invariant along the rotation axis ($\partial_z \langle A_{ij} \rangle = 0$). Because axial translational invariance alone permits uniform nonzero constants, dimensional reduction to the galactic midplane ($z = 0$) additionally invokes equatorial reflection anti-symmetry of the out-of-plane shear components ($A_{xz}(-z) = -A_{xz}(z)$ and $A_{yz}(-z) = -A_{yz}(z)$, forcing $A_{xz}(0) = A_{yz}(0) = 0$) together with the thin-disk plane-stress boundary condition ($\sigma_{zz} = 0$ from vanishing vertical traction at the free disk surfaces).

Under these symmetry and plane-stress boundary conditions, the kinematic response of the continuum is restricted to a 2D cylindrical plane-stress state (r, ϕ).

3. The 2D Laplacian & Logarithmic Shear on the Bounded Annulus:
The trace contraction of the tensor $(\hat{U}_{\mu\nu}\hat{U}^{\mu\nu})$ divides the kinetic energy. The antisymmetric trace filters the transverse shear stress $(P_{shear} = \frac{1}{4}P_c A_{\mu\nu}A^{\mu\nu})$. On the source-free galactic annulus bounded between the inner yield transition radius r_t and the outer capture radius R_{max} $(0 < r_t \leq r \leq R_{max})$, this scalar pressure field satisfies the 2D cylindrical Laplace equation $(\nabla^2_{2D} P_{shear} = 0)$:

$$\nabla^2_{2D} P_{shear} = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial P_{shear}}{\partial r}\right) = 0 \tag{174}$$

Integrating this operator radially across $r \in [r_t, R_{max}]$ subject to the outer vanishing boundary condition $P^{2D}_{shear}(R_{max}) = 0$ constructs the nonnegative logarithmic shear pressure field and its radial stress gradient:

$$P^{2D}_{shear}(r) = k \ln\left(\frac{R_{max}}{r}\right) \geq 0, \qquad \frac{\partial P^{2D}_{shear}}{\partial r} = -\frac{k}{r} \tag{175}$$

The metric generates a logarithmic tension well on the annulus, where the scalar amplitude $(k \geq 0)$ represents the shear load of the galaxy.

4. The Elastic Metric Tension:
By the conservation laws of the thermodynamic budget, this shear load acts as transverse tension upon the Total Scalar Admittance (α_s^2). The symmetric temporal diagonal $(S_{00} = -\alpha_s^2)$ shifts to accommodate both the 3D isotropic Newtonian gravity and polarized 2D shear:

$$S_{00} = -\left(1 - \frac{P^{3D}_{dyn} + P^{2D}_{shear}}{P_c}\right) \tag{176}$$

5. Geodesic Gradient Evaluation:
The coordinate acceleration of a test mass evaluates via the spatial gradient of the time-time diagonal in the weak-field geodesic equation $(a^i \approx \frac{1}{2}c^2\nabla^i S_{00})$. Inserting the admittance components translates the scalar pressure loads into geometric curvature:

$$a^i = \frac{c^2}{2P_c}\left(\nabla P^{3D}_{dyn} + \nabla P^{2D}_{shear}\right) \tag{177}$$

Because the spatial gradient of a pressure well points inward $(\nabla P \propto -\hat{\mathbf{r}})$, taking the derivatives of both the 3D potential $(\rho_\tau G M_{bar}/r)$ and the 2D logarithmic shear $(k\ln(R_{max}/r))$ produces:

$$a^i = \frac{c^2}{2P_c}\left(-\frac{\rho_\tau G M_{bar}}{r^2} - \frac{k}{r}\right)\hat{\mathbf{r}} = \left(\frac{G M_{bar}}{r^2} + \frac{c^2 k}{2P_c r}\right)(-\hat{\mathbf{r}}) \tag{178}$$

Both components pull inward. The 3D component retrieves the Keplerian centripetal bound, while the 2D polarized component supplies the $1/r$ inward acceleration.

6. Macroscopic Plug Flow (Kinematic Plateau Extraction):
Equating the anomalous 2D spatial component to the required inward centripetal acceleration limit (v^2_{rot}/r) cancels the radial coordinate:

$$\frac{v^2_{rot}}{r} = \left(\frac{c^2}{2P_c}\right)\frac{k}{r} \implies v_{rot} = c\sqrt{\frac{k}{2P_c}} \tag{179}$$

The orbital velocity squares to a constant kinematic plateau. In Bingham fluid dynamics, this region of constant velocity and zero internal dynamic shear represents a plug flow. The continuum moves as a unified bulk, sustained by the elastic tension of the vacuum (P_c).

7. The Three-Layer Rheological Coupling & The Macroscopic Yield Limit: The shear amplitude (k) evaluates via boundary matching across the transition interface. The macroscopic yield limit (a_0) is derived from the divergence equilibrium of the tensor components.

Transverse slip is parameterized by the space-time off-diagonal ($\hat{U}_{0i} = -\alpha v_{\perp,i}/c$). The spatial gradient of this component constructs the rotational acceleration ($\partial_j \hat{U}_{0i} \propto a_{rot}/c^2$). Simultaneously, macroscopic cosmological expansion applies a continuous temporal divergence to the volumetric spatial diagonals ($\partial_0 \hat{U}_{ii}$). Because optical depth scales the coordinate wavelength by the redshift ratio ($1+z$), the apparent temporal strain rate attenuates at high redshift ($H_{app}(z) = c/[R_{h,0}(1+z)]$).

To maintain a dynamic slipping state, the applied transverse shear gradient must exceed the baseline temporal strain rate ($\partial_j \hat{U}_{0i} > \partial_0 \hat{U}_{ii}$). Equating these orthogonal derivatives extracts the macroscopic kinematic yield threshold ($a_{rot} = cH_{app}(z)$). Mapping this 1D radial temporal strain onto the 2D azimuthal plane of the galactic disk resolves the rheological limit: $a_0(z) = c^2/[2\pi R_{h,0}(1+z)]$.

This equilibrium constructs a bounded, three-layer architecture for the galactic metric:

Zone 1: *The Inner Shear Layer.* In the inner regions of the galaxy, the transverse spatial gradient overpowers the temporal expansion threshold ($\partial_j \hat{U}_{0i} > \partial_0 \hat{U}_{ii}$). The coordinate layers slip against one another, maintaining a free rotational state structured by Newtonian gravity ($1/r^2$).

Zone 2: *The Trapped Plug Flow (The 3D → 2D Coupling).* As radial acceleration decays, the spatial gradient intersects the temporal limit ($\partial_j \hat{U}_{0i} = \partial_0 \hat{U}_{ii}$). At this transition radius (r_t), the gravitational shear equals the cosmological baseline strain.

When the transverse gradient does not exceed the longitudinal divergence, the exterior derivative of the 3D dynamic slip becomes zero ($d\mathbf{v} = 0$). Because the applied stress is insufficient to cause shear, the torque cannot dissipate via rotation. Instead, this unresolved stress propagates outward as static structural strain.

To conserve angular momentum across the boundary without dynamic slip, the symmetric stress tensor (σ_{ij}) reduces the axial component ($\sigma_{zz} = 0$), into a state of plane stress. This condition collapses the kinematics from a 3D isotropic divergence into a 2D cylindrical stress field (r, ϕ). Mechanically, this 2D planar stress bounding a 3D volume operates as surface tension. The metric locks into a coherent elastic membrane, satisfying the Bingham plug flow condition (Equation 179).

This outward propagation behaves as continuous stress integrated over time, acting as a sustained structural push across the coordinate grid. Consequently, the coordinate frame ($\hat{U}_{\mu\nu}$) rotates with the aggregate mass. The outer stars do not experience centrifugal coordinate acceleration ($d^2x^i/d\tau^2 = 0$); they are embedded at rest within this rotating surface tension. The coordinate space itself carries the rotational momentum outward until it reaches the terminal boundary.

Identifying this inner acceleration boundary constructs the transition radius:

$$\frac{GM_{bar}}{r_t^2} = a_0(z) \implies r_t = \sqrt{\frac{GM_{bar}}{a_0(z)}} \tag{180}$$

Because the boundary acceleration is inversely proportional to optical depth ($a_0(z) \propto 1/[1+z]$), the yield threshold shrinks in the early universe. Evaluating the transition boundary limit as redshift increases ($z \gg 0$):

$$\lim_{a_0 \to 0} r_t = \lim_{a_0 \to 0} \sqrt{\frac{GM_{bar}}{a_0}} \to \infty \tag{181}$$

Because the transition boundary diverges to infinity at high redshift, early galaxies never intersect the yield limit. The metric remains entirely within the shear state (Zone 1). High-redshift galaxies exhibit Newtonian rotation profiles without flat velocity tails.

Zone 3: *The Outer Shear Layer (The Terminal Boundary).* The 2D plug flow does not extend infinitely. While the inner boundary (r_t) couples acceleration, the outer boundary couples velocity. The background universe undergoes 3D volumetric expansion, where outward radial velocity increases with distance ($v_{exp} = H_0 r$). At the terminal Jacobian Capture Radius (R_{max}), the outward expansion velocity of the universe equals the constant rotational velocity of the galactic plug flow:

$$H_0 R_{max} = v_{rot} \implies R_{max} = \frac{v_{rot}}{H_0} \tag{182}$$

At this radial limit, the kinematic slip of the galaxy matches the volumetric divergence of the cosmic background. The 2D planar tension of the trapped plug flow shears against the isotropic tension of the surrounding void. The metric decouples across this phase-locked boundary, releasing the rotational strain back into the quiescent 3D expansion of the intergalactic medium.

Within the plug flow annulus ($r \in [r_t, R_{max}]$), imposing constitutive acceleration continuity between the 2D logarithmic stress gradient ($a_{2D}(r_t) = \frac{c^2 k}{2P_c r_t} = \frac{k}{\rho_\tau r_t}$) and the 3D Newtonian yield acceleration ($a_N(r_t) = \frac{GM_{bar}}{r_t^2} = a_0(z)$) across the transition boundary $r_t = \sqrt{GM_{bar}/a_0(z)}$ (Equation 180) fixes the 2D shear amplitude $k = \rho_\tau a_0(z) r_t = \frac{2P_c}{c^2}\sqrt{a_0(z) GM_{bar}}$ and determines the plateau velocity:

$$\begin{aligned} v_{rot}^2 &= \frac{c^2 k}{2P_c} = a_0(z) r_t = a_0(z)\sqrt{\frac{GM_{bar}}{a_0(z)}} = \sqrt{a_0(z) GM_{bar}} \\ v_{rot}^4 &= a_0(z) GM_{bar} \end{aligned} \tag{183}$$

Under this constitutive boundary-matching model, the Baryonic Tully-Fisher relation emerges from the Bingham plug-flow stress balance on the annulus; a first-principles derivation of the 3D-to-2D phase-transition operator from the microscopic equations of motion, along with the lensing and multi-body entrainment extensions below, remains an open analytical and computational problem.

8. The Macroscopic Resolution (Bypassing the No-Go Theorem):
Substituting the plug-flow plateau relation ($k = 2P_c v_{rot}^2/c^2$, Equation 179) into the 2D logarithmic shear profile (Equation 175) for a galaxy exhibiting a flat rotational velocity of $v_{rot} \approx 200$ km/s yields the radius-dependent dimensionless pressure ratio:

$$\frac{P_{shear}^{2D}(r)}{P_c} = 2\frac{v_{rot}^2}{c^2}\ln\left(\frac{R_{max}}{r}\right) \approx 2\left(\frac{2\times 10^5 \text{ m/s}}{3\times 10^8 \text{ m/s}}\right)^2 \ln\left(\frac{R_{max}}{r}\right) \approx 8.9\times 10^{-7}\ln\left(\frac{R_{max}}{r}\right) \tag{184}$$

Across the mid-to-outer galactic plug-flow domain $r/R_{max} \in [0.05, 0.90]$ (corresponding to a logarithmic radial span of $\ln(R_{max}/r) \in [0.11, 3.00]$), this relation registers as a radius-dependent elastic strain ranging from $\approx 9.4\times 10^{-8} \sim 10^{-7}$ in the outer disk to $\approx 2.7\times 10^{-6} \sim 10^{-6}$ in the inner plug-flow zone, before attenuating to zero at the terminal capture boundary ($r \to R_{max}$).

This radius-dependent ratio quantifies the dimensionless mechanical amplitude of the plug flow. While the asymmetric tensor couples transverse vorticity to electromagnetic gauge fields (Section 9) so that a locked mechanical shear of order 10^{-7} to 10^{-6} in the antisymmetric spatial off-diagonals ($\hat{U}_{ij}$) provides a candidate seed strain for galactic magnetogenesis, translating a dimensionless vacuum metric strain into an observable macroscopic magnetic field strength (such as galactic micro-Gauss fields) is a model-dependent estimate that requires specifying the local baryonic plasma density, ionization fraction, and dynamo transduction efficiency; a direct fixed-amplitude micro-Gauss derivation is therefore not claimed here and is deferred to future work.

By routing the rotational shear back into the symmetric sector, the continuum accommodates the kinematic load, generating flat rotation curves as a projection of the 2D plug flow without requiring non-baryonic particulate mass.

9. Tensor Lensing (Stress Birefringence):
Lensing operates as refraction through the asymmetric metric. The effective refractive index (n_{eff}) projects as the negative inverse of the temporal diagonal (S_{00}):

$$n_{eff}(r) = -\frac{1}{S_{00}(r)} = \left(1 - \frac{P_{dyn}^{3D} + P_{shear}^{2D}}{P_c}\right)^{-1} \tag{185}$$

The 3D isotropic Newtonian core (P_{dyn}^{3D}) provides the spherically symmetric index:

$$n_{iso}(r) = \left(1 - \frac{2GM_{bar}}{rc^2}\right)^{-1} \tag{186}$$

The 2D Bingham plug flow (P_{shear}^{2D}) substitutes the logarithmic shear component into the temporal diagonal, projecting the cylindrical index:

$$n_{cyl}(r) = \left(1 - 2\left(\frac{v_{rot}^2}{c^2}\right)\ln\left(\frac{R_{max}}{r}\right)\right)^{-1} \tag{187}$$

The total deflection angle ($\Delta\theta_{tot}$) evaluates as the line integral of the transverse gradient of the super-positioned refractive index:

$$\Delta\theta_{tot}(I) \propto \int \left(\nabla_r n_{iso} + \sin(I)\nabla_r n_{cyl}\right) dl \tag{188}$$

Because the P_{shear}^{2D} tension operates within the 2D azimuthal plane (r, ϕ), the transverse gradient ($\nabla_\perp n_{cyl}$) projects anisotropically via the inclination angle (I). Edge-on geometries ($I = 90°$) maximize the deflection, whereas face-on geometries ($I = 0°$) nullify the 2D gradient, eliminating the anomalous shear signature.

Furthermore, because the logarithmic shear pressure terminates at the Jacobian Capture Radius ($R_{max} = v_{rot}/H_0$), the 2D component of the temporal diagonal returns to the baseline ($P_{shear}^{2D}(R_{max}) = 0$). The optical strain exhibits a geometric discontinuity at R_{max}, formulating a terminal shear ring where anomalous deflection drops to zero.

10. Isotropic Dispersion & Kinematic Entrainment (Dwarf Morphologies):
While macroscopic angular momentum ($\mathbf{J} \neq 0$) polarizes the metric yield into a 2D cylindrical shear flow, dispersion-supported systems (such as dwarf spheroidal galaxies) lack coherent rotational shear ($\langle A_{ij} \rangle = 0$). These systems, however, universally exist as satellites submerged within the macroscopic Bingham plug flow (Zone 2) of a larger host galaxy.

When high-velocity dispersion stars overshoot the baryonic boundary of the dwarf galaxy, they intercept the host's non-slip metric. If the stellar trajectory lacks the acceleration to shear the surrounding continuum ($a < a_0$), the stars become entrained within the host's bulk flow. Because the satellite is translating through this identical bulk flow, the escaping stars drift longitudinally to form extended structures (stellar streams) rather than escaping to spatial infinity. Excess velocity dispersion observed in dwarf satellites is thus an artifact of stars overshooting the local baryonic pocket and transitioning into the host's entrained macroscopic flow. The full orbital calculus of this multi-body entrainment is deferred to future computational modeling.

12. The Non-Linear Affine Connection (Einstein-Cartan Isomorphism)

The asymmetric strain tensor ($\hat{U}_{\mu\nu} = S_{\mu\nu} + A_{\mu\nu}$) requires an affine connection ($\Gamma^\lambda_{\mu\nu}$) that propagates both longitudinal volumetric deformation and transverse rotational shear across the continuum.

1. Connection Decomposition:
By lower-index symmetry, any affine connection decomposes into its symmetric part $\Gamma^\lambda_{(\mu\nu)}$ and the Cartan torsion tensor $T^\lambda_{\mu\nu} \equiv \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu}$ as $\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{(\mu\nu)} + \frac{1}{2}T^\lambda_{\mu\nu}$. In the Einstein-Cartan formulation, separating the connection into a symmetric Levi-Civita component ($\tilde{\Gamma}^\lambda_{\mu\nu}$) and a contortion tensor ($K^\lambda_{\mu\nu}$) yields:

$$\Gamma^\lambda_{\mu\nu} = \tilde{\Gamma}^\lambda_{\mu\nu} + K^\lambda_{\mu\nu} \tag{189}$$

When the contortion sourced by the axial Clebsch vorticity 2-form is completely antisymmetric in its lower indices ($K^\lambda_{(\mu\nu)} = 0$, so $K^\lambda_{\mu\nu} = \frac{1}{2}T^\lambda_{\mu\nu}$), the symmetrized connection coincides with the Levi-Civita connection ($\Gamma^\lambda_{(\mu\nu)} = \tilde{\Gamma}^\lambda_{\mu\nu}$), and vanishing torsion ($T^\lambda_{\mu\nu} = 0$) reduces $\Gamma^\lambda_{\mu\nu}$ to $\tilde{\Gamma}^\lambda_{\mu\nu}$.

2. The Symmetric Connection (Gravitational Acceleration):
On the subcritical domain where the symmetric sector ($S_{\mu\nu}$) is nondegenerate with inverse $S^{\lambda\sigma}$, the Levi-Civita connection is given by the Christoffel relation $\tilde{\Gamma}^\lambda_{\mu\nu} = \frac{1}{2}S^{\lambda\sigma}(\partial_\mu S_{\nu\sigma} + \partial_\nu S_{\mu\sigma} - \partial_\sigma S_{\mu\nu})$. In a stationary weak field ($\partial_0 S_{0i} = 0$, $S^{ij} \approx \eta^{ij}$), expanding the temporal component ($\mu = \nu = 0, \lambda = i$) relative to the static reserve ($S_{00} = -P_{static}/P_c$) produces the restorative acceleration driven by the static pressure gradient:

$$\tilde{\Gamma}^i_{00} \approx -\frac{1}{2}\eta^{ij}\partial_j S_{00} = +\frac{1}{\rho_\tau c^2}\nabla^i P_{static} \tag{190}$$

The geodesic equation dictates coordinate acceleration ($a^i \approx -c^2\tilde{\Gamma}^i_{00}$). Because the static gradient opposes the dynamic pressure gradient ($\nabla P_{static} = -\nabla P_{dyn}$), this symmetric operator governs the longitudinal transport of the coordinate boundary ($\mathbf{a}_{grav} = -\frac{1}{\rho_\tau}\nabla P_{static} = +\frac{1}{\rho_\tau}\nabla P_{dyn}$). Since the dynamic gradient points inward toward the defect, this replicates the attractive irrotational curvature of General Relativity.

3. Spacetime Torsion:
In Einstein-Cartan theory, spacetime torsion ($T^\lambda_{\mu\nu}$) and kinematic vorticity ($\Omega_{\mu\nu}$) are treated as distinct fields. However, if the medium *is* the spacetime grid, a kinematic vortex in the medium is physically identical to a geometric twist in the coordinate frame.

Following the established differential geometry of topological defects in continuous media [31, 29, 27], the spacetime torsion tensor is isomorphic to the physical dislocation density of the medium. In this standard defect-mapping, dislocation density is defined as the curl of the plastic distortion field. By identifying the asymmetric tensor's antisymmetric strain ($A_{\mu\nu}$) as the plastic distortion, the contortion tensor ($K^\lambda_{\mu\nu}$) is determined. Spacetime torsion is therefore derived as the covariant exterior derivative of the antisymmetric strain ($T^\lambda_{\mu\nu} = \nabla_{[\lambda}A_{\mu\nu]}$).

4. Mapping Clebsch Vorticity to Torsion:
Because the grid is the medium, geometric torsion is sourced by localized rotation. Multiplying the continuum torsion by the macroscopic tension limit (c^4/G) translates the geometric strain gradient into angular momentum density ($s^\lambda_{\mu\nu}$):

$$s^\lambda_{\mu\nu} = \frac{c^4}{8\pi G}T^\lambda_{\mu\nu} = \frac{c^4}{8\pi G}\alpha t_p \nabla^\lambda \Omega_{\mu\nu} \tag{191}$$

The spatial curl of the kinematic velocity field ($\Omega_{\mu\nu}$) induces the macroscopic torsion of the coordinate frame, unifying the kinematic defect with the geometric connection.

5. The Macroscopic Torsion Boundary (Decay Limit):
Because Clebsch vorticity ($\Omega_{\mu\nu} = [d\lambda_\Omega \wedge d\beta_\Omega]_{\mu\nu}$) is confined by the cavitation boundary ($r_c = \hbar/mc$), the spacetime torsion undergoes exponential spatial decay outside the defect core ($T^\lambda_{\mu\nu}(r) = T^\lambda_{0,\mu\nu} e^{-r/r_c}$). At macroscopic distances ($r \gg r_c$), the antisymmetric torsion and contortion components attenuate to zero ($T^\lambda_{\mu\nu} \to 0 \implies K^\lambda_{\mu\nu} \to 0$), and the full affine connection reduces to the symmetric Levi-Civita connection ($\tilde{\Gamma}^\lambda_{\mu\nu}$):

$$r \gg r_c \implies T^\lambda_{\mu\nu}(r) = T^\lambda_{0,\mu\nu} e^{-r/r_c} \to 0 \implies \Gamma^\lambda_{\mu\nu} = \tilde{\Gamma}^\lambda_{\mu\nu} + K^\lambda_{\mu\nu} \to \tilde{\Gamma}^\lambda_{\mu\nu} \tag{192}$$

This macroscopic decay validates the torsion-free Levi-Civita and Minkowski divergence limits utilized in Sections 4 and 5.

6. The Riemann Curvature Expansion:
Applying the asymmetric connection to the Riemann curvature tensor expands the derivative operator:

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma} \tag{193}$$

Substituting the decomposed connection ($\tilde{\Gamma} + K$) partitions the Riemann tensor into a symmetric curvature ($\tilde{R}$) and a torsion-dependent spin. Normalizing the trace contraction with the macroscopic tension limit ($c^4/8\pi G$) determines the non-linear Lagrangian density:

$$\mathcal{L} = \frac{c^4}{8\pi G}\left(\tilde{R} + K_{\mu\nu\lambda}K^{\mu\lambda\nu} - K_{\mu\nu}{}^{\nu}K^{\mu\lambda}{}_{\lambda}\right) \tag{194}$$

This expansion recovers the Einstein-Cartan functional limit. The symmetric and antisymmetric traces partition the Lagrangian into the Einstein-Hilbert baseline ($\mathcal{L}_{EH}$) and the Spin-Spin contact interaction ($\mathcal{L}_{spin}$):

$$\mathcal{L} = \underbrace{\frac{c^4}{8\pi G}\tilde{R}}_{\mathcal{L}_{EH}} + \underbrace{\frac{c^4}{8\pi G}\left(K_{\mu\nu\lambda}K^{\mu\lambda\nu} - K_{\mu\nu}{}^{\nu}K^{\mu\lambda}{}_{\lambda}\right)}_{\mathcal{L}_{spin}(A_{\mu\nu})} \tag{195}$$

Macroscopic gravity (symmetric Riemannian curvature) and quantum spin (antisymmetric Clebsch torsion) propagate as coupled, orthogonal deformations.

13. Transverse Wave Mechanics, Chiral Induction, and Plasma Attenuation

The structural capacity of the metric is derived from the spatial Cauchy stress tensor ($\hat{T}^{ij}$). Applying the field equation against the symmetric volumetric strain (S^{ij}) produces the coordinate stress:

$$\hat{T}^{ij} = P_c\Big(S^{ij} - \delta^{ij}\Big) = P_c\left(\frac{1}{\alpha_s^2} - 1\right)\delta^{ij} \tag{196}$$

Taking the spatial divergence (∇_j) extracts the volumetric force density ($f^i = \nabla_j \hat{T}^{ij}$). Because the scalar admittance is bounded by dynamic pressure ($\alpha_s^2 = 1 - P_{dyn}/P_c$), the continuous restorative force of the metric is governed by the inverse scalar capacity. This establishes a non-linear geometric manifold:

$$f^i = P_c \nabla^i \left(\frac{1}{1 - \frac{P_{dyn}}{P_c}}\right) \tag{197}$$

13.1 Transverse Shear and Strain Propagation (Light)

The continuous metric velocity field partitions into an irrotational compressive mode (dh) and a solenoidal transverse mode ($\mathbf{v}_\perp$).

While classical electrodynamics truncates the compressive scalar wave via gauge fixing, the asymmetric tensor preserves it within the symmetric spatial diagonals (S_{ii}). Because the unperturbed vacuum limit ($P_{dyn} \ll P_c$) dictates that longitudinal dynamic pressure approaches zero, the continuous wave equation is derived from the isolated antisymmetric sector ($A_{\mu\nu}$) of the metric tensor.

The conservation of momentum across the spatial coordinates targets the 4D spacetime divergence of the antisymmetric matrix ($\partial_\mu A^{\mu i} = 0$). Expanding the temporal and spatial components extracts the transverse momentum equivalent:

$$\frac{1}{c^2}\frac{\partial \mathbf{v}_\perp}{\partial t} = t_p(\nabla \cdot \mathbf{\Omega}) \tag{198}$$

To define the temporal evolution of the rotational phase, the tensor must satisfy the Bianchi identity ($\partial_{[\mu} A_{\nu\lambda]} = 0$). Expanding the cyclic permutation of the antisymmetric components results in the induction equation, where the temporal derivative of the 2-form equates to the exterior derivative of the transverse 1-form:

$$t_p \frac{\partial \mathbf{\Omega}}{\partial t} = -d\mathbf{v}_\perp \tag{199}$$

Taking the time derivative of the momentum equation and substituting the rotational induction constraint produces the second-order wave equation:

$$\frac{1}{c^2}\frac{\partial^2 \mathbf{v}_\perp}{\partial t^2} = t_p \nabla \cdot \left(\frac{\partial \mathbf{\Omega}}{\partial t}\right) = -\nabla \cdot (d\mathbf{v}_\perp) \tag{200}$$

Because transverse waves are divergence-free ($\nabla \cdot \mathbf{v}_\perp = 0$), the divergence of the exterior derivative ($\nabla \cdot d\mathbf{v}_\perp = \nabla(\nabla \cdot \mathbf{v}_\perp) - \nabla^2 \mathbf{v}_\perp$) simplifies to the Laplacian limit:

$$\frac{\partial^2 \mathbf{v}_\perp}{\partial t^2} = c^2 \nabla^2 \mathbf{v}_\perp \tag{201}$$

The phase velocity resolves to the acoustic limit of the continuum (c). Light manifests as a linearly polarized, coupled transverse-rotational strain propagating through the baseline metric.

13.2 Metric Depletion and Chiral Induction

When transverse strain propagates into a magnetized plasma, the ambient scalar admittance depletes ($\alpha_s^2 < 1 \implies P_{dyn} = [1 - \alpha_s^2]P_c > 0$). The wave encounters an elevated dynamic pressure and a macroscopic background Clebsch vorticity ($\mathbf{\Omega}_0 = d\lambda_{\Omega,0} \wedge d\beta_{\Omega,0}$) generated by the collective motion of the embedded topological defects.

The kinematic evolution of the propagating wave is governed by the non-inertial covariance transformations derived in Section 10.3. For a wave propagating through a macroscopic background vorticity, the coordinate frame rotates relative to the transverse strain. By Equation 160, this non-inertial rotation couples the spatial vorticity 2-form to the temporal derivative of the transverse velocity vector:

$$\frac{1}{c^2}\frac{\partial^2 \mathbf{v}_\perp}{\partial t^2} - \nabla^2 \mathbf{v}_\perp \propto \frac{1}{c^2}\left(\mathbf{\Omega}_0 \cdot \frac{\partial \mathbf{v}_\perp}{\partial t}\right) \tag{202}$$

Because the inner-product contraction of a 1-form ($\partial_t \mathbf{v}_\perp$) against an antisymmetric 2-form ($\Omega_{0,ij} = -\Omega_{0,ji}$) yields an orthogonal vector, the chiral induction acceleration is strictly orthogonal to the temporal velocity derivative:

$$\frac{\partial \mathbf{v}_\perp}{\partial t} \cdot \left(\mathbf{\Omega}_0 \cdot \frac{\partial \mathbf{v}_\perp}{\partial t}\right) = \Omega_{0,ij} \frac{\partial v_\perp^i}{\partial t}\frac{\partial v_\perp^j}{\partial t} = 0 \tag{203}$$

Consequently, the background vorticity rotates the transverse polarization plane (chiral birefringence / Faraday rotation) without injecting or dissipating scalar wave energy.

13.3 Macroscopic Measurement Boundaries (Deterministic Eigenstates)

A quantum measurement apparatus constitutes an anisotropic boundary condition ($P_{dyn} \to P_c$), imposing an impedance gradient across the metric.

A defect ($r \to r_c, \alpha_s \to 0$) propagates with an extended transverse strain field ($A_{\mu\nu}$). Intercepting the boundary at an incident angle (θ) relative to the axis of minimum resistance ($\mathbf{n}$), the transverse slip vector ($\mathbf{v}_\perp$) undergoes orthogonal decomposition. The boundary projects the amplitude onto the allowed axis:

$$\mathbf{v}_\perp^{trans} = (\mathbf{v}_\perp \cdot \mathbf{n})\mathbf{n} = \mathbf{v}_\perp \cos\theta \tag{204}$$

The observable measurement corresponds to dynamic pressure (P_{dyn}). Substituting the restricted velocity into Equation 5 calculates the transmitted energy limit:

$$\begin{aligned} P_{dyn}^{trans} &= \frac{1}{2}\rho_\tau |\mathbf{v}_\perp \cos\theta|^2 \\ P_{dyn}^{trans} &= \left(\frac{1}{2}\rho_\tau v_\perp^2\right)\cos^2\theta \\ P_{dyn}^{trans} &= P_{dyn}^{(0)}\cos^2\theta \end{aligned} \tag{205}$$

This tensor contraction recovers Malus's Law. Furthermore, the anisotropic projection of $\mathbf{v}_\perp$ generates a spatial gradient across the defect's extended envelope. Satisfying the Bianchi identity ($\partial_{[\mu}A_{\nu\lambda]} = 0$), the restricted slip vector substitutes into the rotational induction equation (Section 13.1), determining the temporal evolution of the core vorticity 2-form ($\mathbf{\Omega}$):

$$t_p \frac{\partial \mathbf{\Omega}}{\partial t} = -d\left(\mathbf{v}_\perp \cos\theta\right) \tag{206}$$

Applying the exterior derivative product rule for a scalar multiplied by a 1-form expands the boundary condition:

$$\begin{aligned} t_p \frac{\partial \mathbf{\Omega}}{\partial t} &= -\left[\cos\theta(d\mathbf{v}_\perp) + d(\cos\theta) \wedge \mathbf{v}_\perp\right] \\ t_p \frac{\partial \mathbf{\Omega}}{\partial t} &= -\cos\theta(d\mathbf{v}_\perp) + \sin\theta(d\theta \wedge \mathbf{v}_\perp) \end{aligned} \tag{207}$$

The expanded equation dictates the transverse shear stress gradient applied to the core. Because the wave propagates transversely, the incident transverse slip ($\mathbf{v}_\perp$) is strictly orthogonal to the phase gradient ($d\theta$). Therefore, their wedge product evaluates to a maximized, non-zero area element ($d\theta \wedge \mathbf{v}_\perp \neq 0$).

Because the boundary is saturated ($P_{dyn} \to P_c$), the defect must reach a steady-state kinematic equilibrium where the temporal acceleration of the transverse slip vanishes ($d\mathbf{v}_\perp \to 0$). Once this primary term stabilizes, fractional angular states ($\sin\theta \neq 0$) generate an unbounded torque that violates the zero-derivative steady-state boundary ($\partial_t \mathbf{\Omega} = 0$). To satisfy this condition against the non-zero wedge product, the remaining angular strain multiplier evaluates to zero:

$$\sin\theta = 0 \implies \theta \in \{0, \pi\} \tag{208}$$

The topology resolves to spatial equilibrium at aligned or anti-aligned orientations. Bounded by the phase limit (c), this discontinuous bifurcation derives quantum eigenstate convergence (wavefunction collapse) from the orthogonal projection limits of the asymmetric tensor.

13.4 Metric Yielding and Attenuation

The spatial attenuation of the transverse slip vector ($v_{\perp,\mu}$) isolates the mechanism of Debye shielding. From Equation 197, the restorative force of the metric scales with the inverse scalar capacity. When a coordinate grid is saturated by background dynamic pressure ($P_{dyn} \to P_c$), the spatial diagonals of the symmetric tensor diverge ($S_{ii} \to \infty$).

The spatial decay of the transverse strain amplitude (A) relative to the radial propagation axis (r) is governed by the metric's stiffness constraint:

$$\nabla^2 A - \left(\frac{1}{\alpha_s^2 L_0^2} \right) A = 0 \tag{209}$$

This second-order differential equation operates as a continuous screened Poisson boundary. Solving this operator extracts the exponential decay envelope of the strain amplitude:

$$A(r) = A_0 \frac{\exp\left(-\frac{r}{\lambda_D}\right)}{r} \tag{210}$$

Where the effective propagation length (λ_D) equates to the inverse of the spatial coordinate expansion:

$$\lambda_D = L_0 \alpha_s = L_0 \sqrt{1 - \frac{P_{dyn}}{P_c}} \tag{211}$$

As dynamic pressure increases, the attenuation length proportionally shrinks ($\lambda_D < L_0$). The transverse wave is not energetically damped; rather, its spatial advection is confined by the diverging coordinate depth. This maps the relativistic length expansion formula directly to the exponential decay envelope of a plasma screening length, deriving macroscopic Debye shielding from continuous geometry rather than particulate Poisson-Boltzmann statistics.

The yield of the metric determines the macroscopic boundary for magnetic reconnection. The Deborah number dictates the rheological state:

$$De = \frac{1}{\alpha_s^2 \alpha t_p \Omega} \tag{212}$$

As dynamic pressure converges on the critical boundary ($P_{dyn} \to P_c$), the scalar admittance collapses ($\alpha_s^2 \to 0$) extracting the glass transition threshold:

$$\lim_{\alpha_s \to 0} De = \infty \tag{213}$$

The spatial stress ($\hat{T}_{ii}$) diverges to infinity, conserving the thermodynamic budget and decoupling the rotational Clebsch phase ($\mathbf{\Omega}$).

The plasma undergoes reconfiguration, lowering the shear stress (P_{shear}), and recovering scalar capacity. This cancels the infinite stress, deriving magnetic reconnection as a mechanical failure of the matrix rather than a fluid anomaly.

14. Continuous Pressure Dynamics (Fundamental Forces)

Macroscopic force interactions operate as geometric projections of a single continuous pressure gradient.

1. The Isochoric Velocity Scaling

The antisymmetric sector ($A_{\mu\nu}$) dictates the trace-free isochoric constraint for the continuous velocity field ($\nabla \cdot \mathbf{v} = 0$). To determine the scaling of this field across a boundary, the divergence operator expands in spherical coordinates (r, θ, ϕ):

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 v_r) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta}(v_\theta \sin\theta) + \frac{1}{r \sin\theta} \frac{\partial v_\phi}{\partial \phi} = 0 \tag{214}$$

For a spherical boundary, the transverse angular gradients cancel ($\partial_\theta = 0$, $\partial_\phi = 0$):

$$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2 v_r) = 0 \tag{215}$$

Multiplying by the squared radius (r^2) and integrating with respect to r:

$$r^2 v_r = C \implies v_r(r) \equiv v(r) = \frac{C}{r^2} \tag{216}$$

Because continuous volumetric flux is conserved, the scalar velocity field ($v(r)$) scales inversely with the boundary area ($A = 4\pi r^2$).

Substituting this velocity into the Bernoulli capacity equation of the temporal diagonal (S_{00}) constructs the dynamic pressure ($P_{dyn} = \frac{1}{2}\rho_\tau v^2$):

$$P_{dyn}(r) = \frac{1}{2}\rho_\tau \left(\frac{C}{r^2}\right)^2 = \frac{\rho_\tau C^2}{2r^4} \tag{217}$$

2. The Geometric Action Boundary (Deriving the Scalar)
To maintain baseline capacity, the background exerts a restorative confinement pressure according to the dynamic deficit ($P(r) = P_{dyn}(r)$).

The geometric Action (S) of a Clebsch twist constitutes an invariant constant (κ). Action becomes the structural energy (E) integrated over the period (T):

$$S = E \cdot T = \kappa \tag{218}$$

The period represents the duration required for the transverse velocity field to complete one closed 2π rotation at the phase limit (c):

$$T = \frac{2\pi r}{c} \tag{219}$$

By integrating the restorative pressure acting across the boundary area to spatial infinity:

$$\begin{aligned} E &= \int_r^\infty F(r')\, dr' = \int_r^\infty P(r') \cdot 4\pi (r')^2\, dr' \\ E &= \int_r^\infty \left(\frac{\rho_\tau C^2}{2(r')^4}\right) 4\pi (r')^2\, dr' = 2\pi\rho_\tau C^2 \int_r^\infty \frac{1}{(r')^2}\, dr' \\ E &= 2\pi\rho_\tau C^2 \left[-\frac{1}{r'}\right]_r^\infty = \frac{2\pi\rho_\tau C^2}{r} \end{aligned} \tag{220}$$

Substituting the period and energy into the Action limit (κ) extracts the integration constant (C):

$$\begin{aligned} \kappa &= \left(\frac{2\pi\rho_\tau C^2}{r}\right)\left(\frac{2\pi r}{c}\right) \\ \kappa &= \frac{4\pi^2\rho_\tau C^2}{c} \implies C^2 = \frac{\kappa c}{4\pi^2\rho_\tau} \end{aligned} \tag{221}$$

The radial dimension cancels, revealing that the integration constant (C) is governed by the invariant action limit (κ) and the baseline density of the vacuum (ρ_τ).

Substituting back into the pressure equation yields the spatial confinement gradient:

$$\begin{aligned} P(r) &= \frac{\rho_\tau}{2r^4}\left(\frac{\kappa c}{4\pi^2\rho_\tau}\right) \\ P(r) &= \frac{\kappa c}{8\pi^2 r^4} \end{aligned} \tag{222}$$

3. The Kinematic Mass Boundary ($\kappa \equiv h$)
Force operates as pressure projected across a bounding cross-section ($F = P \cdot A$). Substituting the area of the spherical boundary ($A = 4\pi r^2$) into the spatial confinement gradient yields:

$$F(r) = \left(\frac{\kappa c}{8\pi^2 r^4}\right)(4\pi r^2) = \frac{\kappa c}{2\pi r^2} \tag{223}$$

In order to determine the value of the action constant (κ), we calculate the mechanical energy (E) required to integrate this restorative force across the characteristic radius:

$$E(r) = \int_r^\infty F(r')\, dr' = \int_r^\infty \frac{\kappa c}{2\pi (r')^2}\, dr' = \frac{\kappa c}{2\pi r} \tag{224}$$

For a boundary to form a stable rest mass (m), this radial energy constructs the kinematic rest energy of the defect (mc^2). Equating this physical limit to the continuous force work isolates the radius (r):

$$\begin{aligned} mc^2 &= \frac{\kappa c}{2\pi r} \\ r &= \frac{\kappa}{2\pi mc} \end{aligned} \tag{225}$$

Because the geometric bounding radius of a defect is defined by the reduced Compton wavelength ($r = \hbar/mc$), equating the two limits reveals the identity of the scalar action constant:

$$\frac{\kappa}{2\pi mc} = \frac{\hbar}{mc} \implies \kappa = 2\pi\hbar = h \tag{226}$$

The invariant action limit (κ) is the unreduced Planck constant (h). Substituting this definition back into the continuum force equation absorbs the reduction factor:

$$F(r) = \frac{hc}{2\pi r^2} \equiv \frac{\hbar c}{r^2} \tag{227}$$

The force scaling of a localized defect inherently differs from the unscaled volumetric expansion of the manifold (Section 4.2) because the reduction of the $8\pi^2$ temporal integration limit by the 4π spatial surface area generates the boundary constant ($\hbar = h/2\pi$).

4. The Interaction Hierarchy (Force Projections)
The boundary interactions map to distinct coordinate radii.

I. Gravity (The Core Boundary):
Applying the spatial gradient at the geometric minimum of the acoustic cavity ($l_p = \sqrt{\hbar G/c^3}$) reproduces the maximum tension of the continuum:

$$F_{grav} = F(l_p) = \frac{\hbar c}{(\sqrt{\hbar G/c^3})^2} = \frac{c^4}{G} \tag{228}$$

II. The Weak Interaction (The Cavitation Yield):
Calculating the continuous pressure across the Compton boundary of the topological W-Boson ($R_W = \hbar/m_W c$) specifies the volumetric rupture limit:

$$F_{weak} = F(R_W) = \frac{\hbar c}{(\hbar/m_W c)^2} = \frac{m_W^2 c^3}{\hbar} \tag{229}$$

III. The Strong Interaction (The Acoustic Confinement):
Projecting the spatial gradient at the viscoelastic relaxation envelope ($L_{crit} = \alpha\hbar/m_e c$) establishes the static boundary for topological phase equilibrium:

$$F_{strong} = F(L_{crit}) = \frac{\hbar c}{(\alpha\hbar/m_e c)^2} = \frac{m_e^2 c^3}{\alpha^2 \hbar} \tag{230}$$

IV. Electromagnetism (The Transverse Leakage):
Applying the spatial gradient at the classical Bohr radius ($a_0 = \hbar/\alpha m_e c$) resolves to the macroscopic electrostatic restoring tension:

$$F_{EM_{base}} = F(a_0) = \frac{\hbar c}{(\hbar/\alpha m_e c)^2} = \frac{\alpha^2 m_e^2 c^3}{\hbar} \tag{231}$$

Because the identical boundary governs the force, the geometric ratio between the macroscopic electromagnetic baseline and the strong boundary is α^4.

5. The Macroscopic Baseline (The Horizon Floor)
Projecting the spatial gradient at the macroscopic scale radius of the universe ($R_{h,0}$) defines the kinematic boundary condition of the cosmic vacuum:

$$F_{cosmic} = F(R_{h,0}) = \frac{\hbar c}{R_{h,0}^2} \equiv \frac{hc}{2\pi R_{h,0}^2} \tag{232}$$

To evaluate the kinematic acceleration floor this tension exerts, the force is divided by the characteristic topological mass of the horizon boundary derived via the unreduced action limit ($m_h = h/R_{h,0}c$):

$$\begin{aligned} a_0 &= \frac{F(R_{h,0})}{m_h} \\ a_0 &= \left(\frac{hc}{2\pi R_{h,0}^2}\right)\left(\frac{R_{h,0}c}{h}\right) = \frac{c^2}{2\pi R_{h,0}} \end{aligned} \tag{233}$$

The action scalar (h) cancels, producing the kinematic acceleration threshold (a_0) dictating galactic rotation curves.

6. Macroscopic Combinatorial Aggregates (Newtonian Gravity)
The unified spatial gradient specifies the continuous state of a singular defect. For a macroscopic object, the physical mass partitions as a combinatorial superposition of boundaries.

The baseline mass required to scale classical aggregate interactions is the Planck mass:

$$M_P = \sqrt{\frac{\hbar c}{G}} \tag{234}$$

The total number of interacting boundaries (N) within a mass boundary (M) equates to the ratio of the aggregate mass to this boundary ($N = M/M_P$). The macroscopic force (F_{macro}) between two aggregate masses (M_1 and M_2) separated by a coordinate radius (r) acts as the combinatorial product of their individual boundaries ($N_1 \cdot N_2$) applied to the unified continuum force:

$$\begin{aligned} F_{macro} &= (N_1 \cdot N_2) \cdot F(r) \\ F_{macro} &= \left(\frac{M_1}{M_P}\right)\left(\frac{M_2}{M_P}\right)\left(\frac{\hbar c}{r^2}\right) \end{aligned} \tag{235}$$

Substituting the squared mass boundary ($M_P^2 = \hbar c/G$) recovers the classical format:

$$\begin{aligned} F_{macro} &= \frac{M_1 M_2}{\left(\frac{\hbar c}{G}\right)}\left(\frac{\hbar c}{r^2}\right) \\ F_{macro} &= M_1 M_2 \left(\frac{G}{\hbar c}\right)\left(\frac{\hbar c}{r^2}\right) = \frac{G M_1 M_2}{r^2} \end{aligned} \tag{236}$$

Universal gravitation scales as the combinatorial accumulation of tension.

15. Macroscopic Lensing (The Optical Pressure Gradient)

The accelerated expansion signal of the universe operates as a longitudinal optical projection of the continuous pressure gradient.

1. Cosmological Redshift as a Metric Capacity Ratio

The nested scalar admittance (α_s) established in Section 2 scales to the macroscopic cosmological domain, providing a geometric derivation for redshift ($1 + z$) independent of metric spatial expansion.

The spatial line element of the symmetric metric dictates how the physical rest wavelength (λ_0) of a photon scales against the background coordinate geometry (dx):

$$\lambda_0^2 = S_{ii} dx^2 = \left(\frac{1}{\alpha_s^2} \right) dx^2 \tag{237}$$

Because the rest wavelength (λ_0) remains invariant, the coordinate wavelength ($\lambda \equiv dx$) scales proportionally to scalar admittance ($dx = \alpha_s \lambda_0$).

When a wave propagates from a saturated emitter coordinate ($\alpha_{emitter}$) into a less perturbed observer coordinate ($\alpha_{observer}$), the wavelength elongates. Redshift determines the geometric ratio of the respective capacities:

$$1 + z = \frac{\lambda_{observed}}{\lambda_{emitted}} = \frac{\alpha_{observer}}{\alpha_{emitter}} \tag{238}$$

Cosmological redshift thus maps to the relative capacity of the vacuum.

2. The Tensor Refractive Index

For a null wave, the coordinate phase velocity (v_c) must satisfy the null geodesic constraint of the asymmetric metric ($ds^2 = S_{00} c^2 dt^2 + S_{ii} dr^2 = 0$). Isolating the phase velocity incorporates both the temporal and spatial tensor capacities:

$$v_c(r) = \frac{dr}{dt} = c \sqrt{\frac{-S_{00}(r)}{S_{ii}(r)}} \tag{239}$$

Because the symmetric diagonals map to inverse thermodynamic states ($S_{00} = -\alpha_s^2$ and $S_{ii} = 1/\alpha_s^2$), substituting the Bernoulli limit ($\alpha_s^2 = 1 - P_{dyn}(r)/P_c$) results in the coordinate velocity:

$$v_c(r) = c \sqrt{\frac{\alpha_s^2}{1/\alpha_s^2}} = c \alpha_s^2 = c \left(1 - \frac{P_{dyn}(r)}{P_c} \right) \tag{240}$$

The refractive index ($n(r)$) scales the ratio of the phase limit (c) to the coordinate velocity:

$$n(r) = \frac{c}{v_c(r)} = \left(1 - \frac{P_{dyn}(r)}{P_c} \right)^{-1} \tag{241}$$

Whenever the dynamic pressure (P_{dyn}) is non-zero, the refractive index exceeds unity, resulting in a continuous temporal retardation of the propagating wavefront.

3. The Macroscopic Pressure Substitution

As derived from the continuity constraints of an expanding manifold with characteristic scale $R_h = ct$, the dynamic pressure approaches P_c asymptotically at large optical depth ($r \gg ct$, with $P_{dyn}(ct) = P_c(1 - e^{-1})^2$). While the boundary constraint governing the continuum limit remains an open derivation, inverting the observable magnitude constraints of the Pantheon+ dataset [9] to this pressure profile produces the following curve:

$$P_{dyn}(r) = P_c \left(1 - e^{-r/ct} \right)^2 \tag{242}$$

Substituting this back into the refractive index equation yields the optical geometry of the background metric:

$$n(r) = \left(1 - \frac{P_c(1 - e^{-r/ct})^2}{P_c}\right)^{-1}$$
$$n(r) = \left(1 - (1 - e^{-r/ct})^2\right)^{-1} \tag{243}$$

Expanding the binomial term $(1 - (1 - 2e^{-r/ct} + e^{-2r/ct}))$ simplifies the operator:

$$n(r) = \left(2e^{-r/ct} - e^{-2r/ct}\right)^{-1}$$
$$n(r) = \frac{1}{e^{-r/ct}(2 - e^{-r/ct})} \tag{244}$$

This function specifies the optical density of the expanding coordinate field.

4. The Longitudinal Path Integral (Shapiro Delay)
According to Fermat's Principle, the apparent optical distance (r_{opt}) a photon travels resolves to the integration of the refractive index along the coordinate path (r_{true}):

$$r_{opt} = \int_0^{r_{true}} n(r')dr' \tag{245}$$

Because the medium maintains $n(r) > 1$, the apparent optical path accumulates continuous delay, guaranteeing $r_{opt} > r_{true}$ (for instance, under the redshift parametrization $v_0/c = z/(1+z)$ with $dr' = R_h dz'/(1+z')$, partial-fraction integration of $n(z') = (1+z')^2/(2z'+1)$ evaluates in closed form via the natural logarithm as $r_{opt}(z) = R_h[z/2 + \frac{1}{4}\ln(2z+1)]$).

5. The Magnitude Anomaly (Optical Expansion)
When assuming an unperturbed background metric where $n(r) = 1$, the optical path gets equated with the geometric coordinate ($r_{opt} = r_{true}$). Because the underlying metric is an expanding volumetric manifold ($R_h = c \cdot t$), standard candles obey the $(1+z)^{-4}$ Tolman surface brightness scaling required by macroscopic energy conservation [52, 38].

However, the observable distance modulus (μ) calculates the luminosity attenuation across the physical boundary via the base-10 Pogson logarithm ($\log_{10} x = \ln x / \ln 10$). The differential magnitude ($\Delta\mu$) between the standard linear expectation (μ_{linear}) and the refractive metric observation (μ_{obs}) determines optical divergence:

$$\Delta\mu = \mu_{obs} - \mu_{linear}$$
$$\Delta\mu = [5\log_{10}(r_{opt}(1+z)) + 25] - [5\log_{10}(r_{true}(1+z)) + 25]$$
$$\Delta\mu = 5\log_{10}\left(\frac{r_{opt}}{r_{true}}\right) = \frac{5}{\ln 10}\ln\left(\frac{r_{opt}}{r_{true}}\right) \tag{246}$$

16. Discussion: Predictive Corollaries & Future Work

Applying the foundational limits established in the preceding sections yields the following secondary corollaries.

Scope Limitation: The following represent hypothesized physical equivalencies between the Asymmetric Metric-Tensor and advanced quantum/relativistic phenomena. While the macroscopic boundaries and geometric bases are derived from the tensor in the preceding sections, the complete mappings required to finalize these proofs fall outside the scope of this text. These entries are presented not as closed derivations, but as a theoretical roadmap for future work.

16.1 Symmetric Diagonals (Bulk Strain & Gravitation)

Formalism	Continuum Tensor Mapping
Lemaître-Tolman-Bondi	Models a radially expanding, pressure-free metric [35, 53, 7]. Substituting the macroscopic velocity field ($v_0(r)$) into the tensor diagonals dictates expanding spatial compliance independent of FLRW homogeneity constraints.
Baryon Acoustic Oscillations	Originates via the spatial derivative of the time-time diagonal ($\hat{U}_{00}$). Depth expands proportionally to the scalar admittance ($dx = \alpha_s ds$). Because the 3D spatial determinant scales as $\det(S_{ij}) = \alpha_s^{-6}$, the volume element scales as its square root ($\sqrt{\det(S_{ij})} = \alpha_s^{-3}$), mapping to the inverse function of redshift ($V \propto (1+z)^3$) and linking boundary acceleration with early cosmological scaling.
The Holographic Bound	Derived from the boundary condition of the symmetric trace ($S_{ii} \to \infty$) [3]. Because the core limit ($r = r_c$) acts as incompressible ($\nabla \cdot \mathbf{u} = 0$), volumetric displacement accumulates across the 2D Dirichlet boundary.
The Casimir Effect	Truncation of the symmetric trace's energy density field ($\hat{T}_{00}$) [12]. Parallel spatial boundaries cause the continuous integration of background pressure ($\int_0^d \hat{T}_{00}\, dz$) to evaluate as a discrete sum, providing the pathway to recover the $1/d^4$ boundary suction.
Wakefield Cavitation	Derived from the spatial gradient of the antisymmetric trace ($\nabla(A_{\mu\nu}A^{\mu\nu})$). An intense transverse shear wave (e.g., a laser pulse) generates a shear stress gradient ($-\nabla P_{shear}$). To conserve the total thermodynamic budget (P_c), this gradient repels the symmetric dynamic pressure (P_{dyn}), structuring the Ponderomotive Force as mass-repulsion yielding metric cavitation.
Alpha Decay	The rupture of a boundary ($P_{dyn} \geq P_c$). Rather than stochastic teleportation, quantum tunneling is governed by the WKB spatial path integral of the temporal strain diagonal ($T \approx \exp(-\int \sqrt{-\hat{U}_{00}}\, dr)$) across the mass gradient. This constructs a phase deficit, providing the basis to derive transmission probability from the compliance of the confining metric well.
Optimal Transport (Wasserstein Metric)	The Benamou-Brenier formulation of the W_2 Wasserstein distance minimizes the integral of kinematic energy ($\int \rho v^2\, dxdt$). Because the continuum tensor solves the integrand as dynamic pressure (P_{dyn}), the geodesic equation is an Optimal Transport plan. Gravitation functions as the minimization of the thermodynamic strain required to displace the metric.

Formalism (Continued)	**Continuum Tensor Mapping**
Michelson-Morley Null Result	Volumetric constraint. Because dynamic pressure collapses scalar admittance (α_s^2), the physical coordinate path dynamically compresses by the fraction ($L_{\parallel} = L_0\alpha_s$). This enforces isotropic acoustic phase invariance ($t_{\parallel} = t_{\perp}$), undetectable to internal interferometry. This affirms the perspective of Lorentz invariance, where relativistic contraction originates from physical continuum forces rather than kinematic postulates [5, 10, 54].
Shapiro Time Delay	Setting the spacetime interval to zero ($ds^2 = 0$) for a radial transit path demonstrates that the apparent phase velocity of light drops proportionally to the squared scalar admittance ($dr/dt = c\alpha_s^2$). Integrating this differential transit time over the path recovers the Shapiro delay as continuous optical refraction through the gradient.
Newton's First Law (Inertia)	Under uniform translation, the dynamic pressure field (P_{dyn}) is symmetric across the longitudinal axis. By the Divergence Theorem, integrating the spatial Cauchy stress gradient over the volume ($\int \nabla P_{dyn}\, dV$) equates to zero. This trace equilibrium recovers d'Alembert's Paradox, conserving momentum.
Newton's Third Law (Action/Reaction)	Bernoulli conservation dictates that an applied kinematic action (∇P_{dyn}) generates an equal and opposite static pressure gradient ($-\nabla P_{static}$). Applying the equilibrium limit ($\partial_t \hat{T}_A^{0i} = -\partial_t \hat{T}_B^{0i}$) transfers this gradient balance into kinematic momentum exchange across the boundary.

16.2 Antisymmetric Off-Diagonals (Transverse Shear & Vorticity)

Formalism	**Continuum Tensor Mapping**
Gravitoelectromagnetism	The transverse slip ($v_{\perp}$) maps to the Gravitoelectric field. Metric vorticity ($\mathbf{\Omega}$) functions as the Gravitomagnetic field. This structures frame-dragging as metric vorticity.

Formalism (Continued)	**Continuum Tensor Mapping**
Dirac Equation & Clifford Algebra	Because the asymmetric tensor operates as a Clifford geometric product ($S + A \equiv u \cdot v + u \wedge v$), spacetime inherently functions as a multivector. The symmetric tensor ($S_{\mu\nu}$) applies the inner product, supplying a moving frame via the tetrad formalism (e_a^μ) to produce the invariant Minkowski signature (η_{ab}) required to span the Clifford algebra ($C\ell_{1,3}$) and construct the anticommuting Dirac vector matrices ($\{\gamma^a, \gamma^b\} = 2\eta^{ab}$). Correspondingly, the 6 antisymmetric off-diagonals ($A_{\mu\nu}$) apply the wedge product, mapping exactly to the 6 bivector spin generators ($\sigma_{\mu\nu}$). At the quantum boundary, a 360° (2π) spatial rotation returns the symmetric volumetric identity to its original state. However, because the defect remains physically tethered to the background via the continuous Clebsch rotational phase ($d\lambda_\Omega \wedge d\beta_\Omega$), this 2π rotation imparts a residual chiral twist into the antisymmetric off-diagonals. A full 720° (4π) rotation "untangles" the geometric tethers and resets the metric state. This mechanical divergence between the symmetric inner product and the antisymmetric wedge product generates the Spin$(1,3)$ double-cover, producing the 4-component Dirac bi-spinor (ψ).
Mott Scattering & Helicity	The symmetric diagonals generate the $1/r$ scalar GRIN lens, providing the basis for macroscopic $1/Q^4$ scattering cross-sections. Furthermore, transverse waves interacting with the antisymmetric vorticity matrix (A_{ij}) structurally conserve kinematic helicity, establishing the geometric baseline to evaluate the scattering amplitude via the tensor contraction ($\frac{d\sigma}{d\Omega} \propto |\langle \mathbf{v}_f | \hat{U}_{\mu\nu} | \mathbf{v}_i \rangle|^2$), capturing the $\cos^2(\theta/2)$ spin-dependent Mott scattering limit.
The Poynting Vector (Convective Energy Flux)	The convective acceleration expands via the vector identity $(\mathbf{v} \cdot \nabla)\mathbf{v} = \frac{1}{2}\nabla(v^2) - \mathbf{v} \cdot d\mathbf{v}$. By Bernoulli conservation, transverse slip ($\mathbf{v}_\perp$) generates a proportional static pressure deficit ($P_{static} = P_c - \frac{1}{2}\rho_\tau v_\perp^2$) causing the surrounding metric (P_c) to generate a restorative spatial gradient ($-\nabla P_{static}$). Transport is controlled by the tensor contraction of the transverse slip 1-form against the vorticity 2-form ($\mathbf{v}_\perp \cdot \mathbf{\Omega}$), defining the Lamb vector. This breaks the spatial symmetry of the pressure gradient, driving the directional transport of kinematic strain. The inner product of the space-time off-diagonals (electric slip) and space-space off-diagonals (magnetic vorticity) maps directly to the Poynting vector ($S^i \propto A^{0j} A_j{}^i$), deriving electromagnetic energy flux as convective metric momentum.

Formalism (Continued)	**Continuum Tensor Mapping**
Quantum Entanglement & Decoherence	Bell's Theorem relies on combining expectation values from mutually exclusive experimental runs onto a 1D scalar ledger to formulate its inequalities ($S = \|E(a,b) - E(a,b') + \dots\|$), mandating a universal Joint Probability Distribution (JPD). This is a geometric category error. An entangled emission does not create independent point-particles governed by a static scalar variable (λ); it establishes a continuous loop. Polarizers establish the geometric constraints for this unified state. Dynamically altering a detector boundary ($b \to b'$) redefines the integration limits of the governing partial differential equations. The integration probability space (Λ) is therefore contextual to the boundaries ($\Lambda_{ab} \neq \Lambda_{ab'}$), invalidating the assumption of statistical independence. Furthermore, this dynamic alteration rotates the relative coordinate basis of the shared space. By the laws of tensor covariance, scalar projections evaluated in rotated reference frames cannot be added or subtracted without applying the requisite transformation matrices. Because the boundary forces the rotational eigenstates into the complex plane (Section 9.6), the correlated measurement evaluates not as a classical statistical integral, but as the sesquilinear inner product of these complex principal eigenvectors against the boundary normals ($E(a,b) \propto \mathbf{v}_A^\dagger \hat{U}_{\mu\nu} \mathbf{v}_B$). This covariant contraction solves the interference cross-terms, deriving correlated measurement. Finally, because the asymmetric metric accommodates viscosity (Section 9.9), interaction with environmental shear dissipates these shared boundary constraints, providing a physical explanation for quantum decoherence.
CP Violation & Baryogenesis	As established in Section 10.6, Parity inversion maps to Charge conjugation ($P \to C$) for an isolated defect. However, the coupling energy between a charged defect ($\mathbf{v}_\perp$) and a chiral background ($\mathbf{\Omega}_0$) solves as the spatial dot product ($E_{CP} = \mathbf{v}_\perp \cdot \mathbf{\Omega}_0$). Under a combined CP transformation, the defect inverts ($\mathbf{v}_\perp \to -\mathbf{v}_\perp$), but the background vacuum remains un-inverted ($\mathbf{\Omega}_0 \to \mathbf{\Omega}_0$). The coupling energy shifts ($E_{CP} \to -E_{CP}$). Because the energy state is not invariant, perfect reversibility ($CP = I$) is prohibited.

Formalism (Continued)	**Continuum Tensor Mapping**
Coulomb Attraction & Positronium Equilibrium	Counter-rotating defects (e.g., an electron and positron) feature opposing rotational phases ($d\lambda_\Omega \wedge d\beta_\Omega$). Because their rotational flow fields align between the cores, their transverse slip 1-forms superimpose, amplifying the local dynamic pressure (P_{dyn}). By Bernoulli conservation, this local kinetic spike depletes the static capacity (P_{static}), causing the unperturbed background metric (P_c) to exert an inward compressive spatial gradient ($-\nabla P_{static}$), mechanically driving charge attraction. As the boundaries translate inward via this advection 1-form ($\mathbf{v}_{adv}$), the translation couples to the internal vorticity 2-form ($\mathbf{\Omega}$). This geometric interaction generates a kinematic helicity 3-form ($\mathcal{H} = \mathbf{v}_{adv} \wedge \mathbf{\Omega}$). Because the invariant contraction ($\star\mathcal{H} \wedge \mathcal{H}$) is bounded by the inverse volumetric capacity (α_s^{-4}), the continuum cannot sustain an infinite kinematic helicity 3-form. To conserve the volumetric trace, the 3-form induces an orthogonal spatial deflection, arresting the linear collapse and locking the defects into a stable rotational orbit prior to geometric intersection.
Atomic Orbital Quantization (The Rydberg Series)	To maintain a stable, non-radiating geometry within an electrostatic pressure gradient, the transverse shear wave operator evaluates to zero ($\Box A_{\mu\nu} = 0$). This boundary constraint forces the kinematic wave to form a continuous acoustic standing wave. Enforcing periodic boundary conditions on the transverse slip 1-form ($\mathbf{v}_\perp$) restricts the continuous spatial path integral to integer multiples of the specific kinematic action ($\oint \mathbf{v}_\perp \cdot d\mathbf{l} = n\frac{h}{m_e}$). Satisfying the resonant nodes of the cavity requires a radial action constraint ($\oint p_r dr \propto n$), forcing the spatial boundary to scale ($r_n \propto n^2$). This combined circulation confines allowed advection to discrete fractional states ($v_n \propto 1/n$). Substituting these phase-locked states into the tensor discretizes the dynamic pressure field ($P_{dyn,n} = \frac{1}{2}\rho_\tau v_n^2$). Multiplying this discrete mechanical load by the effective displacement of the defect (V_{eff}) computes the binding energy of the n-th geometric shell ($E_n \propto 1/n^2$). Discrete atomic orbitals and the Rydberg energy series emerge from the quantization of dynamic pressure required to maintain a structurally stable magnetic geometry, bypassing the probabilistic Schrödinger wave equation.

16.3 Cross-Coupled & Non-Linear Dynamics

Formalism	Continuum Tensor Mapping
Almansi Finite Strain	Near the cavitation limit ($r \to r_c$), deformations require non-linear dynamics. Mapping the spatial diagonals to the Eulerian Almansi strain tensor ($e_{ij} = \frac{1}{2}(\delta_{ij} - S_{ij}^{-1})$) models strain-hardening, reproducing Asymptotic Freedom.
SU(3) Symmetry, The Strong Interaction & The Mass Gap	Isolating the 3×3 spatial sub-matrix of $\hat{U}_{\mu\nu}$ assesses continuous deformation via the Cauchy strain tensor (ε_{ij}). For an incompressible core ($\nabla \cdot \mathbf{v} = 0$), the volumetric rate of change vanishes, isolating the continuous geometry to the deviatoric strain tensor (ε'_{ij}), which is trace-free ($\mathrm{Tr}(\varepsilon') = 0$). This leaves 5 real symmetric deviatoric components and 3 antisymmetric rotational components (A_{ij}). As derived in Section 9.6, at this boundary, the transverse rotational eigenstates are forced into the complex plane ($\lambda = 1 \pm i$). The 3 antisymmetric matrices align with the imaginary axis (iA_{ij}). The superposition of these 5 real symmetric matrices and 3 imaginary antisymmetric matrices spans the 8-dimensional space of traceless Hermitian matrices. This provides the isomorphism to the 8 geometric generators of the $\mathfrak{su}(3)$ Lie algebra. Because this causes the real elastic phase to equal the imaginary viscous phase (Re = Im), the Deborah number (Equation 149) evaluates to unity ($De = 1$). This exact threshold marks a glass transition phase. This yield requirement guarantees that SU(3) structures only manifest as solid-state boundaries, deriving the Yang-Mills mass gap as a phase transition.
The Lorentz Force	Contracting the antisymmetric off-diagonals against a velocity vector ($F_\mu \propto A_{\mu\nu} v^\nu$) produces the Lorentz force. This translates to the Magnus force exerted on a Clebsch vortex advecting through rotational shear.
Z-Pinch & Sausage Instability	Emerges from trace equilibrium under macroscopic rotational shear, generating azimuthal Clebsch vorticity (Ω_ϕ) elevating P_{shear}. To conserve the thermodynamic limit ($P_{static} = P_c - P_{shear}$), the scalar admittance (α_s^2) collapses. The symmetric spatial diagonals ($1/\alpha_s^2$) contract the radial geometry. If boundary pressure exceeds the restoring capacity, the boundary undergoes topological yield and a physical "sausage" fracture.
Acoustic Covariance	As $v \to c$, dynamic pressure drives longitudinal contraction ($L = L_0/\gamma$). Because the macroscopic boundary remains compressible ($\nu < 0.5$), transverse axes do not expand. The symmetric trace constraint forces the total volume ($V = V_0/\gamma$) to shrink, increasing internal frequency (relativistic mass). The zero-sum tensor trace couples longitudinal translational strain directly to transverse rotational strain. Internal Clebsch circulation accelerates proportionally. Kinematic acceleration bounds when the longitudinal axis compresses to the Planck length (l_p).

Formalism (Continued)	**Continuum Tensor Mapping**
Wake Detachment	Resolves the "Bullet Cluster" anomaly [14]. During high-velocity collisions, entrained wakes (β_ϕ) strip from baryonic anchors. Relativistic time variations between interacting frames generate spatial curvature gradients ($\nabla \hat{U}_{00} \neq 0$). Normalizing the antisymmetric transverse components (A_{ij}) against the Total Scalar Admittance (α_s) causes the macroscopic tension gradient to attenuate the amplitude of unanchored rotational shear. This structural cross-coupling stabilizes the detached GRIN lens.
U(1) Gauge Invariance	Translates gauge symmetry via the Clebsch parametrization ($\mathbf{v} = \nabla h + \mathbf{A}$). An arbitrary shift in the irrotational phase ($h \to h + \chi$) leaves the metric invariant if and only if the transverse field simultaneously transforms ($\mathbf{A} \to \mathbf{A} - \nabla\chi$). This Eulerian identity connects to $U(1)$ phase-rotation symmetry.
Viscoelastic Phase Angle	The ratio of the antisymmetric off-diagonals (transverse elastic shear) to the symmetric spatial diagonals (viscous bulk compliance) outputs the rheological phase angle ($\tan\delta$).
Electromagnetic Decoupling (Running Couplings)	Physical measurement of the gauge field isolates the mixed tensor ($A^\mu{}_\nu = S^{\mu\lambda}A_{\lambda\nu}$). Raising the index applies the inverse symmetric metric ($S^{00} = -1/\alpha_s^2$ and $S^{ii} = \alpha_s^2$), separating the effective coupling strength into orthogonal energy-dependent limits. For time-space interactions (Electric Slip), the mixed tensor scales inversely with scalar admittance ($A^0{}_i = (\alpha/\alpha_s^2)v_\perp/c$). As dynamic pressure approaches the metric capacity ($P_{dyn} \to P_c \implies \alpha_s \to 0$), the effective coupling diverges, structuring QED Vacuum Polarization. For space-space interactions (Magnetic Vorticity), the mixed tensor scales proportionally with scalar admittance ($A^i{}_j = (\alpha\alpha_s^2)t_p\Omega$). As dynamic pressure maximizes, the magnetic coupling bounds to zero. This sets the geometric decoupling of electric and magnetic fields across high-density topological boundaries.
Fourier Curvature Decomposition (QFT Momentum States)	Transforming the d'Alembertian wave operator ($\Box\hat{U}_{\mu\nu} = 0$) from coordinate space into momentum space via multidimensional Fourier transform translates spatial curvature (∇^2) as squared wavenumbers ($-k^2$) and temporal evolution (∂_t^2) as squared frequencies ($-\omega^2$). The asymmetric matrix requires this spectral decomposition to partition orthogonally into a symmetric longitudinal spectrum (gravitational potential) and an antisymmetric transverse spectrum (Clebsch vorticity). Because sharp spatial curvature (high k^2) mandates high kinematic energy states (ω^2), this decomposition provides the structural precursor for Quantum Field Theory, deriving discrete momentum states from continuous metric strain.

Formalism (Continued)	**Continuum Tensor Mapping**
Born-Infeld Nonlinear Electrodynamics	Born and Infeld [8] regularized point-charge self-energy by postulating a square-root determinant Lagrangian $\mathcal{L}_{BI} = b^2\big(1 - \sqrt{-\det(\eta_{\mu\nu} + F_{\mu\nu}/b)}\big)$ bounded by an empirical maximum field strength b. Because the generalized determinant (Section 9.7, Equation 146) already couples symmetric compliance and antisymmetric shear inside $\det(\hat{U}_{\mu\nu})$ (with Pfaffian term $(\mathbf{E}\cdot\mathbf{B})^2$ for non-orthogonal fields), the cavitation threshold ($\det(\hat{U}_{\mu\nu}) = 0$) provides the mechanical origin of the Born-Infeld cutoff, identifying the maximum field pressure b^2 with the critical vacuum capacity ($P_c = \frac{1}{2}\rho_\tau c^2$).

Formalization Scope, Postulates, and Lean 4 Parity

To maintain parity between the continuum derivations in this manuscript and the companion Lean 4 codebase (https://github.com/lilalittle/asymmetric-tensor), the operational scope and postulate boundaries are specified below:

1. **Dimensional Audit Basis (Sections 0.3, 16.4):** Physical dimensions are verified across 48 states on the Geometric Curvature Lattice $\langle K, \Omega, E \rangle \in \mathbb{Z}^3$ (`UnitDim`, tracking spatial curvature $k_p = 1/l_p$, temporal curvature $\Omega_p = 1/t_p = c/l_p$, and constitutive energy stiffness $E_P = \hbar\Omega_p$), with a proved unimodular $GL(3, \mathbb{Z})$ isomorphism to the extensive $\langle l_p, c, \hbar \rangle$ basis.

2. **Admissible Elastic Domain and Cavitation Boundary (Sections 8.1, 9.7–9.8):** Continuum compliance expressions containing $1/\alpha_s^2$ and $1/\alpha_s^4$ are defined on the subcritical domain $\alpha_s \in (0, 1]$. The cavitation threshold $\alpha_s \to 0^+$ is a singular boundary where $1/\alpha_s^2 \to +\infty$ while $\hat{U}_{00}\hat{U}_{ii} = -1$ is counterbalanced on the interior; Lean 4 evaluations at $\alpha_s = 0$ operate on denominator-free cleared polynomials to separate physical limits from totalized division conventions.

3. **Unpolarized Coarse-Graining Limit (Sections 4.3, 10.4):** The identification of the volumetric average operator (`VolAvg`) with matrix symmetrization (`symPart`) is scoped to macroscopically unpolarized domains where microscopic vorticity orientations average to zero ($\langle A_{\mu\nu} \rangle_V = 0$).

4. **Constitutive Incompatibility Postulate (Sections 4.3, 6.2, 12):** Section 6.2 formalizes the concrete second-order Fréchet-derivative Saint-Venant operator (`saint_venant_inc`) and its linearized Riemann symmetries (`linearized_riemann`), while Section 4.3 models macroscopic coarse-graining via a linear operator (`Inc`) commuting with `VolAvg` to verify torsion cancellation.

5. **Differential Operators and Action Variation (Sections 1, 5, 6.4–6.5, 13):** First-order partial derivatives (`partial_deriv`), exterior derivatives (`d_scalar_field`), rank-2 Lie derivatives (`lie_derivative`), and the witness-verified divergence-free equation of motion ($\partial_\mu \hat{U}^{\mu\nu} = 0$, `DynamicClebschFields`) are constructed via Fréchet derivatives (`fderiv`), whereas higher-order variational operators (`Laplacian`, `Biharmonic`, `dAlembertian`) are parameterized as linear maps to verify algebraic coupling identities.

6. **Model-Internal Exclusion Scope:** Exclusion statements regarding dark matter, cosmological acceleration, or external field theories apply within the constitutive axioms and boundary conditions of the $\hat{U}_{\mu\nu}$ model. Lean 4 compilation verifies internal algebraic and dimensional consistency, not empirical validation.

References

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